Finite Gaussian-Rational Approximation

Denominator-height bounds and a constructive Hurwitz–Lakein theorem

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Abstract

The finite rational-box Fundamental Theorem of Algebra supplies arbitrarily small polynomial residuals at displayed Gaussian rationals without first postulating completed complex roots. We first make the arithmetic cost of those samples explicit. For a displayed common-denominator fraction $q = (a + ib)/D$, put $H(q) = D^2$. If $f \in \mathbb{Q}(i)[Z]$ is nonzero, let S_f be the least positive integer clearing every reduced coefficient-coordinate denominator. For $f(q) \neq 0$,

$$1 \leq S_f^2 H(q)^{\deg f} \, \mathrm{N}(f(q)) \, .$$

The scaled residual is exactly the quadrance of a computed Gaussian integer, and the family $f(Z) = Z^d/S$, $q = 1/D$ attains equality for every positive S, D. An intrinsic version executes Gaussian Euclidean reduction, canonicalizes the remaining unit, and replaces $H(q)$ by the resulting presentation-independent denominator quadrance. Conversely, nearest-grid rounding constructs a fraction of exact height $(\lfloor (1/2)/r \rfloor + 1)^2$ inside the returned Newton band; this is the least denominator giving the required uniform strict half-mesh. It proves the improved bound $\mathrm{N}(f(q)) \leq 2L_1(f, A)\varepsilon^2$, and a bounded square-threshold scan chooses ε at the expected square-root scale for any displayed residual target. The same argument is extended to protected zero and pole tracks of a rational function, where the nonvanishing opposite polynomial contributes computable upper and lower quadrance bounds.

We then prove a finite Hurwitz–Lakein comparison. Nearest-Gaussian-integer division gives a terminating continued-fraction trace for every displayed Gaussian fraction z. If p_n/q_n is any generated convergent and p/q is any displayed Gaussian-integral competitor with $q \neq 0$ and $\mathrm{N}(q) < \mathrm{N}(q_n)$, then

$$\mathrm{N}(q_n z - p_n) \leq \mathrm{N}(qz - p) \, .$$

The proof uses only the finite Euclidean trace, adjacent-continuant identities, exact fundamental-square inequalities, and a finite Hermitian-form classification of the exceptional quadrance-five case.

We also recover a finite rational analogue of the all-best result. The strict-below/weak-at-equal-height definition of a best approximation is proved equivalent to minimality of the displayed vector $(qz - p, q)$ in its closed error–denominator cylinder. For $z = N/D$, the exact pair (N, D) and the nearest-integer half-square give explicit global denominator and numerator bounds. A terminating finite selector therefore returns all and only the best displayed pairs. A second finite filter returns exactly the pairs at the next occupied best denominator level above any supplied lower pair. Applied to every finite response of any RAT returned by the RFAP-1 FTA or by one of its signed rational-function sites, this becomes a RAT-facing program which emits only full best approximants to the displayed rational anchor of that response. From any supplied best pair, it returns a representative of the next occupied best level, or proves that the pair already represents the response exactly. Exact linear

FTA and rational-function sites receive the same arithmetic package directly at their displayed terminal point. This is deliberately a statement about finite observations, not unnamed limiting roots. Each consecutive pair supplies a finite Dirichlet-window certificate for every natural denominator-quadrance request between its endpoint levels. A normalized nearest-Gaussian-integer calculation gives a rational quadratic boundary, without naming either root. The displayed certificate 102/1189 then bounds the exact window product by 1189/102 times its coefficient-determinant quadrance; that determinant quadrance is exactly one or two. A second finite calculation rotates the normalized error ratio into one of eight rational coordinate octants and tests five small coefficient combinations. Their four-region polynomial certificates force the normalized determinant gap to be at least 205/256, so an index-one window is at most 256/205. In index two, four centered unit certificates first force both normalized coordinate quadrances to be at least 15/56; an oriented coordinate calculation with the affine minorant $2(U+1)/7$ then raises the gap to 205/128 and again bounds the window by 256/205. Thus the universal squared window bound is 256/205. More exactly, every displayed rational τ satisfying

$$\frac{3}{4} \le \tau < 1, \qquad \tau^2 - 12\tau + 9 \ge 0$$

obeys the root-free universal certificate $\tau\mathcal{D}_z(c_-, c_+) \le 1$. These rational cuts approach Chevallier's optimal squared constant from the finite side without constructing the quadratic endpoint. Conversely, every displayed rational τ on the sharper side of that boundary and every natural scale determine, by an explicit Pell recurrence, a displayed rational source and a consecutive index-one window beyond the requested scale for which $1 < \tau\mathcal{D}$. More generally, any requested finite number of such windows is realized at strictly increasing scales by one displayed rational source, and one proper positive-content rational box preserves every window for every displayed rational source inside it. On the universal side, a finite maximum over the complete best-pair table proves that every displayed rational source and every natural squared request $R > 1$ has a nonzero denominator below R satisfying $\tau R\,\mathrm{N}(qz-p)^2 \le 1$ for every admissible rational cut τ. The source used by the converse construction is still allowed to vary with the requested scale and finite run length. An exact parity calculation also identifies the index-two ambient coefficient pairs with the ordinary span together with its centered coset; two putative successive index-two transitions are placed in an explicit unit-and-divisible-coefficient normal form whose remaining Gaussian step has quadrance one, two, or four. Exact normalization and a finite cylinder calculation exclude all twelve values. Hence an index-two transition is necessarily followed by an index-one transition. Two adjacent window products obey $\tau(\mathcal{D}_0 + \mathcal{D}_1) \le 2$, and a chain of n windows obeys $\tau\sum_j \mathcal{D}_j \le n$ for every such τ; 512/205 and $(256/205)n$ remain convenient explicit consequences. From one supplied adjacent pair, a coefficient-local generator contains every pair at the next best level. Exact denominator and error comparisons within that local list return exactly that level, and its candidate list has an explicit quadratic size bound independent of the global numerator cutoff. The terminal Hurwitz pair satisfies the stronger best predicate, although the ordinary Hurwitz trace need not enumerate the whole returned list.

The results assume neither an irrational source nor an infinite continued fraction. All constructions are executable and formally verified in Lean with empty transitive axiom closures, without propositional extensionality, classical choice, or `Quot.sound`.

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1 Introduction

The companion paper [1] constructs the finite objects that replace the usual completed zeros and poles of a polynomial or rational function. A higher-degree site is represented by a root-approximation track (RAT): every displayed positive rational accuracy request produces a smaller rational box with clear boundary, winding one, and a coefficient-controlled residual at its rational anchor. That result establishes arbitrarily accurate finite responses,

but it leaves open two arithmetic questions. How large a denominator is necessary or sufficient for a requested residual? And does a natural finite continued-fraction calculation select samples that compare well with all competitors having smaller denominators?

This paper answers both questions for displayed Gaussian-rational data. The first answer is a coefficient-scaled denominator-height theory. Its coefficient scale is the least common multiple of the reduced coordinate denominators, and its lower obstruction is an exact Gaussian-integral norm identity. The common-denominator bound is attained by an explicit family; an executable Gaussian Euclidean reduction strengthens it to a presentation-independent denominator height. In the other direction, nearest-grid rounding constructs a uniformly least mesh denominator from an RFAP-1 Newton band, retains a sharper residual constant, and converts a displayed residual target into a square-root-scaled request. Protected Bézout bounds extend the same statements to zero and pole tracks of a rational function.

The second answer is a finite Hurwitz–Lakein theorem. Starting with an uncancelled fraction over the Gaussian integers, deterministic nearest-Gaussian-integer division produces a finite Euclidean trace and its adjacent continuants. Every generated convergent has no larger division-free residual quadrance than any displayed competitor of strictly smaller denominator quadrance. The usual language of an infinite continued fraction for a complex number is not needed: the source is finite fraction data, the trace terminates, and the comparison concerns one displayed stage and one displayed competitor at a time.

The stronger question is then answered for the same finite input class. We adopt the usual two-tier definition of a best approximation, identify it coefficientwise with Gaussian-lattice cylinder minimality, and compute the complete finite collection for a displayed rational source. This is not a claim that ordinary Hurwitz digits find them all. Instead the exact source pair supplies a global denominator cutoff, a cleared-error identity supplies a global numerator cutoff, and a bounded comparison pass decides the full predicate. A further bounded pass decides the consecutive-level relation between best pairs, and each returned successor gives an exact finite Dirichlet-window product. Normalizing the endpoint errors and denominators, then testing one nearest Gaussian integer, bounds that product by $1189/102$ times the coefficient index; an exact Cauchy calculation retains the underlying root-free quadratic boundary. Testing five small coefficient combinations in a finitely selected coordinate octant strengthens the gap to $205/256$ and bounds index-one windows by $256/205$. For index two, four centered unit certificates and an oriented affine-minorant calculation raise the normalized gap to $205/128$ and give the same window bound. The same calculations, parameterized by a displayed rational τ with $3/4 \le \tau < 1$ and $\tau^2 - 12\tau + 9 \ge 0$, prove the exact root-free family $\tau\mathcal{D} \le 1$. Hence two adjacent windows satisfy $\tau(\mathcal{D}_0 + \mathcal{D}_1) \le 2$, and a supplied chain of n windows satisfies $\tau \sum_j \mathcal{D}_j \le n$. The concrete bounds $256/205$, $512/205$, and $(256/205)n$ remain simple rational instances. If the quadratic polynomial is instead negative, an explicit Pell family constructs, beyond every supplied finite scale, a displayed rational source and consecutive window with $1 < \tau\mathcal{D}$. The terminal Hurwitz continuant is among these best pairs.

The final statement is compact; making its customary proof obey the same finite discipline was not. References to complete quotients, limiting values, geometric pictures, or cases "following by symmetry" conceal important proof data, especially on the half-open boundary chosen by a deterministic tie rule. The formal proof replaces those shortcuts by exact continuant identities, rational fundamental-square inequalities, and a complete finite classification of exceptional Hermitian forms. The resulting translation difficulties are foreshadowed here and discussed directly in section 12.

After this introduction states the terminology and main results, the remaining sections prove them. Sections 2–4 develop the denominator-height theory. Sections 5– 8 construct the finite Hurwitz trace, its succession geometry, the Hermitian-form invariant, and the Lakein comparison. Section 9 proves the minimal-vector correspondence, the finite all-best selector, and its exact consecutive-level successor filter. Section 10 gives explicit arithmetic envelopes for the RFAP-2 calculations, section 11 retains the exact cubic calculation, and section 12 closes with the constructive and formal lessons of the proof.

1.1 Statement, terminology, and constructive discipline

Every mathematical input in this paper is displayed finite data. Integers, rationals, Gaussian integers, Gaussian rationals, polynomial coefficient lists, rational boxes, continued-fraction digits, and proof certificates are all records on which the stated algorithms operate. For a displayed $z = x + iy \in \mathbb{Q}(i)$, its *quadrance* is the rational number

$$\mathrm{N}(z) = x^2 + y^2. \tag{1}$$

The residual quadrance of a displayed polynomial f at a displayed sample q is therefore $\mathrm{N}(f(q))$, computed by finite Horner evaluation followed by two rational squares.

The polynomial conventions are those of RFAP-1. A displayed polynomial is its normalized finite coefficient list, and polynomial equality means coefficientwise equality of those normalized lists. Evaluation preserves a proved polynomial identity, but agreement at evaluated values is never used to define polynomial equality. The letter f denotes the polynomial under the main approximation argument; P, Q denote a numerator and denominator; A is a fixed ambient rational box; B is a current or returned box; and q is a displayed Gaussian-rational sample.

Two fraction presentations occur and must be distinguished.

- A *common-denominator fraction* is written $q = (a + ib)/D$ with $a, b \in \mathbb{Z}$ and $D \in \mathbb{N}_{>0}$. Its presentation-dependent height is $H(q) = D^2$.
- A *Gaussian fraction* is written $z = N/D$ with $N, D \in \mathbb{Z}[i]$ and $D \neq 0$. Its *displayed Gaussian-denominator height* is the quadrance $h_D(z) = \mathrm{N}(D) \in \mathbb{N}$.

The first presentation is convenient for polynomial coefficient clearing and rational grids. The second is the natural input to Euclidean division over $\mathbb{Z}[i]$. The code applies an executable extended Euclidean calculation to N, D, divides by its displayed common factor, and rotates the result by one of the four Gaussian units so that the reduced denominator has positive real part and nonnegative imaginary part. If the resulting canonical fraction is N_*/D_*, the *intrinsic Gaussian-denominator height* is

$$h(z) = \mathrm{N}(D_*)\,.$$

This height is proved independent of the supplied presentation. Thus no minimization over a quotient is hidden in the notation: cancellation, unit-normalization, and the accompanying Bézout certificate are finite operations performed by the stated program.

No completed zero is introduced in the height argument. Its lower bound is an integer-divisibility statement about the displayed value $f(q)$, not a claim about the distance to a postulated algebraic point. Its upper bound constructs a rational grid point inside a finite Newton band, rather than appealing to density in a completed plane. Similarly, the Hurwitz argument does not introduce an irrational source or an infinite digit stream. A

source Gaussian fraction has a finite Euclidean trace; a *stage* is one finite prefix together with its current complete quotient and continuants; and a comparison is an exact rational inequality between displayed quadrances.

The same convention applies to geometric wording:

- membership in a rational box means four decidable rational inequalities;
- a statement about every $q \in B$ is conditional on a supplied displayed q and its membership inequalities;
- membership in the Hurwitz fundamental square means four displayed rational coordinate inequalities; and
- a finite family, scan, trace, or classification is an explicit list or natural recursion, not a completed collection or an actually infinite process.

1.1.1 Related work

The foundational comparison with constructive FTA work is developed more fully in RFAP-1 [1]. Richman [2] constructs a choice-free completion and represents a monic polynomial by a completed limit multiset of approximate factors. The present height theory instead consumes the finite rational-box interface before any completion is formed. Perrucci and Roy [3] obtain quantitative degree bounds for an effective real-algebraic FTA. Our lower height obstruction is narrower and more elementary: the least common denominator of the displayed coefficient coordinates turns its scaled residual exactly into the quadrance of one Gaussian integer. The sufficient bound, however, depends essentially on the finite Newton-band construction of RFAP-1.

Hurwitz [4] introduced the nearest-Gaussian-integer continued fraction and its square fundamental region. LeVeque's two-part study [5] develops continued fractions and approximation in $\mathbb{Q}(i)$. Lakein [6] isolates the approximation properties needed here: Hurwitz convergents satisfy a second-kind comparison against Gaussian-rational competitors, even though the complex theory does not reproduce every familiar real best-approximation property. The headline inequality of this paper is the finite, calculation-based counterpart of that comparison.

Later work places Hurwitz's algorithm in broader geometric and dynamical settings. Dani and Nogueira [7] treat general complex continued-fraction algorithms and their relation to binary quadratic forms. Ei, Ito, Nakada, and Natsui [8] construct the natural extension of the Hurwitz map and analyze its fundamental-region geometry. Those developments explain why admissible digit succession is subtler than the single bound $\mathrm{N}(\rho) \leq 1/2$. They work with completed complex inputs, infinite expansions, or dynamical spaces; our proof extracts only the finite successor inequalities needed for a displayed rational trace.

Hurwitz convergents need not enumerate every best complex approximation. Chevallier [9] identifies best approximations with minimal vectors in a two-dimensional Gaussian lattice and gives a lattice-successor algorithm reaching all of them. The present paper proves the corresponding minimal-vector equivalence directly for displayed rational data, gives a complete bounded selector, and filters it into the exact consecutive occupied denominator levels. This recovers the all-best conclusion and its finite successor relation for that input class, including the local empty-cylinder characterization. It also supplies a coefficientwise Cramer certificate and proves constructively that the determinant quadrance of consecutive pairs is exactly one or two. This recovers Chevallier's sharp index dichotomy for displayed rational sources without his Minkowski or module-basis route. We

also prove the centered-coset description in the index-two case by displayed coordinate parity. Four explicit centered-coset candidates then give a square-root-free error-ratio bound and reduce the coefficient in a putative repeated-index transition to twelve displayed Gaussian integers. Normalizing the two error and denominator coordinates then reduces these twelve values, by displayed unit associates, to the three coefficients $1+i$, 2, and $2+2i$. A finite exact cylinder calculation excludes all three, so an index-two transition is followed by an index-one transition. This proves the displayed-rational counterpart of Chevallier's no-repeated-index lemma. We also formalize its coefficient-local candidate generation consequence, give it a coefficient-local validation pass, and prove that the resulting executable update skips no next best level. It is a bounded coefficient enumeration rather than Chevallier's more economical structural formula. We do not treat arbitrary completed complex inputs or an infinite minimal-vector sequence. For displayed rational inputs, however, five small coefficient tests in one of eight finite coordinate octants, together with four oriented centered unit tests in index two, give the exact rational-cut family

$$\frac{3}{4} \le \tau < 1, \quad \tau^2 - 12\tau + 9 \ge 0 \quad \Longrightarrow \quad \tau\mathcal{D} \le 1.$$

The smaller boundary of this cut is conventionally $6 - 3\sqrt{3}$, whose reciprocal is Chevallier's optimal squared constant $(2+\sqrt{3})/3 \approx 1.24$. Thus the present finite result recovers the full upper boundary without constructing that endpoint. In the reverse direction, for every displayed rational τ satisfying the same range and $\tau^2 - 12\tau + 9 < 0$, and every natural scale, the Pell recurrence produces an explicit displayed rational source and a consecutive window beyond that scale with $1 < \tau\mathcal{D}$. The strengthened finite form returns an arbitrary finite same-source run and a proper positive-content rational box preserving it. For every displayed rational source in that box, the bounded coefficient-local successor table gives a stopped regenerative kernel: genuine successor edges strictly increase denominator quadrance, a terminal edge resets to the selected critical seed, and the seed is then held. Natural-rank absorption therefore gives probability one of a bad-window hit after any burn-in and one positive finite block horizon. The source and box may depend on the requested scale and finite run length, and the stop and reset are explicit boundary rules rather than Haar invariance. This does not recover Chevallier's almost-everywhere fixed-source sharpness statement, nor his formulation for arbitrary completed complex inputs and an infinite minimal-vector sequence. Separately, our Hurwitz theorem still claims only that each generated convergent is at least as good as every competitor of strictly smaller denominator quadrance.

Formal continued-fraction developments also exist. Eberl's Isabelle formalization [10] treats real continued fractions, periodic quadratic irrationals, approximation, and executable applications such as Pell equations. The present formalization has a different boundary: it treats Gaussian fractions and the Hurwitz–Lakein comparison, and it audits the public theorem specifically to exclude classical choice and completed infinite objects from its proof dependency.

1.2 Machine verification

The complete machinery described in this paper has been implemented and verified in Lean. The accompanying development contains the executable coefficient-clearing, grid, Gaussian Euclidean, continuant, and stage-trace programs together with machine-checked proofs of the paper-facing guarantees. The *Formal verification* markers throughout the paper name the corresponding Lean declarations.

This is a particularly appropriate use of an automated theorem prover. Every construction is finite and explicitly checkable: recursive calls decrease a displayed natural measure, searches traverse displayed finite lists, polynomial identities are coefficientwise, and geometric assertions reduce to integer or rational equalities and inequalities. The proof assistant can check the algorithms at the same level at which the theorem is stated, including their termination and the exact dependency closure of the result.

RFAP paper 1 [1] records the same formal-verification problems. In particular, apparently elementary generic Mathlib routes can introduce `Classical.choice` through an elaborated instance or inherited theorem. The present development again uses project-owned constructive rational and Gaussian algebra, direct order lemmas, and a repository-wide dependency audit. The public RFAP-2 declarations have empty audited transitive axiom closures; neither `propext`, `Classical.choice`, `Quot.sound`, nor any other axiom occurs. Independent nanoda checking enforces the same empty-axiom policy. The scope and separate compiler-helper obligation are described in RFAP-1's formalization appendix. The paper-specific problem—turning customary Hurwitz–Lakein prose into finite machine-checkable certificates—belongs to the mathematical discussion in section 12.

Formal verification:

`Purelib.Papers.RFAP.Approximation.scaledHeightResidual_eq_integralNorm;`
`Purelib.Papers.RFAP.Approximation.one_le_scaled_gaussianHeight_mul_`
`residual;Purelib.Papers.RFAP.Approximation.denominatorHeightBound_isSharp;`
`Purelib.Papers.RFAP.Approximation.approximateWithTarget_residual_le;`
`Purelib.Papers.RFAP.Approximation.hurwitzConvergent_atLeastAsGoodAs;`
`Purelib.Papers.RFAP.Approximation.mem_allBestGaussianApproximations_iff`.

1.3 Definitions and theorem statements

We now give, in dependency order, the definitions needed to read the main results. This subsection is deliberately a fast path to the main theorem statements. It records the essential finite interfaces and defers intuition, derivation, and the larger calculations to the later proof sections.

1.3.1 Height data and finite RFAP input

Definition 1.1 (least coefficient scale and displayed heights)**.** *For a displayed polynomial* $f(Z) = \sum_{j=0}^{d} c_j Z^{d-j} \in \mathbb{Q}(i)[Z]$*, let* $s(c_j)$ *be the least common multiple of the reduced denominators of* $\operatorname{Re} c_j$ *and* $\operatorname{Im} c_j$*. The* coefficient scale *is*

$$S_f = \operatorname*{lcm}_j s(c_j).$$

Thus $S_f c_j \in \mathbb{Z}[i]$ *for every* j*, and* S_f *divides every natural scale with this property. For a common-denominator fraction* $q = (a + ib)/D$*, put* $H(q) = D^2$*. For a displayed Gaussian fraction* $q = U/V$*, put* $h_V(q) = \mathrm{N}(V)$*. The extended Gaussian Euclidean calculation divides* U, V *by their displayed common factor, and a deterministic choice among* $1, -1, i, -i$ *makes the reduced denominator* V_* *have positive real part and nonnegative imaginary part. Write the canonical result as* $\operatorname{red}(q) = U_*/V_*$ *and put* $h(q) = \mathrm{N}(V_*)$*. With* $A_j = S_f c_j \in \mathbb{Z}[i]$*, define the integral homogeneous numerator*

$$W_f(U, V) = \sum_{j=0}^{d} A_j U^{d-j} V^j \in \mathbb{Z}[i].$$

Formal verification:

`Purelib.Polynomial.DenominatorHeight.coefficientScale;Purelib.Polynomial.DenominatorHeight.coefficientScale_dvd_of_clears;Purelib.Data.Gaussian.Height.Fraction.height;Purelib.NumberTheory.Approximation.HurwitzRational.Fraction.height;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduceCanonical;Purelib.Polynomial.GaussianDenominatorHeight.intrinsicHeight;Purelib.Papers.RFAP.Approximation.intrinsicGaussianDenominatorHeight`.

The geometric input imported from RFAP-1 is the following finite interface. Its hypotheses are displayed coefficient data for a squarefree polynomial f of degree at least two, a certified winding state inside an ambient box A, and the finite axis-elimination data required for boundary-clear cuts.

Proposition 1.2 (RFAP-1 Newton-band interface)**.** *For every displayed $\varepsilon > 0$, finite Newton extraction computes a centre $c = x + iy \in \mathbb{Q}(i)$, a radius $r > 0$, and a rational box B such that*

(i) the left, right, bottom, and top sides of B lie respectively at coordinate distances between r and $2r$ from c;
(ii) B has clear boundary, winding one, and $\operatorname{diag}^2(B) \le \frac{1}{2}\varepsilon^2$; and
(iii) every displayed $q \in B$ satisfies

$$\mathrm{N}(f(q)) \le 2L_1(f, A)\varepsilon^2, \tag{2}$$

where $L_1(f, A) > 0$ is computed from the displayed coefficients and endpoints by finite Horner recursions.

The returned box, centre, radius, winding calculation, and coefficient bound are output data. The proposition does not quantify over an unshown point.

Formal verification:

`Purelib.RootIsolation.Newton.Band.extract;Purelib.RootIsolation.Newton.Band.Output.diagonalSq_le_half;Purelib.Polynomial.CoprimeBounds.normSq_eval_le_four_lipschitz_diagonal;Purelib.Polynomial.HeightApproximation.approximate`.

Theorem 1.3 (finite denominator-height package)**.** *Let $f \in \mathbb{Q}(i)[Z]$ be displayed and nonzero, of degree d.*

(i) For $q = u/D$ in common-denominator form,

$$S_f^2 H(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(u, D))\,.$$

If $f(q) \neq 0$, this natural number is at least one.
(ii) For $q = U/V$ in displayed Gaussian-denominator form,

$$S_f^2 h_V(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(U, V))\,,$$

and again a nonzero residual makes the common value at least one.
(iii) For the canonical reduction $\mathrm{red}(q) = U_*/V_*$,

$$S_f^2 h(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(U_*, V_*))\,.$$

The height $h(q)$ is positive, no larger than $h_V(q)$, and equal for any two displayed fractions representing the same exact value. A nonzero residual again makes the common value at least one.

(iv) *For every $d \in \mathbb{N}$ and positive S, D, the displayed family $f(Z) = Z^d/S$, $q = 1/D$ has $S_f = S$, $H(q) = D^2$, nonzero residual, and makes the common value in (i) exactly one. The unit constant and the displayed powers of S_f and $H(q)$ are therefore jointly sharp.*

(v) *At a protected numerator or denominator site of a rational function, the lower obstruction transfers to the shrinking quotient residual after including the displayed upper majorant of the opposite polynomial.*

Formal verification:

`Purelib.Papers.RFAP.Approximation.scaledHeightResidual_eq_integralNorm; Purelib.Papers.RFAP.Approximation.one_le_scaled_height_mul_residual; Purelib.Papers.RFAP.Approximation.one_le_scaled_gaussianHeight_mul_residual;Purelib.Papers.RFAP.Approximation.intrinsicGaussianDenominatorHeight_le_displayed;Purelib.Papers.RFAP.Approximation.intrinsicGaussianDenominatorHeight_eq_of_value_eq;Purelib.Papers.RFAP.Approximation.scaledIntrinsicGaussianHeightResidual_eq_integralNorm;Purelib.Papers.RFAP.Approximation.one_le_scaled_intrinsicGaussianHeight_mul_residual;Purelib.Papers.RFAP.Approximation.denominatorHeightBound_isSharp;Purelib.Papers.RFAP.Approximation.one_le_scaled_height_mul_quotientResidual`.

Theorem 1.4 (finite sufficient-height package)**.** *In the Newton extraction of Proposition 1.2, put*

$$D_r = \left\lfloor \frac{1/2}{r} \right\rfloor + 1.$$

Nearest-grid rounding returns a displayed $q_r \in B$ with

$$H(q_r) = D_r^2, \qquad \mathrm{N}(f(q_r)) \le 2L_1(f, A)\varepsilon^2.$$

The number D_r is the least positive denominator whose uniform half-mesh is strictly smaller than r. For every displayed target $\delta > 0$, a bounded scan computes a reciprocal-integer request ε at square-root scale and makes the last residual at most δ. At a protected rational-function site the same construction applies to the shrinking quotient residual after division by the positive protection margin of the opposite polynomial.

Formal verification:

`Purelib.Polynomial.HeightApproximation.gridDenominator_minimal;Purelib.Papers.RFAP.Approximation.approximateWithHeight;Purelib.Papers.RFAP.Approximation.approximateWithTarget_residual_le;Purelib.Papers.RFAP.Approximation.approximateQuotientWithHeight;Purelib.Papers.RFAP.Approximation.approximateQuotientWithTarget_residual_le`.

The two packages are proved in sections 2 to 4.

1.3.2 Finite Hurwitz data

Definition 1.5 (nearest Gaussian integer and fundamental square)**.** *For $x \in \mathbb{Q}$, put*

$$[x]_{\mathrm{n}} = \left\lfloor x + \frac{1}{2} \right\rfloor.$$

Thus a half-integer tie is sent upward. For $z = x + iy \in \mathbb{Q}(i)$, define

$$[z]_{\mathbb{Z}[i]} = [x]_{\mathrm{n}} + i[y]_{\mathrm{n}}, \qquad \rho(z) = z - [z]_{\mathbb{Z}[i]}.$$

The chosen half-open fundamental square is

$$\mathfrak{F} = \left\{ u + iv : -\frac{1}{2} \le u < \frac{1}{2}, -\frac{1}{2} \le v < \frac{1}{2} \right\}. \tag{3}$$

Finite floor inequalities certify $\rho(z) \in \mathfrak{F}$ and $\mathrm{N}(\rho(z)) \le 1/2$.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzNearest.nearestCoordinate;`
`Purelib.NumberTheory.Approximation.HurwitzNearest.nearest;Purelib.`
`NumberTheory.Approximation.HurwitzNearest.InFundamentalSquare;Purelib.`
`NumberTheory.Approximation.HurwitzNearest.remainder_mem;Purelib.`
`NumberTheory.Approximation.HurwitzNearest.remainder_normSq_le_half`.

Definition 1.6 (finite Hurwitz transition and encoding)**.** *Let $z = N/D$ be a displayed Gaussian fraction. Its digit is $a = [z]_{\mathbb{Z}[i]}$ and its integral remainder is*

$$R = N - aD \in \mathbb{Z}[i].$$

If $R = 0$, the trace is terminal. If $R \neq 0$, the next displayed fraction is

$$T(z) = \frac{D}{R} = \rho(z)^{-1}.$$

The exact norm identity

$$\mathrm{N}(R) = \mathrm{N}(\rho(z))\,\mathrm{N}(D)$$

and $\mathrm{N}(\rho(z)) \le 1/2$ give $\mathrm{N}(R) < \mathrm{N}(D)$. Recursion on this strictly decreasing natural denominator quadrance therefore returns a finite, nonempty digit list and an exact reconstruction certificate for z.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.digit;Purelib.`
`NumberTheory.Approximation.HurwitzRational.remainder;Purelib.NumberTheory.`
`Approximation.HurwitzRational.next;Purelib.NumberTheory.Approximation.`
`HurwitzRational.remainder_norm_lt_denominator;Purelib.NumberTheory.`
`Approximation.HurwitzRational.encodeResult;Purelib.NumberTheory.`
`Approximation.HurwitzRational.encode_reconstruct`.

Definition 1.7 (continuants, generated stages, and denominator ratio)**.** *For finite digits $a_0, \ldots, a_n \in \mathbb{Z}[i]$, define adjacent continuants by*

$$(p_{-1}, q_{-1}) = (1, 0), \qquad (p_0, q_0) = (a_0, 1),$$

and

$$(p_{j+1}, q_{j+1}) = a_{j+1}(p_j, q_j) + (p_{j-1}, q_{j-1}). \tag{4}$$

A Hurwitz stage *records one displayed complete quotient, the adjacent pairs (p_{j-1}, q_{j-1}) and (p_j, q_j), their unit-determinant certificate, and the exact reconstruction identity for the source. A stage is* generated *when it is the initial stage or is obtained from a generated stage by one nonterminal transition. Its* denominator ratio *is the displayed Gaussian rational*

$$r_j = \frac{q_{j-1}}{q_j}. \tag{5}$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.ConvergentPair;Purelib.NumberTheory.Approximation.HurwitzTrace.advancePair;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage;Purelib.NumberTheory.Approximation.HurwitzGeometry.Generated;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.denominatorRatio`.

Definition 1.8 (division-free error and Lakein comparison)**.** *For a displayed source z and an integral pair $(p, q) \in \mathbb{Z}[i]^2$, define its division-free error and error quadrance by*

$$E_z(p, q) = zq - p, \qquad \mathcal{E}_z(p, q) = \mathrm{N}(E_z(p, q)) .$$

A convergent pair (p_n, q_n) is at least as good as *a competitor (p, q) when*

$$\mathcal{E}_z(p_n, q_n) \le \mathcal{E}_z(p, q).$$

This is a second-kind comparison: denominator size is used only to select the competitors, while the compared quantity has been cleared of division.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzApproximation.error;Purelib.NumberTheory.Approximation.HurwitzApproximation.AtLeastAsGoodAs`.

Theorem 1.9 (finite Hurwitz–Lakein inequality)**.** *Let $z = N/D$ be a displayed Gaussian fraction, let a generated stage have current continuant (p_n, q_n), and let $(p, q) \in \mathbb{Z}[i]^2$ be any displayed competitor. If*

$$q \neq 0, \qquad \mathrm{N}(q) < \mathrm{N}(q_n) ,$$

then

$$\mathrm{N}(zq_n - p_n) \le \mathrm{N}(zq - p) . \tag{6}$$

At a terminal stage the left side is zero. Every earlier generated stage is covered by the finite fundamental-square and Hermitian-form invariant proved below.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.atLeastAsGoodAs_of_generated;Purelib.Papers.RFAP.Approximation.hurwitzConvergent_atLeastAsGoodAs`.

The proof is completed in section 8.

Definition 1.10 (best approximation and source-lattice minimality)**.** *Let $c = (p, q) \in \mathbb{Z}[i]^2$ with $q \neq 0$. It is a* best approximation *to the displayed source z when every $d = (a, b) \in \mathbb{Z}[i]^2$ with $b \neq 0$ satisfies the two-tier comparison*

$$\mathrm{N}(b) < \mathrm{N}(q) \Longrightarrow \mathcal{E}_z(c) < \mathcal{E}_z(d), \tag{7}$$

$$\mathrm{N}(b) \le \mathrm{N}(q) \Longrightarrow \mathcal{E}_z(c) \le \mathcal{E}_z(d). \tag{8}$$

The associated displayed source-lattice vector is $(E_z(c), q)$. It is minimal *when every nonzero displayed pair d in its closed cylinder,*

$$\mathcal{E}_z(d) \le \mathcal{E}_z(c), \qquad \mathrm{N}(b) \le \mathrm{N}(q) ,$$

has equality in both coordinates. This is a definition on coefficient records; it does not first form a completed lattice carrier.

Two best pairs c_- and c_+ occupy consecutive best levels *when*

$$\mathrm{N}(q_-) < \mathrm{N}(q_+)$$

and no best pair has denominator quadrance strictly between these two values. All unit associates and any other ties remain at the same level. For a displayed rational source, this is equivalently the local condition that no nonzero coefficient pair c satisfies both

$$\mathcal{E}_z(c) < \mathcal{E}_z(c_-), \qquad \mathrm{N}(q_c) < \mathrm{N}(q_+)\,.$$

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.IsBest;Purelib.NumberTheory.Approximation.BestApproximation.sourceVector;Purelib.NumberTheory.Approximation.BestApproximation.sourceVector_injective;Purelib.NumberTheory.Approximation.BestApproximation.pairNonzero_iff_sourceVector_ne_zero;Purelib.NumberTheory.Approximation.BestApproximation.InClosedCylinder;Purelib.NumberTheory.Approximation.BestApproximation.IsGaussLatticeMinimal;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest`.

Theorem 1.11 (finite minimal-vector correspondence and all-best selector)**.** *For every displayed Gaussian fraction $z = N/D$ and every displayed pair c, the following are equivalent:*

1. *c is a best approximation in the sense of (7)–(8);*
2. *the source-lattice vector $(E_z(c), q)$ is minimal in its closed error–denominator cylinder;*
3. *c occurs in the executable finite list* $\mathrm{AllBest}(z)$.

If $B = \mathrm{N}(D)$ and

$$P = 2(\mathrm{N}(N)\ B + B),$$

the selector need inspect only denominators whose two component magnitudes are at most B and numerators whose two component magnitudes are at most P. Its rectangular candidate list has exactly

$$16(B+1)^2(P+1)^2$$

entries before filtering; duplicated zero components are harmless. The terminal pair of every generated Hurwitz trace is best. Every generated pair, terminal or not, retains the weaker conclusion of theorem 1.9.

For any supplied lower pair c_-, a second executable list $\mathrm{NextBest}(z, c_-)$ *satisfies*

$$c_+ \in \mathrm{NextBest}(z, c_-) \quad \Longleftrightarrow \quad c_-, c_+ \text{ occupy consecutive best levels.} \tag{9}$$

Thus the finite all-best collection also has an exact computed successor relation. This first implementation filters the global rectangle; the later coefficient-local update computes the same relation from an adjacent state.

For the rational anchor z_R carried by any finite response of an RFAP-1 RAT, the same calculation emits all and only best approximants to z_R. Selecting the first member of the next occupied level defines a total optional step: if the supplied pair c is best, the step returns a consecutive best pair, or else $\mathcal{E}_{z_R}(c) = 0$. Conversely, the option is empty exactly

when this error is zero. Thus no provisional or merely nearby candidate can be emitted by the RAT-response interface. The target here is the displayed rational response carried by the finite call; no completed root is introduced. This applies to every degree-indexed RAT returned by the finite FTA and to every higher-degree signed numerator or denominator site returned by RFAP-1; their exact linear branches use the same calculation at the displayed terminal point.

For every returned consecutive pair, the displayed coefficient determinant satisfies the sharp finite alternative

$$\mathrm{N}(p_- q_+ - p_+ q_-) \in \{1, 2\}.$$

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.isBest_iff_isGaussLatticeMinimal;Purelib.NumberTheory.Approximation.BestApproximation.candidates_length;Purelib.NumberTheory.Approximation.BestApproximation.mem_allBest_iff_isBest;Purelib.NumberTheory.Approximation.BestApproximation.mem_consecutiveSuccessors_iff;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.coefficientIndexQuadrance_eq_one_or_two;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.isWeakLowerBest_of_generated;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.isBest_of_generated_of_terminal;Purelib.Papers.RFAP.Approximation.mem_allBestGaussianApproximations_iff;Purelib.Papers.RFAP.Approximation.mem_consecutiveBestSuccessors_iff;Purelib.NumberTheory.Approximation.RATBestApproximation.respondToPoint;Purelib.NumberTheory.Approximation.RATBestApproximation.respondToTrack;Purelib.Papers.RFAP.Approximation.strengthenFTAResult;Purelib.Papers.RFAP.Approximation.strengthenRationalSite;Purelib.Papers.RFAP.Approximation.consecutiveBest_coefficientIndexQuadrance_eq_one_or_two`.

The proof is completed in section 9.

1.4 Proof-route glossary

The entries below are a dependency-ordered map of the proof. They summarize the formal definitions rather than replacing them.

Displayed data and certificate. An object is *displayed* when it is finite integer or rational data on which the stated calculation runs. A *certificate* is displayed data together with finite checks that imply its advertised property; later algorithms consume certificates and return new ones.

Quadrance and residual. The *quadrance* of $x + iy$ is $x^2 + y^2$. A polynomial *residual* is $\mathrm{N}(f(q))$ at a displayed sample. A continued-fraction *division-free residual* is $\mathrm{N}(zq - p)$ for a displayed integral pair (p, q).

Common-denominator height and Gaussian denominator height. For $q = (a + ib)/D$ with positive integer D, the *displayed common-denominator height* is $H(q) = D^2$. For $z = N/D$ with nonzero Gaussian-integer D, the *displayed Gaussian denominator*

height is $h_D(z) = \mathrm{N}(D)$. Gaussian Euclidean reduction and a four-unit normalization produce canonical coordinates N_*/D_*; the *intrinsic Gaussian denominator height* is $h(z) = \mathrm{N}(D_*)$. Equal displayed Gaussian fractions have identical canonical coordinates and hence the same intrinsic height.

Coefficient scale and height obstruction. The *coefficient scale* S_f is the least common multiple of all reduced real and imaginary coefficient-coordinate denominators. Homogeneous Horner evaluation then produces a Gaussian integer W_f satisfying $W_f = D^d S_f f(q)$ for a displayed common denominator D, or $W_f = V^d S_f f(q)$ for a displayed Gaussian denominator V. Applying the same calculation to the canonical reduced denominator D_* replaces the presentation-dependent height by $h(q)$. Its quadrance is exactly the corresponding scaled residual. If the displayed residual is nonzero, $\mathrm{N}(W_f) \geq 1$ is the *height obstruction*.

RAT, Newton band, and grid fraction. A *RAT* is the RFAP-1 finite request-and-response program returning strictly refining, boundary-clear, winding-one rational boxes. A *Newton band* is its coefficient-certified box around a displayed Newton centre. The *grid fraction* rounds both centre coordinates with the nearest-integer rule and denominator $\lfloor (1/2)/r \rfloor + 1$, and lies strictly inside that band. This denominator is least for the uniform strict half-mesh condition. A *square-threshold request* uses a bounded scan for the least reciprocal-integer ε sufficient for a displayed quadratic residual target.

Protected quotient residual. At a numerator site for P/Q, a protected RFAP-1 state supplies a positive lower bound and a finite upper majorant for $\mathrm{N}(Q(q))$. These calibrate the *quotient residual* $\mathrm{N}(P(q)) \,/\, \mathrm{N}(Q(q))$. At a pole the roles are exchanged, so the shrinking quantity is the reciprocal quotient residual.

Nearest Gaussian integer and fundamental square. The *nearest Gaussian integer* rounds each rational coordinate by $\lfloor x + 1/2 \rfloor$. Its residual lies in the half-open *fundamental square* $[-1/2, 1/2)^2$. Retaining which sides are open and closed is essential to the strict successor inequalities.

Integral remainder, complete quotient, and finite trace. For $z = N/D$ and digit a, the *integral remainder* is $R = N - aD$. If it is nonzero, the next *complete quotient* is D/R. The strict decrease $\mathrm{N}(R) < \mathrm{N}(D)$ makes iteration a terminating *finite trace*.

Continuant and generated stage. A *continuant* is an integral pair of a numerator and denominator produced by the recurrence (4). A *generated stage* retains two adjacent continuants, their unit determinant, the current complete quotient, and an exact reconstruction of the original source.

Competitors and Lakein factor. The adjacent unit determinant expresses every displayed competitor (p, q) with explicit Gaussian-integer coefficients s, t. Its error factors as the current convergent error times an explicit *Lakein factor*. Proving the competitor is no better therefore reduces to proving that factor has quadrance at least one.

Denominator growth and denominator ratio. The *denominator-growth certificate* is $\mathrm{N}(q_{j-1}) < \mathrm{N}(q_j)$. The *denominator ratio* $r_j = q_{j-1}/q_j$ is consequently a displayed Gaussian rational of quadrance less than one. Its finite backward succession is the state variable used by the exceptional proof.

Successor geometry and forbidden pair. The fundamental-square inequalities imply exact exterior-disc conditions for successive digits. A *forbidden pair* is one of the finitely classified small adjacent-digit pairs contradicted by those conditions, including the direction of every half-open boundary tie.

Hermitian root form and pullback. The remaining quadrance-five comparison is one of eight displayed integral *Hermitian root forms* evaluated at r_j. A digit *pulls back* a form by substituting the exact backward-ratio transformation and clearing its positive denominator quadrance. Coefficientwise calculation proves the pullback identity.

Exceptional form and root-form invariant. Most pulled-back root forms have an immediate nonnegative slack certificate. The remaining twenty-four *exceptional forms* are checked across one more finite predecessor step; every unsafe result would require a forbidden digit pair. Induction on the displayed generated-stage proof yields the *root-form invariant*.

Finite Lakein inequality. The root-form invariant supplies the last quadrance-five Lakein-factor case; the smaller coefficient cases follow from elementary fundamental-square geometry and the larger cases from denominator estimates. The factor has quadrance at least one, so multiplying by the current error proves (6). A terminal trace instead gives exact zero error.

Best approximation, minimal vector, and finite successors. A *best approximation* is strictly better than every lower-height pair and weakly better than every pair of no greater height. Its displayed source-lattice vector is *minimal* when no nonzero vector enters both the interior error range and the no-greater denominator range of its closed cylinder. For a rational source, the *all-best selector* checks this condition over one source-bounded rectangle; its output contains precisely the best pairs, including the terminal Hurwitz pair but not only Hurwitz pairs. Two best pairs occupy *consecutive best levels* when their denominator quadrances are strictly ordered and no best pair has an intermediate denominator quadrance. The *finite successor selector* returns every pair at the consecutive level above a supplied lower pair. A finite lexicographic descent proves that this global level condition is equivalent to emptiness of the corresponding *mixed open cylinder*. The coefficient determinant of two consecutive pairs has quadrance exactly one or two; a computed Bézout complement and one reduced Gaussian remainder prove this without constructing a quotient lattice.

Centered coefficient coset. When the consecutive determinant δ has quadrance two, divisibility by δ is exactly even parity of the sum of the displayed real and imaginary coordinates. The *centered coefficient coset* consists of the pairs x for which $\delta x = au + bv$ with both cleared coefficients nondivisible by δ. Together with the ordinary integral span it contains every ambient displayed pair, and every pair of nondivisible coefficients constructs

such an x. For two putative successive index-two transitions, this classification gives the *repeated-index normal form* $w = \varepsilon u + \delta k v$ with $\mathrm{N}(\varepsilon) = 1$ and $\mathrm{N}(k) \in \{1, 2, 4\}$. The bound on k comes from four explicit centered candidates and the square-root-free error estimate $\mathcal{E}_z(u) < 4\mathcal{E}_z(v)$. Dividing the two recurrences by the displayed nonzero error and denominator references gives *normalized cylinder data*. Unit rotation reduces its transition coefficient to $1 + i$, 2, or $2 + 2i$; the corresponding finite cylinder exclusions rule out all three cases.

Coefficient-local successor generator. Given adjacent best pairs u, v, the *local successor candidates* are computed from the eight leading coefficients of quadrance one or two and a bounded trailing coefficient. They use either the integral form $au+bv$ or the centered form obtained by exact quadrance-two division. Every actual next best pair occurs in this list. The *local successor output* retains the improving candidates having least denominator quadrance in this list and, among candidates on that level, least error quadrance. This is a local finite calculation, and its output is proved to be exactly the next best level.

Dirichlet window and finite sharpness witness. For consecutive best pairs c_-, c_+, the *squared Dirichlet-window product* is $\mathcal{D}_z(c_-, c_+) = \mathrm{N}(q_+)\,\mathcal{E}_z(c_-)$. A displayed rational threshold τ on the certified side of the quadratic boundary satisfies $\tau\mathcal{D} \le 1$ for every such window. A *finite sharpness witness* for a threshold on the other side consists of an explicitly computed displayed rational source and consecutive pair whose denominator quadrance exceeds a supplied natural scale and for which $1 < \tau\mathcal{D}$. The witness source is output data and may change when the requested scale changes.

Arithmetic envelope. An *arithmetic envelope* is a natural-number recurrence computed from the displayed input before the calculation it bounds. It limits recursive clauses or the bit widths of the integral records produced by the grid, homogeneous-evaluation, and Hurwitz programs. It is not a maximum recovered by inspecting an already completed trace.

2 The denominator-height obstruction

This section turns the glossary's coefficient scale into the universal height obstruction. Finite coefficient clearing and homogeneous Horner evaluation construct one displayed Gaussian integer. If the residual is nonzero, that integer is nonzero and hence has quadrance at least one. The entire lower bound is the quadrance of this coefficientwise identity. We first make the scale least, then prove both common- and Gaussian-denominator forms of the identity, and finally exhibit equality cases.

Definition 2.1 (least coefficient-denominator scale)**.** *Write a rational coordinate in its reduced numerator–positive-denominator form and denote its denominator by* $\mathrm{den}(-)$*. For* $c \in \mathbb{Q}(i)$*, put*

$$s(c) = \mathrm{lcm}\big(\mathrm{den}(\mathrm{Re}\, c), \mathrm{den}(\mathrm{Im}\, c)\big).$$

For the normalized displayed coefficient list $(c_0, \dots, c_d)$*, define*

$$S_f = \mathrm{lcm}_{0 \le j \le d}\, s(c_j). \tag{10}$$

The finite fold starts with 1*, so* S_f *is positive even for an empty intermediate list.*

Lemma 2.2 (coefficient-scale minimality)**.** *The number S_f clears every coefficient: $S_f c_j \in \mathbb{Z}[i]$. If a natural number T clears every reduced real and imaginary coefficient denominator, then $S_f \mid T$.*

Proof. For one coefficient, each coordinate denominator divides its least common multiple. At each step of the finite list fold, the current least common multiple and the remaining-list scale both divide the new one, proving that S_f clears every coordinate. Conversely, if T is divisible by the two coordinate denominators of every coefficient, it is divisible by each $s(c_j)$. Induction through the same fold and the defining universal property of the least common multiple give $S_f \mid T$. □

Formal verification:

`Purelib.Polynomial.DenominatorHeight.coefficientDenominator;Purelib.Polynomial.DenominatorHeight.lcmScale;Purelib.Polynomial.DenominatorHeight.lcmScale_clears;Purelib.Polynomial.DenominatorHeight.coefficientScale_dvd_of_clears`.

Definition 2.3 (homogeneous residual numerator)**.** *Let*

$$f(Z) = \sum_{j=0}^{d} c_j Z^{d-j},$$

and put $A_j = S_f c_j \in \mathbb{Z}[i]$. For $U, V \in \mathbb{Z}[i]$, the homogeneous residual numerator *is the Gaussian integer*

$$W_f(U, V) = \sum_{j=0}^{d} A_j U^{d-j} V^j \in \mathbb{Z}[i]. \tag{11}$$

It is computed by a high-to-low homogeneous Horner recursion on the displayed coefficient list; the sum is notation for the resulting finite fold.

Formal verification:

`Purelib.Polynomial.DenominatorHeight.homogeneousEval;Purelib.Polynomial.DenominatorHeight.residualNumerator;Purelib.Polynomial.GaussianDenominatorHeight.homogeneousEval;Purelib.Polynomial.GaussianDenominatorHeight.residualNumerator`.

Lemma 2.4 (homogeneous evaluation identity)**.** *For a common-denominator fraction $q = u/D$, with D regarded as the Gaussian integer $D+0i$, the displayed numerator satisfies*

$$W_f(u, D) = D^d S_f f(q) \tag{12}$$

after embedding both sides in $\mathbb{Q}(i)$. More generally, a Gaussian fraction $q = U/V$ with $V \neq 0$ satisfies

$$W_f(U, V) = V^d S_f f(q). \tag{13}$$

Proof. Coefficient clearing first proves by induction on the normalized coefficient list that ordinary Horner evaluation of the cleared list is $S_f f(q)$. Independently, induction on that same list proves that homogeneous Horner evaluation at (U, V) is V^d times ordinary evaluation at U/V. Composing the two coefficientwise identities gives (13); taking $V = D + 0i$ gives (12). No conclusion is inferred from equality on a collection of evaluated points. □

Formal verification:

`Purelib.Polynomial.DenominatorHeight.eval_lcmClear;Purelib.Polynomial.DenominatorHeight.homogeneousEval_cons_eq_scaled;Purelib.Polynomial.DenominatorHeight.residualNumerator_identity;Purelib.Polynomial.GaussianDenominatorHeight.residualNumerator_identity`.

Lemma 2.5 (nonzero residual gives nonzero integral numerator)**.** *If f is nonzero, $q = u/D$ is in common-denominator form, and the displayed evaluation $f(q)$ is nonzero, then $W_f(u, D) \neq 0$ in $\mathbb{Z}[i]$.*

Proof. The positive integers D and S_f embed as nonzero Gaussian rationals. Their product with the nonzero displayed value $f(q)$ is therefore nonzero by direct Gaussian multiplication. The identity (12) then prevents $W_f(u, D)$ from being zero. □

Formal verification:

`Purelib.Polynomial.DenominatorHeight.residualNumerator_ne_zero`.

Lemma 2.6 (Gaussian-integral quadrance floor)**.** *For every displayed $w \in \mathbb{Z}[i]$, if $w \neq 0$, then $1 \leq \mathrm{N}(w)$.*

Proof. Write $w = x + iy$ with $x, y \in \mathbb{Z}$. Its quadrance $x^2 + y^2$ is a natural number. If it were zero, the two nonnegative integer squares would both be zero and hence $w = 0$. Positivity of the nonzero natural quadrance gives the stated lower bound. □

Formal verification:

`Purelib.Data.Gaussian.Height.GaussianInt.normNat_pos`.

Theorem 2.7 (exact common-denominator height identity)**.** *If f is nonzero and $q = u/D$ is a displayed common-denominator fraction, then*

$$S_f^2 H(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(u, D))\,. \tag{14}$$

If the displayed evaluation $f(q)$ is nonzero, then

$$1 \leq S_f^2 H(q)^d \,\mathrm{N}(f(q))\,. \tag{15}$$

Consequently a nonzero residual is at least $1/(S_f^2 H(q)^d)$, with equality exactly when $\mathrm{N}(W_f(u, D)) = 1$.

Proof. Taking quadrances in (12) gives the exact calculation

$$\mathrm{N}(W_f(u, D)) = D^{2d} S_f^2 \,\mathrm{N}(f(q)) = S_f^2 H(q)^d \,\mathrm{N}(f(q))\,,$$

which is (14). If $f(q) \neq 0$, then lemmas 2.5 and 2.6 give $1 \leq \mathrm{N}(W_f(u, D))$, proving (15). The equality criterion follows from the exact identity. □

Formal verification:

`Purelib.Polynomial.DenominatorHeight.scaled_height_mul_residual_eq_normNat;Purelib.Polynomial.DenominatorHeight.scaled_height_mul_residual_eq_one_iff;Purelib.Polynomial.DenominatorHeight.one_le_scaled_height_mul_residual;Purelib.Papers.RFAP.Approximation.scaledHeightResidual_eq_integralNorm;Purelib.Papers.RFAP.Approximation.one_le_scaled_height_mul_residual`.

Theorem 2.8 (displayed Gaussian-denominator height identity)**.** *Let $q = U/V$ be a displayed Gaussian fraction with $V \neq 0$, and put $h_V(q) = \mathrm{N}(V)$. If f is nonzero, then*

$$S_f^2 h_V(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(U,V))\,. \tag{16}$$

If $f(q) \neq 0$, the common value is at least one, and hence

$$\mathrm{N}(f(q)) \geq \frac{1}{S_f^2 h_V(q)^d}. \tag{17}$$

Proof. Take quadrances in (13). The rules $\mathrm{N}(ab) = \mathrm{N}(a)\,\mathrm{N}(b)$ and $\mathrm{N}\left(V^d\right) = \mathrm{N}(V)^d$ give (16). If $f(q) \neq 0$, the nonzero factors V^d and S_f make $W_f(U,V)$ nonzero. Its Gaussian-integral quadrance is therefore at least one by lemma 2.6. □

Formal verification:

`Purelib.Polynomial.GaussianDenominatorHeight.scaled_height_mul_residual_eq_normNat;Purelib.Polynomial.GaussianDenominatorHeight.one_le_scaled_height_mul_residual;Purelib.Papers.RFAP.Approximation.one_le_scaled_gaussianHeight_mul_residual`.

Definition 2.9 (canonical Gaussian reduction and intrinsic height)**.** *For a displayed Gaussian fraction $q = U/V$, run the extended Gaussian Euclidean algorithm on (U,V). Its output includes a displayed common factor G, quotients U_0, V_0, and Bézout coefficients satisfying*

$$U = GU_0, \qquad V = GV_0, \qquad 1 = AU_0 + BV_0.$$

Multiply U_0, V_0 by the uniquely selected unit in $\{1, -1, i, -i\}$ for which the new denominator V_* has $\mathrm{Re}(V_*) > 0$ and $\mathrm{Im}(V_*) \geq 0$. Write the resulting fraction as

$$\mathrm{red}(q) = U_*/V_*$$

and define its intrinsic Gaussian-denominator height *by*

$$h(q) = \mathrm{N}(V_*)\,.$$

The calculation preserves the exact value and gives

$$\mathrm{N}(V) = \mathrm{N}(G)\; h(q), \qquad 0 < h(q) \leq \mathrm{N}(V)\,.$$

If two displayed fractions have equal exact values, cross multiplication and their two displayed Bézout identities show that their reduced denominators are unit associates. The chosen denominator sector then forces the unit to be 1, after a finite classification of the four quadrance-one Gaussian integers. Their canonical numerators and denominators, and therefore their intrinsic heights, are equal. This proves presentation independence by an executable normalization rather than by selecting a representative from a quotient.

Formal verification:

`Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduce;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduce_value;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduce_height_factor;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduce_height_le;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.`

`canonicalize;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.canonicalize_value;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.canonicalize_height;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduceCanonical;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduceCanonical_value;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduceCanonical_coordinates_eq_of_value_eq;Purelib.NumberTheory.Approximation.ReducedGaussianFraction.reduceCanonical_height_eq_of_value_eq;Purelib.Polynomial.GaussianDenominatorHeight.intrinsicHeight;Purelib.Polynomial.GaussianDenominatorHeight.intrinsicHeight_le_height;Purelib.Polynomial.GaussianDenominatorHeight.intrinsicHeight_eq_of_value_eq;Purelib.Papers.RFAP.Approximation.canonicalReducedGaussianFraction;Purelib.Papers.RFAP.Approximation.intrinsicGaussianDenominatorHeight`.

Theorem 2.10 (intrinsic Gaussian-denominator height identity)**.** *Let* $\mathrm{red}(q) = U_*/V_*$ *be the canonical reduction from definition 2.9. If* f *is nonzero, then*

$$S_f^2 h(q)^d \,\mathrm{N}(f(q)) = \mathrm{N}(W_f(U_*, V_*))\,. \tag{18}$$

If $f(q) \neq 0$*, the common value is at least one, and hence*

$$\mathrm{N}(f(q)) \geq \frac{1}{S_f^2 h(q)^d}. \tag{19}$$

Proof. The executable reduction preserves q exactly. Apply theorem 2.8 to its canonical displayed coordinates (U_*, V_*). Their displayed denominator height is, by definition, the intrinsic height $h(q)$. The exact identity and the nonzero-residual lower bound follow unchanged. □

Formal verification:

`Purelib.Polynomial.GaussianDenominatorHeight.intrinsicResidualNumerator;Purelib.Polynomial.GaussianDenominatorHeight.scaled_intrinsicHeight_mul_residual_eq_normNat;Purelib.Polynomial.GaussianDenominatorHeight.one_le_scaled_intrinsicHeight_mul_residual;Purelib.Papers.RFAP.Approximation.scaledIntrinsicGaussianHeightResidual_eq_integralNorm;Purelib.Papers.RFAP.Approximation.one_le_scaled_intrinsicGaussianHeight_mul_residual`.

Theorem 2.11 (sharpness of the common-denominator obstruction)**.** *For every* $d \in \mathbb{N}$ *and positive* $S, D \in \mathbb{N}$*, let*

$$f(Z) = \frac{1}{S} Z^d, \qquad q = \frac{1}{D}.$$

Then $S_f = S$*,* $\deg f = d$*,* $H(q) = D^2$*,* $f(q) \neq 0$*, and*

$$S_f^2 H(q)^d \,\mathrm{N}(f(q)) = 1. \tag{20}$$

Thus the unit constant, the square on S_f*, and the degree-th power of* $H(q)$ *are jointly sharp for a uniform theorem depending only on these displayed parameters.*

Proof. The only nonzero coefficient is the leading coordinate $1/S$, whose reduced denominator is S; every zero coordinate has denominator 1. The finite least-common-multiple calculation therefore gives $S_f = S$. Finite monomial evaluation gives $f(1/D) = 1/(SD^d) \neq 0$. Taking its quadrance and substituting $H(q) = D^2$ makes every factor cancel, giving (20). □

Formal verification:

`Purelib.Polynomial.DenominatorHeightSharp.sharpPolynomial;Purelib.Polynomial.DenominatorHeightSharp.sharpFraction;Purelib.Polynomial.DenominatorHeightSharp.sharpPolynomial_coefficientScale;Purelib.Polynomial.DenominatorHeightSharp.sharpPolynomial_natDegree;Purelib.Polynomial.DenominatorHeightSharp.scaled_height_mul_residual_eq_one;Purelib.Papers.RFAP.Approximation.denominatorHeightBound_isSharp`.

The two displayed height identities have an exact relation. A positive common denominator D is the Gaussian denominator $D + 0i$, with the same value and displayed height. Conversely, multiplying a Gaussian fraction U/V by $\overline{V}/\overline{V}$ produces the common-denominator record

$$\frac{U\overline{V}}{\mathrm{N}(V)},$$

with the same value; displayed Gaussian height $h_V = \mathrm{N}(V)$ becomes common-denominator height h_V^2. These remain useful finite conversions. Canonical Gaussian reduction is stronger: it cancels every displayed common Gaussian factor, removes the remaining unit ambiguity, and supplies the presentation-independent height used in theorem 2.10. Scaling an unreduced presentation can weaken either displayed bound, while the intrinsic bound is unchanged. Finally, theorem 2.11 shows that the common-denominator identity cannot be uniformly improved from its displayed scale, height, and degree alone.

Formal verification:

`Purelib.Polynomial.GaussianDenominatorHeight.ofCommon_value;Purelib.Polynomial.GaussianDenominatorHeight.ofCommon_height;Purelib.Polynomial.GaussianDenominatorHeight.toCommon_value;Purelib.Polynomial.GaussianDenominatorHeight.toCommon_height;Purelib.Polynomial.GaussianDenominatorHeight.intrinsicHeight_le_height`.

3 A constructive sufficient height

This section follows the sufficient-height branch of the glossary. From the displayed Newton centre and band radius it computes one grid denominator, rounds the two centre coordinates, and proves strict box membership. The RFAP-1 residual certificate can then be applied to that displayed fraction.

Definition 3.1 (Newton grid denominator and fraction)**.** *For a displayed centre $c = x + iy \in \mathbb{Q}(i)$ and radius $r > 0$, put*

$$D_r = \left\lfloor \frac{1/2}{r} \right\rfloor + 1, \qquad q_r = \frac{[D_r x]_{\mathrm{n}} + i[D_r y]_{\mathrm{n}}}{D_r}.$$

The same positive denominator is used in both coordinates, so q_r is a displayed common-denominator fraction of exact height $H(q_r) = D_r^2$.

Formal verification:

`Purelib.Polynomial.HeightApproximation.gridDenominator;Purelib.Polynomial.HeightApproximation.gridCoordinate;Purelib.Polynomial.HeightApproximation.gridFraction;Purelib.Polynomial.HeightApproximation.gridFraction_height`.

Lemma 3.2 (strict and least uniform grid mesh)**.** *For every displayed $r > 0$, the denominator of definition 3.1 satisfies*

$$\frac{1/2}{D_r} < r.$$

Moreover, if $E > 0$ is a natural number and $(1/2)/E < r$, then $D_r \le E$.

Proof. Put $t = (1/2)/r$. The floor certificate gives $\lfloor t \rfloor \le t < \lfloor t \rfloor + 1 = D_r$. Clearing the positive factors r and D_r in the strict inequality gives $(1/2)/D_r < r$.

Conversely, $(1/2)/E < r$ gives $t < E$. Since E is integral, $\lfloor t \rfloor < E$, and hence $D_r = \lfloor t \rfloor + 1 \le E$. □

Formal verification:

`Purelib.Polynomial.HeightApproximation.gridDenominator_pos;Purelib.Polynomial.HeightApproximation.half_reciprocal_gridDenominator_lt;Purelib.Polynomial.HeightApproximation.gridDenominator_minimal`.

Lemma 3.3 (nearest coordinate bracket)**.** *For every displayed $t \in \mathbb{Q}$,*

$$t - \frac{1/2}{D_r} < \frac{[D_r t]_{\mathrm{n}}}{D_r} \le t + \frac{1/2}{D_r}.$$

Consequently the grid fraction satisfies

$$x - r < \operatorname{Re} q_r < x + r, \qquad y - r < \operatorname{Im} q_r < y + r.$$

Proof. The tie-up nearest-integer certificate is

$$-\frac{1}{2} \le D_r t - [D_r t]_{\mathrm{n}} < \frac{1}{2}.$$

Rearrange it, divide by the positive D_r, and retain the resulting strict lower and weak upper half-mesh endpoints. The strict mesh inequality of lemma 3.2 makes both final band inequalities strict, independently in the real and imaginary coordinates. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzNearest.nearestCoordinate_error_bounds;Purelib.Polynomial.HeightApproximation.gridCoordinate_bounds`.

Lemma 3.4 (strict membership in the Newton band)**.** *The fraction q_r lies strictly inside every Newton band with centre c and radius r returned by Proposition 1.2.*

Proof. A Newton band places its left and bottom sides at least r below the corresponding centre coordinate and its right and top sides at least r above it. The strict coordinate bounds of lemma 3.3 therefore place both coordinates strictly between the certified sides. □

Formal verification:

`Purelib.Polynomial.HeightApproximation.gridFraction_strictly_inside`.

Lemma 3.5 (exact Newton-band geometric coefficient)**.** *Every Newton band of radius r has*

$$\operatorname{diag}^2(B) \le 32r^2.$$

The band using all four outer allowed sides has equality. In particular, when $\varepsilon = 8r$, that band satisfies

$$\operatorname{diag}^2(B) = \frac{1}{2}\varepsilon^2. \tag{21}$$

Thus the factor $1/2$ retained by Proposition 1.2 cannot be improved from the Newton-band side constraints and radius budget alone.

Proof. Each side length is at most $4r$, so the sum of their squares is at most $16r^2 + 16r^2 = 32r^2$. Choosing the left, right, bottom, and top sides at distance exactly $2r$ from the centre gives both side lengths $4r$ and hence equality. Substitution of $\varepsilon = 8r$ gives $32r^2 = (1/2)(8r)^2$. □

Formal verification:

`Purelib.RootIsolation.Newton.Band.NewtonBandBox.diagonalSq_le;Purelib.RootIsolation.Newton.Band.widestNewtonBand;Purelib.RootIsolation.Newton.Band.widestNewtonBand_diagonalSq;Purelib.RootIsolation.Newton.Band.widestNewtonBand_diagonalSq_eq_half_request`.

Theorem 3.6 (constructive height–residual calibration)**.** *In the Newton extraction of Proposition 1.2, the algorithm constructs the fraction q_r of definition 3.1 and certifies*

$$H(q_r) = \left(\left\lfloor \frac{1/2}{r} \right\rfloor + 1\right)^2, \qquad \mathrm{N}(f(q_r)) \le 2L_1(f, A)\varepsilon^2. \tag{22}$$

Proof. Proposition 1.2 returns a Newton band with squared diagonal at most $(1/2)\varepsilon^2$. Definition 3.1 gives the displayed height identity, and Lemma 3.4 puts q_r strictly inside the returned band. The RFAP-1 bound first gives

$$\mathrm{N}(f(q_r)) \le 4L_1(f, A)\operatorname{diag}^2(B).$$

Substituting $\operatorname{diag}^2(B) \le (1/2)\varepsilon^2$ gives the stated factor 2. □

Formal verification:

`Purelib.Polynomial.HeightApproximation.approximate;Purelib.Polynomial.HeightApproximation.Output.height_eq;Purelib.Polynomial.HeightApproximation.Output.residual_le;Purelib.Papers.RFAP.Approximation.approximateWithHeight`.

Corollary 3.7 (a displayed residual target)**.** *Let $\delta > 0$ be displayed, put $C = 2L_1(f, A) > 0$, and compute $n = \lceil C/\delta \rceil$. Scan the positive integers from 1 through the certified endpoint $n + 2$, let m be the first for which $n < m^2$, and put $\varepsilon_\delta = 1/m$. It certifies*

$$C\varepsilon_\delta^2 \le \delta. \tag{23}$$

Calling Theorem 3.6 with this square-root-scaled request returns a displayed fraction of exact sufficient height and residual quadrance at most δ.

Proof. The endpoint $n+2$ works because $n < (n+2)^2$, so the finite scan terminates; checking candidates in ascending order proves its minimality. The ceiling certificate and the selected strict square threshold give

$$\frac{C}{\delta} \le n < m^2.$$

Multiplying by the displayed positive δ/m^2 yields $C/m^2 \le \delta$, which is (23). The residual conclusion follows from (22). □

Formal verification:

`Purelib.Polynomial.HeightApproximation.squareThreshold;Purelib.Polynomial.HeightApproximation.squareThreshold_spec;Purelib.Polynomial.HeightApproximation.squareThreshold_minimal;Purelib.Polynomial.HeightApproximation.targetDenominator;Purelib.Polynomial.HeightApproximation.targetEpsilon;Purelib.Polynomial.HeightApproximation.coefficient_mul_targetEpsilon_sq_le;Purelib.Polynomial.HeightApproximation.approximateTarget_residual_le;Purelib.Papers.RFAP.Approximation.approximateWithTarget_residual_le`.

The lower and upper bounds answer different questions. Theorem 2.7 rules out too small a residual for a specified displayed height. Theorem 3.6 constructs one sufficient height from the actual band radius. It does not claim that this height is least for the particular centre or polynomial. Lemma 3.2 proves only the precise uniform statement available: D_r is least among denominators whose half-mesh is strictly smaller than r. A particular centre may happen to admit a smaller denominator, and the nearest grid point need not minimize the polynomial residual at its height.

The Lean development also retains a checked exhaustive residual minimiser for a window with both numerator coordinates and the positive common denominator bounded. This routine finite-list calculation is not part of the height theorem and no later result depends on it; unlike Hurwitz–Lakein, it imposes an independent numerator bound.

Formal verification:

`Purelib.Polynomial.BestResidual.best_is_optimal`.

4 Protected rational-function sites

This section applies the height package to the glossary's protected quotient residual. The argument is symmetric: call the polynomial being made small the *source* g and the other polynomial the *opposite* h. A zero site of P/Q uses $(g,h) = (P,Q)$; a pole site uses $(g,h) = (Q,P)$. A displayed Bézout certificate first keeps h uniformly away from zero on a protected source box. Its lower margin and upper Horner majorant then transfer the polynomial height bounds to the quotient.

Definition 4.1 (quotient residual)**.** *For displayed polynomials $g, h \in \mathbb{Q}(i)[Z]$ and a displayed sample q at which $h(q) \neq 0$, define*

$$\mathcal{R}_{g/h}(q) = \mathrm{N}(g(q)/h(q)) = \frac{\mathrm{N}(g(q))}{\mathrm{N}(h(q))}. \tag{24}$$

The nonvanishing proof is part of every quantitative use of this expression.

Formal verification:

`Purelib.RationalFunctions.ResidualChordal.quotientResidualSq;Purelib.Papers.RFAP.Paper.quotientResidualSq`.

Definition 4.2 (Bézout protection data)**.** *Let A be a rational ambient box. A displayed coprimality certificate for g, h supplies polynomials $U, V \in \mathbb{Q}(i)[Z]$ with the coefficientwise identity*

$$Ug + Vh = 1. \tag{25}$$

Let $\mathsf{M}_A(R) > 0$ denote the finite Horner majorant for $\mathrm{N}(R(q))$ on A, and put

$$\theta_g = \frac{1}{16\mathsf{M}_A(U)}, \qquad \eta_h = \frac{1}{16\mathsf{M}_A(V)}, \qquad M_h = \mathsf{M}_A(h). \tag{26}$$

The protected source state *is obtained by asking the source RAT for the explicit positive radius*

$$\rho_{g,h} = \min\left\{1, \frac{\theta_g}{8L_1(g,A)}, \frac{\eta_h}{2L_1(h,A)}\right\}. \tag{27}$$

A local quotient context records this certificate, the source RAT, its current state, and the resulting protected state.

Formal verification:

`Purelib.Polynomial.Bounds.denseEvalNormMajorant;Purelib.Polynomial.CoprimeBounds.leftThreshold;Purelib.Polynomial.CoprimeBounds.rightMargin;Purelib.Polynomial.CoprimeBounds.protectionRadius;Purelib.Polynomial.CoprimeBounds.protectedState;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext`.

Lemma 4.3 (the opposite polynomial is protected)**.** *Every displayed q in the protected source state satisfies*

$$0 < \eta_h \le \mathrm{N}(h(q)) \le M_h. \tag{28}$$

In particular, $h(q) \neq 0$ and the quotient residual (24) is defined there.

Proof. The source residual estimate and the first radius budget in (27) give $\mathrm{N}(g(q)) \le \theta_g$. Suppose the decidable rational comparison for the lower bound instead returned $\mathrm{N}(h(q)) < \eta_h$. Evaluating (25), using multiplicativity of quadrance, and applying the Horner bounds would give

$$\mathrm{N}(U(q)g(q)) \le \mathsf{M}_A(U)\theta_g = \frac{1}{16}, \qquad \mathrm{N}(V(q)h(q)) \le \mathsf{M}_A(V)\eta_h = \frac{1}{16}.$$

The exact two-term quadrance estimate then calculates

$$1 = \mathrm{N}(1) = \mathrm{N}(U(q)g(q) + V(q)h(q)) \le 2\left(\frac{1}{16} + \frac{1}{16}\right) = \frac{1}{4},$$

which the rational order decision rejects. Hence $\eta_h \le \mathrm{N}(h(q))$. Positivity of η_h gives nonvanishing. The upper bound is the defining Horner-majorant property for h on A, applied after box refinement. □

Formal verification:

`Purelib.Polynomial.CoprimeBounds.leftThreshold_pos;Purelib.Polynomial.CoprimeBounds.rightMargin_pos;Purelib.Polynomial.CoprimeBounds.rightMargin_le_of_left_small;Purelib.Polynomial.CoprimeBounds.protectedState_left_small;Purelib.Polynomial.CoprimeBounds.protectedState_right_lower;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.oppositeLower;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.oppositeNonzero;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.oppositeUpper`.

Lemma 4.4 (protected quotient transfer)**.** *For every displayed q in a state refining the protected source state,*

$$\mathrm{N}(g(q)) \le M_h \mathcal{R}_{g/h}(q), \qquad \mathcal{R}_{g/h}(q) \le \frac{\mathrm{N}(g(q))}{\eta_h}. \tag{29}$$

Proof. By lemma 4.3, $h(q) \neq 0$. Taking quadrances in the displayed identity

$$(g(q)/h(q))h(q) = g(q)$$

therefore gives

$$\mathcal{R}_{g/h}(q)\,\mathrm{N}(h(q)) = \mathrm{N}(g(q))\,.$$

Multiplication by the nonnegative quotient residual preserves the upper bound $\mathrm{N}(h(q)) \le M_h$, proving the first inequality. Multiplication by the positive inverse of η_h preserves the lower bound $\eta_h \le \mathrm{N}(h(q))$, proving the second. □

Formal verification:

`Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.quotientResidualSq_mul_oppositeNorm;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.sourceResidual_le_quotientResidualSq_mul_majorant;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.quotientResidualSq_le_diagonal`.

Theorem 4.5 (protected quotient height obstruction)**.** *Let q be a displayed common-denominator fraction in a state refining the protected g-state. If $g(q) \neq 0$, then*

$$1 \le S_g^2 H(q)^{\deg g} M_h \mathcal{R}_{g/h}(q). \tag{30}$$

Proof. Theorem 2.7 gives the same lower bound with $\mathrm{N}(g(q))$ as its last factor. Substitute the first inequality of (29) and multiply by the nonnegative displayed scale and height factors. □

Formal verification:

`Purelib.RationalFunctions.DenominatorHeight.one_le_scaled_height_mul_quotientResidual;Purelib.Papers.RFAP.Approximation.one_le_scaled_height_mul_quotientResidual`.

Theorem 4.6 (constructive protected quotient height)**.** *For every displayed $\varepsilon > 0$, Newton-grid extraction within the protected g-state returns a displayed q_r such that*

$$H(q_r) = \left(\left\lfloor \frac{1/2}{r} \right\rfloor + 1\right)^2, \qquad \mathcal{R}_{g/h}(q_r) \le \frac{2L_1(g,A)\varepsilon^2}{\eta_h}. \tag{31}$$

Proof. Theorem 3.6 constructs q_r in the protected source box, with the displayed height and $\mathrm{N}(g(q_r)) \le 2L_1(g,A)\varepsilon^2$. Apply the second inequality of (29). The only division is by the explicitly positive rational margin η_h. □

Formal verification:

`Purelib.RationalFunctions.DenominatorHeight.Output;Purelib.RationalFunctions.DenominatorHeight.approximate;Purelib.Papers.RFAP.Approximation.approximateQuotientWithHeight`.

Corollary 4.7 (a displayed protected quotient target)**.** *For a displayed target $\delta > 0$, put*

$$C_{g/h} = \frac{2L_1(g,A)}{\eta_h}.$$

Apply the bounded square-threshold scan of corollary 3.7 to $(C_{g/h}, \delta)$. Its reciprocal-integer request ε_δ makes Newton-grid extraction return a protected sample with

$$\mathcal{R}_{g/h}(q_r) \le \delta.$$

Proof. The coefficient $C_{g/h}$ is positive because both the Lipschitz bound and the protection margin are explicitly positive. The scan proves $C_{g/h}\varepsilon_\delta^2 \le \delta$, while (31) gives $\mathcal{R}_{g/h}(q_r) \le C_{g/h}\varepsilon_\delta^2$. □

Formal verification:

`Purelib.RationalFunctions.DenominatorHeight.targetCoefficient;Purelib.RationalFunctions.DenominatorHeight.targetCoefficient_pos;Purelib.RationalFunctions.DenominatorHeight.approximateTarget;Purelib.RationalFunctions.DenominatorHeight.approximateTarget_quotientResidual_le;Purelib.Papers.RFAP.Approximation.approximateQuotientWithTarget_residual_le`.

Corollary 4.8 (zero and pole specialisations)**.** *For a protected zero track of P/Q, take $(g,h) = (P,Q)$ in theorems 4.5 and 4.6 and corollary 4.7. For a protected pole track take $(g,h) = (Q,P)$. The shrinking quantity at a pole is therefore $\mathcal{R}_{Q/P}$, with bounds*

$$1 \le S_Q^2 H(q)^{\deg Q} M_P \mathcal{R}_{Q/P}(q), \qquad \mathcal{R}_{Q/P}(q_r) \le \frac{2L_1(Q,A)\varepsilon^2}{\eta_P}.$$

The opposite polynomial genuinely affects the rational-function estimate. At a pole the numerator majorant M_P occurs in the necessary lower bound, while its Bézout protection margin η_P occurs in the sufficient upper bound and in the square-root target coefficient $2L_1(Q,A)/\eta_P$ for the reciprocal residual. If finite evaluation instead finds $Q(q_r) = 0$, it has found an exact displayed pole. Otherwise an upper bound on $\mathcal{R}_{Q/P}$ is equivalently a lower bound on the ordinary quotient residual $\mathcal{R}_{P/Q}$.

5 Finite Hurwitz expansion and continuants

This section constructs the glossary's finite trace and generated-stage algebra. Deterministic coordinatewise rounding puts each residual in the half-open fundamental square.

Clearing the division produces an integral remainder of strictly smaller denominator quadrance, so recursion terminates on a natural number. Along the resulting trace, adjacent continuants retain a unit determinant and an exact reconstruction of the original displayed source. These identities express every competitor in the adjacent basis and factor its error by the current convergent error.

Definition 5.1 (nearest-integer residual)**.** *For $x \in \mathbb{Q}$, $z = x + iy \in \mathbb{Q}(i)$, define*

$$[x]_{\mathrm{n}} = \left\lfloor x + \frac{1}{2} \right\rfloor, \qquad [z]_{\mathbb{Z}[i]} = [x]_{\mathrm{n}} + i[y]_{\mathrm{n}}, \qquad \rho(z) = z - [z]_{\mathbb{Z}[i]}.$$

The chosen tie rule sends a half-integer upward in each coordinate. Thus the associated fundamental square is the half-open region

$$\mathfrak{F} = \{u + iv : -\tfrac{1}{2} \le u < \tfrac{1}{2}, -\tfrac{1}{2} \le v < \tfrac{1}{2}\}.$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzNearest.nearestCoordinate; Purelib.NumberTheory.Approximation.HurwitzNearest.nearest;Purelib. NumberTheory.Approximation.HurwitzNearest.remainder;Purelib.NumberTheory. Approximation.HurwitzNearest.InFundamentalSquare`.

Lemma 5.2 (fundamental-square remainder)**.** *For every displayed $z \in \mathbb{Q}(i)$,*

$$\rho(z) \in \mathfrak{F}, \qquad \mathrm{N}(\rho(z)) \le \frac{1}{2}.$$

Proof. Put $n = \lfloor x + 1/2 \rfloor$. The defining floor inequalities are

$$n \le x + \frac{1}{2} < n + 1.$$

Subtracting $n + 1/2$ gives $-1/2 \le x - n < 1/2$. Apply the same calculation to the imaginary coordinate. Each coordinate square is at most $1/4$, so their sum is at most $1/2$. The strict upper sides and weak lower sides are retained exactly; no symmetry argument changes the tie convention. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzNearest.nearestCoordinate_error_ bounds;Purelib.NumberTheory.Approximation.HurwitzNearest.remainder_mem; Purelib.NumberTheory.Approximation.HurwitzNearest.remainder_normSq_le_ half`.

Lemma 5.3 (nearest Gaussian integer is quadrance-minimising)**.** *For every $z \in \mathbb{Q}(i)$ and every displayed $c \in \mathbb{Z}[i]$,*

$$\mathrm{N}\Big(z - [z]_{\mathbb{Z}[i]}\Big) \le \mathrm{N}(z - c)\,. \tag{32}$$

Proof. It is enough to prove the statement in one rational coordinate and add the two inequalities. Let $n = [x]_{\mathrm{n}}$, let $e = x - n$, and write an integral competitor as $m = n + k$. By lemma 5.2, $-1/2 \le e < 1/2$, while

$$(x - m)^2 - (x - n)^2 = k(k - 2e).$$

If $k = 0$ this is zero. If $k \ge 1$, then $k - 2e > 0$; if $k \le -1$, both k and $k - 2e$ are nonpositive. The product is therefore nonnegative in every decidable integer case. Apply this to the real and imaginary coordinates and add. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzNearest.nearestCoordinate_sq_le`; `Purelib.NumberTheory.Approximation.HurwitzNearest.remainder_normSq_le_competitor`.

Definition 5.4 (displayed Gaussian fraction and Hurwitz transition)**.** *A displayed Gaussian fraction is a record*

$$z = \frac{N}{D}, \qquad N, D \in \mathbb{Z}[i], \quad D \neq 0,$$

whose denominator height is $\mathrm{N}(D) \in \mathbb{N}_{>0}$*. Put*

$$a = [z]_{\mathbb{Z}[i]}, \qquad R = N - aD, \qquad \rho = z - a. \tag{33}$$

If $R = 0$*, the fraction is terminal. If* $R \neq 0$*, its next complete quotient is the displayed fraction*

$$T(z) = \frac{D}{R} = \rho^{-1}. \tag{34}$$

No cancellation is implicit in either fraction record.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.Fraction`;`Purelib.NumberTheory.Approximation.HurwitzRational.Fraction.value`;`Purelib.NumberTheory.Approximation.HurwitzRational.Fraction.height`;`Purelib.NumberTheory.Approximation.HurwitzRational.digit`;`Purelib.NumberTheory.Approximation.HurwitzRational.residual`;`Purelib.NumberTheory.Approximation.HurwitzRational.remainder`;`Purelib.NumberTheory.Approximation.HurwitzRational.next`.

Lemma 5.5 (one Hurwitz step decreases denominator quadrance)**.** *The data of definition 5.4 satisfy*

$$\mathrm{N}(R) = \mathrm{N}(\rho)\ \mathrm{N}(D)\,. \tag{35}$$

Consequently $\mathrm{N}(R) < \mathrm{N}(D)$*.*

Proof. Cross-multiplication in (33) gives the displayed Gaussian identity $R = \rho D$. Multiplicativity of quadrance proves (35). Since $D \neq 0$, its integral quadrance is positive. Lemma 5.2 therefore gives

$$\mathrm{N}(R) \leq \frac{1}{2}\,\mathrm{N}(D) < \mathrm{N}(D)\,.$$

The cast comparison between the two natural quadrances reflects this exact rational calculation. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.digit_mul_denominator_add_remainder`;`Purelib.NumberTheory.Approximation.HurwitzRational.remainder_toGaussian`;`Purelib.NumberTheory.Approximation.HurwitzRational.remainder_norm_identity`;`Purelib.NumberTheory.Approximation.HurwitzRational.remainder_norm_lt_denominator`;`Purelib.NumberTheory.Approximation.HurwitzRational.next_value`.

Theorem 5.6 (terminating finite Hurwitz encoding)**.** *Every displayed Gaussian fraction $z = N/D$ computes a nonempty finite digit list $[a_0, \ldots, a_m]$ and a displayed Gaussian-integral pair (p_m, q_m) with $q_m \neq 0$ such that*

$$zq_m = p_m. \tag{36}$$

The list is obtained solely by repeating definition 5.4 until the integral remainder is zero.

Proof. Recurse on the natural denominator quadrance $\mathrm{N}(D)$. A zero remainder returns the one-element list $[a_0]$, and the integral division identity $a_0 D = N$ proves reconstruction. A nonzero remainder calls the recursion on D/R; this is structurally permitted by lemma 5.5. If the recursive tail represents D/R, prepend a_0 and use $z = a_0 + R/D$ to obtain the new continuant identity. The tail numerator is nonzero because its represented complete quotient is nonzero, so it becomes the nonzero denominator of the prepended pair. The recursion therefore returns both the list and the certificate (36). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.Encoding;Purelib.NumberTheory.Approximation.HurwitzRational.encodeResult;Purelib.NumberTheory.Approximation.HurwitzRational.encode;Purelib.NumberTheory.Approximation.HurwitzRational.encode_nonempty;Purelib.NumberTheory.Approximation.HurwitzRational.encode_denominatorNonzero;Purelib.NumberTheory.Approximation.HurwitzRational.encode_reconstruct`.

The theorem should be read as a Euclidean algorithm, not as a truncation of an already existing infinite expansion. Its recursive argument is the visible natural number $\mathrm{N}(D)$; for rational input the terminal digit and the exact reconstruction are computed data.

Definition 5.7 (chronological continuants)**.** *For a displayed digit prefix $a_0, \ldots, a_n$, set*

$$(p_{-1}, q_{-1}) = (1, 0), \qquad (p_0, q_0) = (a_0, 1),$$

and advance chronologically by

$$(p_{j+1}, q_{j+1}) = a_{j+1}(p_j, q_j) + (p_{j-1}, q_{j-1}). \tag{37}$$

The finite-list continuant used by the encoder is the equivalent front recursion with empty tail $(1, 0)$ and

$$\mathrm{cont}(a :: L) = (ap_L + q_L, p_L), \qquad \mathrm{cont}(L) = (p_L, q_L).$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.ConvergentPair;Purelib.NumberTheory.Approximation.HurwitzRational.prepend;Purelib.NumberTheory.Approximation.HurwitzRational.continuant;Purelib.NumberTheory.Approximation.HurwitzTrace.advancePair`.

Definition 5.8 (determinant and adjacent basis)**.** *For integral pairs $u = (p, q)$ and $v = (p', q')$, put*

$$\Delta(u, v) = pq' - p'q \in \mathbb{Z}[i].$$

They have a unit determinant *when the displayed alternative $\Delta(u, v) = 1$ or $\Delta(u, v) = -1$ is supplied.*

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzApproximation.determinant;`
`Purelib.NumberTheory.Approximation.HurwitzApproximation.`
`HasUnitDeterminant`.

Lemma 5.9 (adjacent continuants have unit determinant)**.** *Every two adjacent continuants produced by* (37) *have unit determinant.*

Proof. The initial determinant is $\Delta((a_0,1),(1,0)) = -1$. Direct coefficient calculation gives

$$\Delta\big(a_{j+1}(p_j,q_j)+(p_{j-1},q_{j-1}),(p_j,q_j)\big) = -\Delta((p_j,q_j),(p_{j-1},q_{j-1})).$$

Thus each displayed step merely flips the explicit sign alternative. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzApproximation.determinant_`
`prepend;Purelib.NumberTheory.Approximation.HurwitzApproximation.adjacent_`
`hasUnitDeterminant;Purelib.NumberTheory.Approximation.HurwitzTrace.`
`determinant_advancePair;Purelib.NumberTheory.Approximation.HurwitzTrace.`
`unitDeterminant_advancePair`.

Definition 5.10 (Hurwitz stage and finite stage trace)**.** *At prefix n, let x_n be the current complete quotient, let $\rho_n = x_n - [x_n]_{\mathbb{Z}[i]}$, and let the adjacent pairs be $c_{n-1} = (p_{n-1},q_{n-1})$ and $c_n = (p_n,q_n)$. A* Hurwitz stage *stores these displayed objects, a unit-determinant certificate, and the exact source reconstruction*

$$z(q_n + \rho_n q_{n-1}) = p_n + \rho_n p_{n-1}. \tag{38}$$

The first stage uses $x_0 = z$, $c_{-1} = (1,0)$ and $c_0 = (a_0,1)$. At a nonterminal stage, advance the complete quotient by $x_{n+1} = \rho_n^{-1}$ and the pair by (37)*. Repeating this transition until the integral remainder vanishes is the finite stage trace.*

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage;Purelib.`
`NumberTheory.Approximation.HurwitzTrace.firstStage;Purelib.NumberTheory.`
`Approximation.HurwitzTrace.advanceStage;Purelib.NumberTheory.`
`Approximation.HurwitzTrace.stagesFrom;Purelib.NumberTheory.Approximation.`
`HurwitzTrace.stages`.

Lemma 5.11 (stage reconstruction is preserved)**.** *The first stage satisfies* (38)*. Every nonterminal transition preserves that identity and the unit-determinant certificate. The stage recursion terminates.*

Proof. For the first stage, (38) reduces to $z = a_0+\rho_0$. At a nonterminal step use $x_{n+1}\rho_n = 1$ and $x_{n+1} = a_{n+1}+\rho_{n+1}$. Multiplying the old reconstruction by x_{n+1} and regrouping gives

$$z\big((a_{n+1}q_n + q_{n-1}) + \rho_{n+1}q_n\big) = (a_{n+1}p_n + p_{n-1}) + \rho_{n+1}p_n,$$

which is precisely the next reconstruction. The determinant certificate is preserved by lemma 5.9; termination is again the strict natural decrease of lemma 5.5. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.firstStage;Purelib.NumberTheory.Approximation.HurwitzTrace.advanceStage;Purelib.NumberTheory.Approximation.HurwitzTrace.advanceStage_quotient;Purelib.NumberTheory.Approximation.HurwitzRational.remainder_norm_lt_denominator`.

Lemma 5.12 (explicit adjacent-basis decomposition)**.** *Let c_n, c_{n-1} be the adjacent pairs of a stage, let $c = (p, q)$ be any displayed Gaussian-integral competitor, and write $\delta = \Delta(c_n, c_{n-1}) \in \{1, -1\}$. Define*

$$s = \delta\Delta(c, c_{n-1}), \qquad t = \delta\Delta(c_n, c). \tag{39}$$

Then, coefficientwise in $\mathbb{Z}[i]^2$,

$$c = sc_n + tc_{n-1}. \tag{40}$$

Consequently its division-free error satisfies

$$E_z(c) = sE_z(c_n) + tE_z(c_{n-1}). \tag{41}$$

Proof. The explicit unit alternative gives $\delta^2 = 1$. Expanding the two determinants in (39), first in the numerator coordinate and then in the denominator coordinate, cancels the mixed terms and leaves $\delta^2 p = p$ and $\delta^2 q = q$. This proves (40) coefficientwise. Substitute those two identities into $E_z(p, q) = zq - p$ and distribute to obtain (41). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzApproximation.determinant_mul_self_eq_one;Purelib.NumberTheory.Approximation.HurwitzApproximation.currentCoefficient;Purelib.NumberTheory.Approximation.HurwitzApproximation.previousCoefficient;Purelib.NumberTheory.Approximation.HurwitzApproximation.competitor_numerator_decomposition;Purelib.NumberTheory.Approximation.HurwitzApproximation.competitor_denominator_decomposition;Purelib.NumberTheory.Approximation.HurwitzApproximation.competitor_error_decomposition`.

Lemma 5.13 (stage error recurrence and Lakein factor)**.** *At a nonterminal stage,*

$$E_z(c_n) + \rho_n E_z(c_{n-1}) = 0, \qquad E_z(c_{n-1}) = -x_{n+1}E_z(c_n). \tag{42}$$

For the coefficients s, t of lemma 5.12,

$$E_z(c) = (s - tx_{n+1})E_z(c_n). \tag{43}$$

Thus the second-kind comparison follows whenever $1 \le \mathrm{N}(s - tx_{n+1})$.

Proof. Subtract the two sides of the stored reconstruction (38); the result is the first identity in (42). Multiply it by $x_{n+1} = \rho_n^{-1}$ to obtain the second. Substitution into (41) gives (43). Taking quadrances, using multiplicativity, and multiplying the factor bound by the nonnegative $\mathrm{N}(E_z(c_n))$ proves the comparison. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.error_add_residual_mul_previous_eq_zero;Purelib.NumberTheory.Approximation.HurwitzTrace.`

`Stage.previous_error_eq_neg_next_mul_current;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.competitor_error_eq_factor_mul_current;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.atLeastAsGoodAs_of_one_le_factor_normSq`.

The algebra has now isolated the only difficult quantity: the Lakein factor $s - tx_{n+1}$. The denominator budget constrains the integral coefficients s, t, while the Hurwitz trace constrains x_{n+1} and the backward denominator ratio. The next three sections turn those constraints into the quadrance-one exterior inequality without introducing a limiting value or an infinite tail.

6 Fundamental-square and succession geometry

This section supplies the finite geometric constraints used by the last factor cases. The fundamental square gives exterior-disc laws for every noninitial complete quotient. A two-part stage invariant then proves strict growth of adjacent denominator quadrances. Normalising those displayed denominators produces the backward ratio $r_n = q_{n-1}/q_n$, whose transition is an exact finite fractional transformation. Finally, cleared exterior laws exclude a displayed list of small adjacent digit pairs; these are the forbidden successors consumed by the Hermitian-form induction in the next section.

Definition 6.1 (Gaussian units, diagonals, and corner turn)**.** *The four Gaussian units and four diagonal Gaussian integers are the finite classes*

$$\mathcal{U} = \{1, -1, i, -i\}, \qquad \mathcal{D} = \{1 + i, 1 - i, -1 + i, -1 - i\}.$$

Thus $u \in \mathcal{U}$ exactly in the cases explicitly listed by the predicate Gaussian unit*, and $c \in \mathcal{D}$ exactly in the cases listed by* diagonal*. For a diagonal c, define its corner turn by*

$$c^\sharp = -\overline{c}.$$

The transform permutes $\mathcal{D}$ and is the direction in which a failed corner configuration propagates one step backward.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.imaginaryUnit;Purelib.NumberTheory.Approximation.HurwitzGeometry.IsGaussianUnit;Purelib.NumberTheory.Approximation.HurwitzGeometry.IsDiagonal;Purelib.NumberTheory.Approximation.HurwitzGeometry.conjugate;Purelib.NumberTheory.Approximation.HurwitzGeometry.cornerTurn;Purelib.NumberTheory.Approximation.HurwitzGeometry.cornerTurn_isDiagonal`.

Lemma 6.2 (small Gaussian-integral quadrances)**.** *For $w \in \mathbb{Z}[i]$:*

(i) if $\mathrm{N}(w) < 4$, then $\mathrm{N}(w) \in \{0, 1, 2\}$;
(ii) if $\mathrm{N}(w) < 6$, then $\mathrm{N}(w) \in \{0, 1, 2, 4, 5\}$;
(iii) quadrance one means $w \in \mathcal{U}$, quadrance two means $w \in \mathcal{D}$, quadrance four means $w \in \{2, -2, 2i, -2i\}$, and quadrance five gives the eight values $\pm 1 \pm 2i$, $\pm 2 \pm i$.

Proof. Write $w = a + ib$. The assumptions bound each integral square a^2, b^2 by the same displayed natural threshold. For quadrance below four, each coordinate is therefore $-1, 0$, or 1; for quadrance below six it is one of $-2, -1, 0, 1, 2$. The finite Cartesian products give (i) and (ii), and sorting the same checked pairs by $a^2 + b^2$ gives (iii). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.normNat_eq_zero_or_one_or_two_of_lt_four;Purelib.NumberTheory.Approximation.HurwitzGeometry.normNat_eq_zero_or_one_or_two_or_four_or_five_of_lt_six;Purelib.NumberTheory.Approximation.HurwitzGeometry.isGaussianUnit_of_normNat_eq_one;Purelib.NumberTheory.Approximation.HurwitzGeometry.isDiagonal_of_normNat_eq_two;Purelib.NumberTheory.Approximation.HurwitzGeometry.axisDouble_cases_of_normNat_eq_four;Purelib.NumberTheory.Approximation.HurwitzGeometry.normFive_cases_of_normNat_eq_five`.

Lemma 6.3 (complete-quotient exterior laws)**.** *If a displayed fraction x is nonterminal and $x^+ = T(x) = \rho(x)^{-1}$, then*

$$2 \le \mathrm{N}\Big(x^+\Big), \qquad 2 \le \mathrm{N}\Big([x^+]_{\mathbb{Z}[i]}\Big). \tag{44}$$

For every $u \in \mathcal{U}$,

$$1 \le \mathrm{N}\Big(x^+ - u\Big). \tag{45}$$

At the positive real unit the half-open tie convention strengthens this to $1 < \mathrm{N}(x^+ - 1)$.

Proof. Since $x^+\rho(x) = 1$, multiplicativity and $\mathrm{N}(\rho(x)) \le 1/2$ give $2 \le \mathrm{N}(x^+)$. For a unit u, the fundamental-square inequalities give

$$\mathrm{N}(\rho(x)) \le \mathrm{N}(u\rho(x) - 1).$$

For $u = 1$ the upper real inequality $\mathrm{Re}\,\rho(x) < 1/2$ makes this strict. Multiplying by the positive $\mathrm{N}(x^+)$ and using $x^+\rho(x) = 1$ gives (45) and its strict case. If the nearest digit of x^+ had quadrance zero or one, subtracting that digit would simultaneously have quadrance at most $1/2$ by the fundamental square and at least one by the exterior law (or x^+ itself would contradict (44)). Hence the digit quadrance is at least two. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.residual_normSq_le_unit_mul_sub_one;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_next_sub_unit_normSq;Purelib.NumberTheory.Approximation.HurwitzGeometry.next_normSq_ge_two;Purelib.NumberTheory.Approximation.HurwitzGeometry.two_le_next_digit_normNat;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_lt_next_sub_one_normSq`.

Definition 6.4 (stage geometry)**.** *For a Hurwitz stage with adjacent denominators q_{n-1}, q_n and next complete quotient x_{n+1}, its* stage geometry *consists of:*

(i) strict denominator growth

$$\mathrm{N}(q_{n-1}) < \mathrm{N}(q_n); \tag{46}$$

(ii) for every diagonal $c \in \mathcal{D}$, exclusion of the joint corner failure

$$\mathrm{N}(cq_n + q_{n-1}) \le \mathrm{N}(q_n) \quad \textit{and} \quad \mathrm{N}(x_{n+1} - c) < 1. \tag{47}$$

The second clause is a finite natural-extension state: it retains precisely the information needed to prove the first clause again after advancing.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.StageGeometry`.

Lemma 6.5 (generated stages have stage geometry)**.** *The first Hurwitz stage has stage geometry, and every nonterminal advance preserves it. Consequently every generated stage satisfies* (46)*.*

Proof. At the first stage the denominators are $0, 1$, so growth is immediate, and a diagonal multiplier has quadrance two and cannot make $\mathrm{N}(c \cdot 1 + 0) \leq \mathrm{N}(1)$.

Assume the invariant at stage n. If the next denominator

$$q_{n+1} = a_{n+1} q_n + q_{n-1}$$

were no longer than q_n, the small-coefficient geometry forced by (46) classifies a_{n+1} as a diagonal. Its own fundamental-square residual is strictly inside the unit disc, so these two facts are exactly the corner failure (47) at stage n. This proves the new strict growth.

For corner safety at stage $n + 1$, expand a proposed failure using the two successive recurrences and the identities $x_{n+1} = a_{n+1} + x_{n+2}^{-1}$ and $c^\sharp = -\overline{c}$. The cleared quadrance comparison transfers the failure to stage n at $c^\sharp$. Since $c^\sharp \in \mathcal{D}$, the old corner-safety clause rejects it. Induction on the displayed generation certificate proves the result for every stage. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.next_digit_isDiagonal_of_combination_norm_le;Purelib.NumberTheory.Approximation.HurwitzGeometry.corner_failure_transfers_backward;Purelib.NumberTheory.Approximation.HurwitzGeometry.firstStage_geometry;Purelib.NumberTheory.Approximation.HurwitzGeometry.advanceStage_geometry;Purelib.NumberTheory.Approximation.HurwitzGeometry.geometry_of_generated;Purelib.NumberTheory.Approximation.HurwitzGeometry.stage_denominatorGrowth;Purelib.NumberTheory.Approximation.HurwitzGeometry.generated_of_mem_stages`.

This is the point at which the familiar picture of expanding convergent denominators becomes an actual finite proof. The fundamental-square bound alone does not make the chronological recurrence monotone term by term; the corner-safety memory is what rules out the diagonal cancellation that could otherwise spoil the next step.

Definition 6.6 (backward denominator ratio)**.** *At a generated stage define the displayed Gaussian rational*

$$r_n = \frac{q_{n-1}}{q_n}. \tag{48}$$

The denominator is nonzero by strict growth. On advancing,

$$r_{n+1} = (a_{n+1} + r_n)^{-1}. \tag{49}$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.denominatorRatio;Purelib.NumberTheory.Approximation.HurwitzGeometry.current_denominator_ne_zero_of_generated;Purelib.NumberTheory.Approximation.HurwitzGeometry.denominatorRatio_mul_current;Purelib.NumberTheory.Approximation.HurwitzGeometry.advanceStage_denominatorRatio`.

Lemma 6.7 (the backward ratio lies in the open unit disc)**.** *Every generated stage satisfies*

$$\mathrm{N}(r_n) < 1. \tag{50}$$

Proof. Multiply (48) by the nonzero displayed q_n and take quadrances:

$$\mathrm{N}(r_n)\,\mathrm{N}(q_n) = \mathrm{N}(q_{n-1})\,.$$

The strict denominator growth of lemma 6.5 and positivity of $\mathrm{N}(q_n)$ give (50) by exact rational division. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.denominatorRatio_normSq_lt_one_of_generated`.

Lemma 6.8 (cleared successor exterior law)**.** *Let a_j and a_{j+1} be two displayed noninitial adjacent digits, and let x_{j+1} be the complete quotient beginning with a_{j+1}. For every $u \in \mathcal{U}$,*

$$\mathrm{N}(x_{j+1}) \le \mathrm{N}((a_j - u)x_{j+1} + 1)\,. \tag{51}$$

For $u = 1$ the inequality is strict.

Proof. The preceding complete quotient is $x_j = a_j + x_{j+1}^{-1}$. Apply (45) to $x_j - u$ and multiply by the positive $\mathrm{N}(x_{j+1})$. The cleared identity

$$(x_j - u)x_{j+1} = (a_j - u)x_{j+1} + 1$$

gives (51). The strict exterior law at $u = 1$ preserves strictness under the same positive multiplication. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.successor_exterior_cleared;Purelib.NumberTheory.Approximation.HurwitzGeometry.successor_strict_exterior_cleared`.

Definition 6.9 (forbidden small successors)**.** *The relation $\mathcal{F}(a_j, a_{j+1})$ consists of the following twenty-eight ordered pairs. To keep the finite table readable, the left column fixes the following digit a_{j+1} and the right column lists its forbidden preceding digits a_j.*

a_{j+1}	*forbidden* a_j
-2	$1-i,\ 1+i$
$-1-i$	$-2i,\ 1-2i,\ 1-i,\ 2-i,\ 2$
$-1+i$	$2i,\ 1+i,\ 1+2i,\ 2,\ 2+i$
$-2i$	$-1-i,\ 1-i$
$2i$	$-1+i,\ 1+i$
$1-i$	$-2-i,\ -2,\ -1-2i,\ -1-i,\ -2i$
$1+i$	$-2,\ -2+i,\ -1+i,\ -1+2i,\ 2i$
2	$-1-i,\ -1+i$

It is an inductive finite relation with one constructor for each displayed entry, not a predicate obtained from an unbounded search.

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.ForbiddenSmallSuccessor`.

Lemma 6.10 (forbidden successors cannot occur)**.** *No two adjacent noninitial digits of a generated finite Hurwitz trace belong to* $\mathcal{F}$*.*

Proof. Write the following quotient as $x_{j+1} = a_{j+1} + \sigma$ with $\sigma \in \mathfrak{F}$. For a proposed table entry, choose the displayed unit u attached to that entry and subtract the two sides of (51). Expansion gives one of a small collection of signed gaps, including

$$\mathrm{N}(\sigma) - 1, \qquad -1 \pm 2\,\mathrm{Re}\,\sigma, \qquad -1 \pm 2\,\mathrm{Im}\,\sigma, \qquad -3 \pm 2\,\mathrm{Re}\,\sigma, \qquad -3 \pm 2\,\mathrm{Im}\,\sigma.$$

The fundamental-square inequalities make the appropriate gap negative. Where a lower boundary could make a gap zero, the strict $u = 1$ exterior law or the preceding inversion supplies the needed strict inequality. Thus the proposed pair makes the cleared exterior inequality point the wrong way. Elimination on the twenty-eight constructors performs these exact polynomial calculations one by one. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.successorGap;Purelib.NumberTheory.Approximation.HurwitzGeometry.successor_signs_of_one_add_i;Purelib.NumberTheory.Approximation.HurwitzGeometry.residual_eq_next_value_inv;Purelib.NumberTheory.Approximation.HurwitzGeometry.not_forbiddenSmallSuccessor`.

The table is less elegant than the phrase "and similarly in the other quadrants", but it records what that phrase omits: which boundary is strict, which unit exterior disc is used, and which ordered digit pair is actually excluded. Section 7 uses precisely this finite relation to close the second pullback step.

7 Hermitian root forms and the exceptional invariant

This section proves the glossary's root-form invariant. The remaining comparison between quadrance-five and diagonal coefficients is encoded by eight integral Hermitian forms evaluated at the backward ratio. Pulling a form through the exact recurrence $r_{n+1} = (a_{n+1} + r_n)^{-1}$ is a coefficient calculation. Most pullbacks have an immediate sum-of-squares certificate; twenty-four first-step exceptions require one preceding digit. Their second pullbacks are either certified nonnegative or demand one of the forbidden successors already excluded in lemma 6.10. A two-stage induction then establishes all eight inequalities at every generated stage.

Definition 7.1 (integral Hermitian form)**.** *For displayed integers* A, B, C, D *define*

$$\mathcal{H}_{A,B,C,D}(z) = A + B\,\mathrm{N}(z) + 2\big(C\,\mathrm{Re}\,z + D\,\mathrm{Im}\,z\big). \tag{52}$$

The associated integral coefficient record is written $F = (A, B, C, D)$*, with evaluation* $F[z] = \mathcal{H}_{A,B,C,D}(z)$ *and determinant*

$$\det F = AB - (C^2 + D^2). \tag{53}$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.hermitianForm;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.evaluate;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.determinant`.

Lemma 7.2 (finite unit-disc certificate)**.** *Let* $F = (A, B, C, D)$ *and suppose* $\mathrm{N}(z) \le 1$*. If*

$$B + 1 > 0, \qquad (A - 1)(B + 1) - (C^2 + D^2) \ge 0, \tag{54}$$

then $F[z] \ge 0$*.*

Proof. Put $z = x + iy$. Direct expansion gives the displayed identity

$$\begin{aligned}(B+1)F[z] = {} & ((B+1)x + C)^2 + ((B+1)y + D)^2 \\ & + (B+1)(1 - \mathrm{N}(z)) + \big((A-1)(B+1) - (C^2 + D^2)\big).\end{aligned}$$

Every term on the right is nonnegative by the supplied rational inequalities. Exact division by the positive $B + 1$ proves the result. The formal lemma also permits a displayed nonnegative slack other than one; the form above is the only instance needed by the finite classification. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.hermitianForm_nonnegative_of_certificate;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.evaluate_nonnegative_of_slack_one`.

Definition 7.3 (integral pullback)**.** *For* $F = (A, B, C, D)$ *and* $a = u + iv \in \mathbb{Z}[i]$*, its pullback through* $z \mapsto (a + z)^{-1}$ *is the integral form*

$$F \triangleleft a = \big(A\,\mathrm{N}(a) + B + 2(Cu - Dv),\ A,\ Au + C,\ Av - D\big). \tag{55}$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.pullback`.

Lemma 7.4 (pullback identity and determinant)**.** *If* $a + z \ne 0$*, then*

$$F[(a + z)^{-1}]\,\mathrm{N}(a + z) = (F \triangleleft a)[z]. \tag{56}$$

Moreover

$$\det(F \triangleleft a) = \det F. \tag{57}$$

Proof. The inverse has coordinates

$$(a + z)^{-1} = \frac{\mathrm{Re}(a + z) - i\,\mathrm{Im}(a + z)}{\mathrm{N}(a + z)}.$$

Substitute these two displayed rational coordinates into (52) and multiply by $\mathrm{N}(a + z)$. Collecting the constant, radial, real-cross, and imaginary-cross coefficients gives exactly (55). Expanding (53) for those four new coefficients cancels all u, v terms and leaves $AB - (C^2 + D^2)$. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.hermitianForm_inverse_add_mul_normSq;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.evaluate_pullback_mul_normSq;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.pullback_determinant`.

Lemma 7.5 (transport through one generated advance)**.** *Let r_n be the backward ratio at a generated nonterminal stage and let a_{n+1} be its next digit. If*

$$(F \triangleleft a_{n+1})[r_n] \geq 0,$$

then $F[r_{n+1}] \geq 0$.

Proof. The affine denominator $a_{n+1} + r_n$ cannot vanish. Otherwise $r_n = -a_{n+1}$, but lemma 6.7 gives $\mathrm{N}(r_n) < 1$, whereas (44) gives $\mathrm{N}(a_{n+1}) \geq 2$. Now use $r_{n+1} = (a_{n+1}+r_n)^{-1}$ in (56). Its other factor $\mathrm{N}(a_{n+1} + r_n)$ is positive, so the pullback's nonnegativity divides through to $F[r_{n+1}] \geq 0$. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.denominatorRatio_step_sum_ne_zero;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.evaluate_advance_nonnegative_of_pullback`.

Definition 7.6 (Lakein root forms)**.** *Let*

$$\mathcal{X} = \{(-3,-1),(-3,1),(-1,-3),(-1,3),(1,-3),(1,3),(3,-1),(3,1)\}.$$

For $(C, D) \in \mathcal{X}$, the corresponding Lakein root form *is*

$$L_{C,D} = (4, 2, C, D), \qquad L_{C,D}[z] = 4 + 2\,\mathrm{N}(z) + 2(C\,\mathrm{Re}\,z + D\,\mathrm{Im}\,z). \tag{58}$$

All eight have determinant -2. The root-form condition *at z means $L_{C,D}[z] \geq 0$ for every displayed pair $(C, D) \in \mathcal{X}$.*

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.IsLakeinRootCross;Purelib.NumberTheory.Approximation.HurwitzGeometry.lakeinRootForm;Purelib.NumberTheory.Approximation.HurwitzGeometry.LakeinRootFormsNonnegative`.

Lemma 7.7 (root forms are the quadrance-five exterior test)**.** *Let $s, t \in \mathbb{Z}[i]$ satisfy $\mathrm{N}(s) = 5$ and $\mathrm{N}(t) = 2$. Then the two integral cross terms*

$$C = s_{\mathrm{re}}t_{\mathrm{re}} + s_{\mathrm{im}}t_{\mathrm{im}}, \qquad D = -s_{\mathrm{re}}t_{\mathrm{im}} + s_{\mathrm{im}}t_{\mathrm{re}}$$

form one of the eight pairs in $\mathcal{X}$, and

$$\mathrm{N}(s + tr) = 1 + L_{C,D}[r]. \tag{59}$$

Consequently the root-form condition at r implies $1 \leq \mathrm{N}(s + tr)$ for every such s, t.

Proof. Lemma 6.2 lists the eight possible s and four possible t. Substitution gives exactly the eight cross pairs in $\mathcal{X}$. Expanding the real and imaginary coordinates of $s + tr$, and then using $\mathrm{N}(s) = 5$ and $\mathrm{N}(t) = 2$, gives

$$\mathrm{N}(s + tr) = 5 + 2\,\mathrm{N}(r) + 2(C\,\mathrm{Re}\,r + D\,\mathrm{Im}\,r),$$

which is (59). Add the certified nonnegative root form to one. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_normSq_normFive_add_diagonal_mul_of_rootForms`.

Definition 7.8 (first exceptional pullbacks)**.** *For a noninitial digit a with $\mathrm{N}(a) \geq 2$, pull back each of the eight root forms once. A constant-coefficient gap certifies every result except the following twenty-four displayed form–digit pairs:*

a	*exceptional* $F = L_{C,D} \triangleleft a$
$1 + i$	$(6, 4, 1, 5)$, $(2, 4, 1, 3)$, $(2, 4, 3, 1)$, $(6, 4, 5, 1)$
$1 - i$	$(2, 4, 1, -3)$, $(6, 4, 1, -5)$, $(2, 4, 3, -1)$, $(6, 4, 5, -1)$
2	$(6, 4, 5, 1)$, $(6, 4, 5, -1)$
$-1 - i$	$(6, 4, -5, -1)$, $(2, 4, -3, -1)$, $(2, 4, -1, -3)$, $(6, 4, -1, -5)$
$-2i$	$(6, 4, -1, -5)$, $(6, 4, 1, -5)$
$-1 + i$	$(6, 4, -5, 1)$, $(2, 4, -3, 1)$, $(6, 4, -1, 5)$, $(2, 4, -1, 3)$
$2i$	$(6, 4, -1, 5)$, $(6, 4, 1, 5)$
-2	$(6, 4, -5, 1)$, $(6, 4, -5, -1)$

The inductive relation first exceptional Hermitian *records these pairs with one constructor per row entry.*

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.FirstExceptionalHermitian`.

Lemma 7.9 (first-pullback classification)**.** *For every root form L and every Gaussian integer a with $\mathrm{N}(a) \geq 2$, either*

$$7 \leq (L \triangleleft a)_{\mathrm{constant}}, \tag{60}$$

or $(L \triangleleft a, a)$ is one of the twenty-four exceptions in definition 7.8. In the first case the pulled-back form is nonnegative throughout the closed unit disc.

Proof. Every root form has constant four and determinant -2. The pullback determinant is still -2. If its new constant is at least seven, the determinant identity rearranges the second inequality of (54); its radial coefficient is four, so the first inequality is positive. Lemma 7.2 applies.

It remains to classify failure of (60). The explicit pullback constant has the lower estimate

$$(A - 1)\,\mathrm{N}(a) + B - (C^2 + D^2) \leq (F \triangleleft a)_{\mathrm{constant}}.$$

For $F = L$ this settles $\mathrm{N}(a) \geq 5$. Under $2 \leq \mathrm{N}(a) < 5$, Lemma 6.2 leaves only the four diagonal and four axial-double digits. Substituting those eight digits into the eight root forms gives either the gap or exactly the displayed twenty-four exceptions. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.pullback_constant_lower;Purelib.NumberTheory.Approximation.HurwitzGeometry.IntegralHermitianForm.pullback_evaluate_nonnegative_of_constant_gap;Purelib.NumberTheory.Approximation.HurwitzGeometry.FirstExceptionalHermitian`.

Lemma 7.10 (exceptional second pullback)**.** *Let (F, a) be one of the twenty-four first-step exceptions. Let b be the preceding noninitial digit, so $\mathrm{N}(b) \geq 2$, and suppose $\mathcal{F}(b, a)$ does not hold. If $\mathrm{N}(r) \leq 1$ and the eight root forms are nonnegative at r, then*

$$(F \triangleleft b)[r] \geq 0. \tag{61}$$

Proof. If $\mathrm{N}(b) \geq 11$, the pullback-constant lower estimate gives the same slack-one certificate directly. If $2 \leq \mathrm{N}(b) < 11$, the two coordinate squares show that b belongs to an explicit list of thirty-two Gaussian integers with coordinates between -3 and 3. Substitution of those digits into the twenty-four exceptional forms classifies each of the 768 displayed cases as one of:

(i) a forbidden pair $\mathcal{F}(b, a)$;
(ii) one of the original eight root forms;
(iii) one of the eight augmented forms $(8, 6, \pm 7, \pm 1)$ or $(8, 6, \pm 1, \pm 7)$; or
(iv) a form satisfying the slack-one certificate (54).

Case (i) is excluded by hypothesis, (ii) by the root-form condition, and (iv) by lemma 7.2. Each augmented form is one root form plus $4\,\mathrm{N}(r + u)$ for a displayed Gaussian unit u, so case (iii) is also nonnegative. These finite alternatives prove (61). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.FirstExceptionalHermitian.pullback_nonnegative`.

The 768 substitutions are intentionally kernel-checked coefficient calculations, but reproducing them as a paper table would hide rather than clarify the argument. The meaningful finite interface is the four-way classification above: only its first branch depends on actual Hurwitz succession, and Section 6 has already ruled that branch out.

Theorem 7.11 (root-form invariant)**.** *For every generated Hurwitz stage and every $(C, D) \in \mathcal{X}$,*

$$L_{C,D}[r_n] \geq 0. \tag{62}$$

Proof. The first ratio is $r_0 = 0$, where all eight forms have value four. At the first advance, apply the first-pullback classification at r_0. A regular pullback is nonnegative on the unit disc; an exceptional pullback is still nonnegative at zero because its displayed constant is two or six.

For the induction step, retain the root-form condition both at the current stage and after its next possible advance. To prove it two advances later, pull a target root form back through the following digit. A regular first pullback is nonnegative at the middle ratio by lemma 7.2. For an exceptional first pullback, pull once more through the preceding digit. The previous ratio has quadrance below one by lemma 6.7, its root forms are nonnegative by the induction hypothesis, and the digit pair is not forbidden by lemma 6.10; hence lemma 7.10 applies. Transport forward once and then once more using lemma 7.5. The

two-component invariant aligns this two-step calculation with ordinary induction on the one-step generated-stage certificate. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.lakeinRootFormsNonnegative_of_generated`.

Corollary 7.12 (quadrance-five denominator safety)**.** *At every generated stage, for all* $s, t \in \mathbb{Z}[i]$ *with* $\mathrm{N}(s) = 5$ *and* $\mathrm{N}(t) = 2$,

$$\mathrm{N}(q_n) \le \mathrm{N}(sq_n + tq_{n-1})\,. \tag{63}$$

Proof. By theorem 7.11 and lemma 7.7, $1 \le \mathrm{N}(s + tr_n)$. Multiply by $\mathrm{N}(q_n) > 0$ and use

$$(s + tr_n)q_n = sq_n + tq_{n-1}.$$

The resulting quadrance inequality is (63). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.NormFiveDenominatorSafe;Purelib.NumberTheory.Approximation.HurwitzGeometry.normFiveDenominatorSafe_of_rootForms;Purelib.NumberTheory.Approximation.HurwitzGeometry.normFiveDenominatorSafe_of_generated`.

This corollary is the sole fact exported from the long exceptional classification into the final Lakein comparison. Its form is deliberately denominator-only: if a competitor coefficient pair ever has the exceptional quadrances $(5, 2)$, the required smaller-denominator hypothesis contradicts (63) before any appeal to the source value.

8 The finite Lakein inequality

This section assembles the glossary's final comparison. The exact complete denominator makes the current error a reciprocal in quadrance. A large previous-basis coefficient is then excluded by a determinant estimate. If that test fails, the coefficient has quadrance below four and hence is zero, a unit, or diagonal. The first two cases use elementary exterior laws; the diagonal case leaves a current coefficient of quadrance below six, where the root-form invariant removes the sole exceptional value five. The resulting Lakein-factor bound completes every nonterminal stage, while a terminal stage has exact zero error.

Definition 8.1 (complete denominator)**.** *At a nonterminal stage put*

$$\Lambda_n = x_{n+1}q_n + q_{n-1} \in \mathbb{Q}(i). \tag{64}$$

This is an unnormalised displayed Gaussian rational, not a new continuant. The stage reconstruction gives

$$z\Lambda_n = x_{n+1}p_n + p_{n-1}. \tag{65}$$

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.completeDenominator;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.scaled_represents`.

Lemma 8.2 (exact current-error product)**.** *At every nonterminal stage,*

$$E_z(c_n)\Lambda_n = -\Delta(c_n, c_{n-1}), \qquad \mathrm{N}(E_z(c_n))\,\mathrm{N}(\Lambda_n) = 1. \tag{66}$$

Both factors in the second identity are positive.

Proof. Multiply $E_z(c_n) = zq_n - p_n$ by (64), substitute (65), and expand:

$$\begin{aligned}(zq_n - p_n)\Lambda_n &= q_n(x_{n+1}p_n + p_{n-1}) - p_n(x_{n+1}q_n + q_{n-1}) \\ &= q_n p_{n-1} - p_n q_{n-1} = -\Delta(c_n, c_{n-1}).\end{aligned}$$

The adjacent determinant is explicitly 1 or -1, so taking quadrances gives the second identity. A product equal to one cannot contain a zero-quadrance factor. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.current_error_mul_completeDenominator`;`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.current_error_norm_product_eq_one`;`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.current_error_normSq_pos`;`Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.completeDenominator_normSq_pos`.

Lemma 8.3 (complete-denominator growth)**.** *Every generated nonterminal stage satisfies*

$$\mathrm{N}(q_n) < \mathrm{N}(\Lambda_n)\,. \tag{67}$$

Proof. At the first stage, $\Lambda_0 = x_1$ and $\mathrm{N}(x_1) \geq 2 > 1 = \mathrm{N}(q_0)$. For the induction step, the exact recurrence

$$\Lambda_{n+1} = x_{n+2}\Lambda_n$$

transfers the preceding strict inequality after multiplication by the positive $\mathrm{N}(x_{n+2})$. Suppose nevertheless that $\mathrm{N}(\Lambda_{n+1}) \leq \mathrm{N}(q_{n+1})$. The resulting two rational quadrance estimates force $\mathrm{N}(x_{n+2}) < 3$. The four unit exterior laws then classify its nearest digit as diagonal. Expanding the corresponding fundamental-square lune shows $\mathrm{N}(q_{n+2}) \leq \mathrm{N}(q_{n+1})$, contradicting the strict denominator growth of lemma 6.5. Hence the proposed failure is rejected at each generated step. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.advanceStage_completeDenominator`;`Purelib.NumberTheory.Approximation.HurwitzGeometry.quotient_normSq_lt_three_of_complete_failure`;`Purelib.NumberTheory.Approximation.HurwitzGeometry.next_digit_isDiagonal_of_normSq_lt_three`;`Purelib.NumberTheory.Approximation.HurwitzGeometry.diagonal_combination_norm_le_of_complete_failure`;`Purelib.NumberTheory.Approximation.HurwitzGeometry.completeDenominatorGrowth_of_generated`.

Lemma 8.4 (large previous coefficient)**.** *Let a displayed competitor $c = (p, q)$ have $q \neq 0$ and $\mathrm{N}(q) < \mathrm{N}(q_n)$. Let s, t be its adjacent-basis coefficients from (39). If*

$$4\,\mathrm{N}(q_n) < \mathrm{N}(t)\;\mathrm{N}(\Lambda_n)\,, \tag{68}$$

then

$$\mathrm{N}(E_z(c_n)) \leq \mathrm{N}(E_z(c))\,.$$

Proof. For any current pair and competitor, direct expansion of the cross determinant gives

$$\Delta(c_n, c) = q_n E_z(c) - q E_z(c_n).$$

The two-term quadrance estimate therefore gives

$$\mathrm{N}(\Delta(c_n, c)) \le 2\big(\mathrm{N}(q_n)\,\mathrm{N}(E_z(c)) + \mathrm{N}(q)\,\mathrm{N}(E_z(c_n))\big). \tag{69}$$

If the claimed comparison failed strictly, the smaller-error and smaller-denominator hypotheses would make the right side strictly less than $4\,\mathrm{N}(q_n)\,\mathrm{N}(E_z(c_n))$. Since the adjacent determinant is a unit, $\mathrm{N}(t) = \mathrm{N}(\Delta(c_n, c))$. Multiply the strict estimate by the positive $\mathrm{N}(\Lambda_n)$ and use (66); the result is

$$\mathrm{N}(t)\ \mathrm{N}(\Lambda_n) < 4\,\mathrm{N}(q_n)\,,$$

contrary to (68). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.determinant_eq_denominator_mul_error_sub;Purelib.NumberTheory.Approximation.HurwitzTrace.determinant_normSq_le;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.previousCoefficient_normSq;Purelib.NumberTheory.Approximation.HurwitzTrace.Stage.atLeastAsGoodAs_of_large_previousCoefficient`.

Lemma 8.5 (small previous-coefficient reduction)**.** *Under the hypotheses of lemma 8.4, if* (68) *does not hold, then*

$$\mathrm{N}(t) < 4. \tag{70}$$

Hence $\mathrm{N}(t)$ *is* 0, 1, *or* 2.

Proof. If $\mathrm{N}(t) \ge 4$, the complete-denominator growth (67) gives

$$4\,\mathrm{N}(q_n) < 4\,\mathrm{N}(\Lambda_n) \le \mathrm{N}(t)\ \mathrm{N}(\Lambda_n)\,,$$

which is exactly the supposedly false large-coefficient test. Therefore $\mathrm{N}(t) < 4$, and the three displayed possibilities follow from lemma 6.2. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.previousCoefficient_normNat_lt_four_of_not_large;Purelib.NumberTheory.Approximation.HurwitzGeometry.normNat_eq_zero_or_one_or_two_of_lt_four`.

Lemma 8.6 (small-coefficient Lakein factors)**.** *Assume*

$$q = sq_n + tq_{n-1}, \qquad q \neq 0, \qquad \mathrm{N}(q) < \mathrm{N}(q_n)\,. \tag{71}$$

Then every case in (70) *either is impossible or satisfies*

$$1 \le \mathrm{N}(s - tx_{n+1})\,. \tag{72}$$

Proof. If $\mathrm{N}(t) = 0$, then $t = 0$ and $q = sq_n$. A nonzero s makes this denominator at least as large as q_n, while $s = 0$ contradicts $q \neq 0$.

Suppose $\mathrm{N}(t) = 1$. Conjugating by the unit t rewrites $s = tc$ with $c = \bar{t}s$, without changing quadrances. The strict denominator inequality becomes $\mathrm{N}(cq_n + q_{n-1}) < \mathrm{N}(q_n)$ and forces $\mathrm{N}(c) < 4$. Thus c is zero, a unit, or diagonal. The first two cases follow from $\mathrm{N}(x_{n+1}) \geq 2$ and the unit exterior laws. In the diagonal case, failure of (72) would put x_{n+1} strictly within the corresponding corner disc while the displayed denominator combination does not grow. This is exactly the corner failure excluded by stage geometry.

Finally suppose $\mathrm{N}(t) = 2$, so t is diagonal. The strict denominator combination and (46) give the finite estimate $\mathrm{N}(s) < 6$. Lemma 6.2 leaves $\mathrm{N}(s) = 0, 1, 2, 4$, or 5. The first three cases follow respectively from the complete-quotient size, a unit exterior estimate, and association of two diagonals. Quadrance four factors through a displayed diagonal corner and is again excluded by corner safety if its Lakein factor enters the unit disc. In the remaining quadrance-five case, corollary 7.12 says

$$\mathrm{N}(q_n) \leq \mathrm{N}(sq_n + tq_{n-1}) = \mathrm{N}(q)\,,$$

contradicting (71). All cases are therefore impossible or satisfy (72). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.coefficient_normNat_lt_four_of_combination_norm_le;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_factor_normSq_of_unit_previousCoefficient;Purelib.NumberTheory.Approximation.HurwitzGeometry.currentCoefficient_normNat_lt_six_of_diagonal_previous;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_factor_normSq_of_unit_and_diagonal;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_factor_normSq_of_diagonal_coefficients;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_factor_normSq_of_four_and_diagonal;Purelib.NumberTheory.Approximation.HurwitzGeometry.one_le_factor_normSq_of_diagonal_previousCoefficient_of_not_five`.

Theorem 8.7 (finite Lakein comparison at a nonterminal stage)**.** *Let $c_n = (p_n, q_n)$ be the current pair of a generated nonterminal stage. For every displayed $c = (p, q) \in \mathbb{Z}[i]^2$ satisfying*

$$q \neq 0, \qquad \mathrm{N}(q) < \mathrm{N}(q_n)\,,$$

one has

$$\mathrm{N}(zq_n - p_n) \leq \mathrm{N}(zq - p)\,. \tag{73}$$

Proof. Use the explicit adjacent-basis coefficients s, t of lemma 5.12; its denominator identity is (71). Decide the rational large-coefficient comparison (68). If it holds, lemma 8.4 proves the result directly. If it does not, lemma 8.5 and lemma 8.6 prove $1 \leq \mathrm{N}(s - tx_{n+1})$. The error factorisation (43) then gives

$$\mathrm{N}(zq - p) = \mathrm{N}(s - tx_{n+1})\,\mathrm{N}(zq_n - p_n) \geq \mathrm{N}(zq_n - p_n)\,.$$

□

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.atLeastAsGoodAs_of_generated_of_normFiveDenominatorSafe;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.atLeastAsGoodAs_of_generated_nonterminal`.

Theorem 8.8 (finite Hurwitz–Lakein inequality)**.** *At every generated stage of the finite Hurwitz trace, every displayed Gaussian-integral competitor* (p, q) *with* $q \neq 0$ *and* $\mathrm{N}(q) < \mathrm{N}(q_n)$ *satisfies*

$$\mathrm{N}(zq_n - p_n) \leq \mathrm{N}(zq - p)\,. \tag{74}$$

Proof. If the stage is nonterminal, apply theorem 8.7. If its integral remainder is zero, then (35) and $D \neq 0$ give $\rho_n = 0$. The stage error recurrence (42) reduces to $E_z(c_n) = 0$. Its quadrance is therefore zero and hence no larger than the nonnegative quadrance of any displayed competitor error. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.atLeastAsGoodAs_of_generated;Purelib.Papers.RFAP.Approximation.hurwitzConvergent_atLeastAsGoodAs`.

The theorem compares division-free residuals. If both displayed denominators are nonzero, dividing by their quadrances converts each to the ordinary squared approximation error, but that changes the comparison and is not the theorem proved here. The second-kind form is exactly what the adjacent determinant and smaller-denominator budget control; it also remains meaningful without choosing reduced representatives for the competitor.

9 Finite all-best approximations

This section proves the stronger conclusion available for a displayed rational source. It first identifies the glossary's best pairs with minimal source-lattice vectors. The exact source representation then supplies a global denominator cutoff, while nearest-integer geometry and denominator clearance supply a global numerator cutoff. Those bounds turn the universal definition into an executable finite test. The same rectangle then decides whether two best pairs occupy consecutive denominator levels and returns the complete next level above a supplied pair. These are rectangular exhaustive procedures rather than Chevallier's local structural minimal-vector update; the proof nevertheless recovers his local empty-cylinder characterization and the sharp index-one/index-two determinant dichotomy. It also gives the centered-coset classification in the index-two case, the resulting four-candidate error-ratio bound, a finite twelve-value normal form for two putative successive index-two transitions, and the exact cylinder calculation excluding every value in that normal form. Finally it uses these structural results to generate the next level from bounded coefficients of one adjacent pair, with a proved no-skipped-level specification and candidate-count bound. A local denominator-and-error comparison pass then returns exactly the next occupied level without scanning the global source rectangle.

Lemma 9.1 (best approximation equals cylinder minimality)**.** *For a displayed source* z *and pair* $c = (p, q)$ *with* $q \neq 0$*, the two-tier conditions* (7)–(8) *hold if and only if the vector* $(E_z(c), q)$ *is minimal in its closed error–denominator cylinder.*

Proof. First restrict attention to competitors with nonzero denominator. Suppose c is best and $d = (a, b)$ lies in its closed cylinder. If $\mathrm{N}(b) < \mathrm{N}(q)$, the cylinder inequality $\mathcal{E}_z(d) \le \mathcal{E}_z(c)$ contradicts the strict best comparison. Hence $\mathrm{N}(b) = \mathrm{N}(q)$; the weak best comparison supplies the reverse error inequality, so both cylinder coordinates are equal.

Conversely, suppose the cylinder condition holds. For a lower-height competitor, failure of the strict error comparison would put that competitor in the closed cylinder, whose boundary conclusion would falsely equate the two denominator quadrances. For a no-higher competitor, failure of the weak comparison would again put it in the cylinder and would falsely equate its strictly smaller error with the reference error. Thus both tiers follow.

It remains to admit the lattice vectors with $b = 0$. At the fixed denominator q, take the coordinatewise nearest Gaussian integer to zq. The weak equal-height comparison and the fundamental-square estimate give

$$\mathcal{E}_z(c) \le \frac{1}{2}. \tag{75}$$

A nonzero pair $(a, 0)$ has error $-a$, whose integral quadrance is at least one. It therefore cannot lie in the cylinder of c. This proves the full source-lattice statement without postulating a carrier beyond the displayed coefficient pairs. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.isBest_iff_isDenominatorMinimal;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.errorQuadrance_le_half;Purelib.NumberTheory.Approximation.BestApproximation.isBest_iff_isGaussLatticeMinimal`.

Lemma 9.2 (global finite rectangle)**.** *Let $z = N/D$ be the supplied Gaussian fraction and put*

$$B = \mathrm{N}(D)\,, \qquad P = 2(\mathrm{N}(N)\ B + B). \tag{76}$$

Every best pair (p, q) satisfies

$$\mathrm{N}(q) \le B, \qquad \mathrm{N}(p) \le P. \tag{77}$$

Consequently each real and imaginary component of q has magnitude at most B, and each component of p has magnitude at most P.

Proof. The exact pair (N, D) is a displayed competitor with zero error. If $B < \mathrm{N}(q)$, the strict lower-height clause for a best pair would say $\mathcal{E}_z(p, q) < 0$, contrary to nonnegativity. Hence $\mathrm{N}(q) \le B$.

Equation (75) supplies the other ingredient. Define the cleared Gaussian-integer error

$$W = Nq - pD \in \mathbb{Z}[i]. \tag{78}$$

Direct multiplication by D gives

$$\mathrm{N}(W) = \mathcal{E}_z(p, q)\,\mathrm{N}(D) \le B. \tag{79}$$

Since $pD = Nq - W$, multiplicativity and the two-term quadrance estimate give

$$\mathrm{N}(p)\ B = \mathrm{N}(pD) \le 2\big(\mathrm{N}(N)\,\mathrm{N}(q) + \mathrm{N}(W)\big) \le 2(\mathrm{N}(N)\,B + B) = P.$$

The nonzero integral D has $B \ge 1$, so $\mathrm{N}(p) \le \mathrm{N}(p)\,, B \le P$. Finally, for a Gaussian integer $u + iv$, each of $|u|$ and $|v|$ is bounded by $u^2 + v^2$; applying this elementary displayed inequality gives the component bounds. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.IsBest.denominatorQuadrance_le_source;Purelib.NumberTheory.Approximation.BestApproximation.clearedError_norm_identity;Purelib.NumberTheory.Approximation.BestApproximation.clearedError_normNat_le_denominator;Purelib.NumberTheory.Approximation.BestApproximation.numeratorQuadrance_le_bound;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.mem_candidates`.

Lemma 9.3 (finite descent and the mixed-cylinder criterion)**.** *Let d be a displayed pair with nonzero denominator, $\mathrm{N}(q_d) \le B$, and $\mathcal{E}_z(d) \le 1/2$. There is a best pair c such that*

$$\mathrm{N}(q_c) \le \mathrm{N}(q_d)\,, \qquad \mathcal{E}_z(c) \le \mathcal{E}_z(d). \tag{80}$$

Consequently, two best pairs c_-, c_+ with $\mathrm{N}(q_-) < \mathrm{N}(q_+)$ occupy consecutive best levels if and only if no nonzero displayed pair lies in the mixed open cylinder

$$\mathring{C}_z(c_-, c_+) = \{d : \mathcal{E}_z(d) < \mathcal{E}_z(c_-),\ \mathrm{N}(q_d) < \mathrm{N}(q_+)\}. \tag{81}$$

In particular, for Gaussian integers g, h not both zero,

$$\mathcal{E}_z(c_-) \le \mathcal{E}_z(gc_- + hc_+) \quad \text{or} \quad \mathrm{N}(q_+) \le \mathrm{N}(gq_- + hq_+)\,. \tag{82}$$

Proof. Restrict the global candidate rectangle to nonzero denominators of quadrance at most $\mathrm{N}(q_d)$. The supplied pair d belongs to that finite list. Recursively select the lexicographically least pair, first by error quadrance and then, among exact error ties, by denominator quadrance. The selected pair has no greater error or denominator than d.

To verify that it is best, consider a competitor below the selected height. If its error exceeds $1/2$, the selected error is already strictly smaller. Otherwise the same global bounds put the competitor in the checked list. Lexicographic minimality gives the weak error inequality, and equality would force the selected denominator to be no greater, contradicting strict lower height. The no-higher weak clause follows by the same two cases without the last contradiction. This proves (80) by an executable finite selection.

Now suppose c_-, c_+ are consecutive and that a nonzero d lies in (81). A vertical pair has integral error quadrance at least one, whereas $\mathcal{E}_z(c_-) \le 1/2$, so $q_d \neq 0$. Bestness of c_- forces $\mathrm{N}(q_-) < \mathrm{N}(q_d)$. Apply the descent just proved to obtain a best c no higher and no worse than d. Bestness of c_- again gives $\mathrm{N}(q_-) < \mathrm{N}(q_c)$, while $\mathrm{N}(q_c) \le \mathrm{N}(q_d) < \mathrm{N}(q_+)$, contradicting consecutiveness.

Conversely, a best pair at an intermediate level has strictly smaller error than c_- and therefore lies in the mixed open cylinder. This proves the equivalence. Finally, coefficient independence of c_-, c_+ makes $gc_- + hc_+$ nonzero whenever $(g, h) \neq (0, 0)$; applying cylinder emptiness and deciding which strict inequality fails gives (82). □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.RankLE;Purelib.NumberTheory.Approximation.BestApproximation.selectRankMinimum;Purelib.NumberTheory.Approximation.BestApproximation.exists_best_below_of_small;Purelib.NumberTheory.Approximation.BestApproximation.InOpenMixedCylinder;Purelib.NumberTheory.Approximation.BestApproximation.MixedCylinderEmpty;Purelib.NumberTheory.Approximation.BestApproximation.areConsecutiveBest_`

`iff_mixedCylinderEmpty;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.linearCombination_pairNonzero;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.linearCombination_outside_mixedCylinder;Purelib.Papers.RFAP.Approximation.consecutiveBest_iff_mixedCylinderEmpty;Purelib.Papers.RFAP.Approximation.exists_bestBelow_of_small`.

Theorem 9.4 (executable finite all-best selector)**.** *For a displayed source $z = N/D$, form the component rectangles from lemma 9.2. For every candidate reference pair in that rectangle, check (75) and both comparisons (7)–(8) against every pair in the same rectangle. Retain precisely the successful references. The resulting finite list* $\mathrm{AllBest}(z)$ *satisfies*

$$c \in \mathrm{AllBest}(z) \quad \Longleftrightarrow \quad c \text{ is a best approximation to } z. \tag{83}$$

Its unfiltered candidate list has exactly $16(B+1)^2(P+1)^2$ *entries, and the output has no more entries than this.*

Proof. Soundness is the only point at which a candidate outside the rectangle could appear to matter. If the strict comparison failed, the offending competitor would have no greater error than the reference; if the weak comparison failed, it would have strictly smaller error. In either case its error is at most $1/2$, and its denominator is no higher than the reference denominator, which is at most B. The proof of lemma 9.2 therefore places its numerator and denominator inside the same checked rectangle. The finite test has already rejected the proposed failure.

For completeness, every genuinely best reference satisfies (75) and the bounds of lemma 9.2, so it occurs in the rectangle and passes every checked comparison by definition. Each signed component list contains both signs of the $B+1$ or $P+1$ possible magnitudes, including a harmless second copy of zero. The two denominator coordinates and two numerator coordinates therefore give

$$\big(2(B+1)\big)^2\big(2(P+1)\big)^2 = 16(B+1)^2(P+1)^2$$

displayed entries. Structural filtering cannot increase their number. □

Formal verification:

`Purelib.Data.Gaussian.Height.GaussianInt.componentSquare;Purelib.NumberTheory.Approximation.BestApproximation.PassesBestTest;Purelib.NumberTheory.Approximation.BestApproximation.passesBestTest_iff_isBest;Purelib.NumberTheory.Approximation.BestApproximation.candidates_length;Purelib.NumberTheory.Approximation.BestApproximation.allBest;Purelib.NumberTheory.Approximation.BestApproximation.allBest_length_le;Purelib.NumberTheory.Approximation.BestApproximation.mem_allBest_iff_isBest`.

Theorem 9.5 (executable consecutive-best successor)**.** *Fix a displayed source z and a displayed lower pair c_-. Filter the same candidate rectangle by retaining a candidate c_+ exactly when both endpoint pairs pass the best test, their denominator quadrances satisfy $\mathrm{N}(q_-) < \mathrm{N}(q_+)$, and every candidate at an intermediate denominator quadrance fails the best test. The resulting finite list* $\mathrm{NextBest}(z, c_-)$ *satisfies*

$$c_+ \in \mathrm{NextBest}(z, c_-) \quad \Longleftrightarrow \quad c_-, c_+ \text{ occupy consecutive best levels.} \tag{84}$$

Its length is at most $16(B+1)^2(P+1)^2$.

Proof. Suppose first that the finite test accepts c_+. Exactness of the endpoint best tests proves that c_- and c_+ are best. If a genuinely best pair lay at an intermediate denominator quadrance, the global rectangle lemma would place it among the checked candidates, contradicting the final clause of the test. The two endpoint levels are therefore consecutive.

Conversely, suppose the endpoints occupy consecutive best levels. Both endpoint tests pass by (83). No candidate passing the best test can have intermediate denominator quadrance, because exactness of that test would turn it into the best pair excluded by consecutiveness. Thus the finite conjunction passes and the filter retains c_+. Filtering the same candidate rectangle also gives the stated length bound. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest; Purelib.NumberTheory.Approximation.BestApproximation. PassesConsecutiveTest;Purelib.NumberTheory.Approximation. BestApproximation.passesConsecutiveTest_iff;Purelib.NumberTheory. Approximation.BestApproximation.consecutiveSuccessors;Purelib. NumberTheory.Approximation.BestApproximation.consecutiveSuccessors_length_ le;Purelib.NumberTheory.Approximation.BestApproximation.mem_ consecutiveSuccessors_iff;Purelib.Papers.RFAP.Approximation.mem_ consecutiveBestSuccessors_iff.`

Theorem 9.6 (total next-best step for a finite RAT response)**.** *Let T be any RFAP-1 RAT, let σ be one of its displayed current states, and execute a positive-accuracy request $\sigma' = T(\sigma, \varepsilon)$. Let z_R be the canonical common-denominator presentation of the rational anchor of σ'. Then z_R belongs to the returned winding-one box,*

$$\operatorname{diag}^2(B') \le \varepsilon^2, \qquad \mathrm{N}(f(z_R)) \le L_1(f, A) \operatorname{diag}^2(B'),$$

and B' refines the current box. The executable list $\mathrm{AllBest}(z_R)$ *contains all and only best approximants to z_R and is nonempty.*

If c is any member of this list, a terminating optional calculation returns one c_+ satisfying

$$c, c_+ \text{ occupy consecutive best levels},$$

or returns empty. It returns empty if and only if $\mathcal{E}_{z_R}(c) = 0$. Consequently every nonterminal call produces the next occupied best level, and every emitted pair is a genuine best approximant to the displayed RAT response.

Proof. The RFAP-1 track call supplies the refined winding-one state, its squared diagonal bound, and its rational anchor. Canonical common-denominator conversion preserves that anchor exactly. Nonzero winding gives the stated anchor residual bound, while the track record gives refinement and accuracy. Conversion from the positive natural common denominator to a real Gaussian-integer denominator again preserves the value exactly, so theorems 9.4 and 9.5 apply. Finite descent from the exact pair representing z_R proves that the all-best table is nonempty.

For totality, the same descent produces a best pair of zero error. If c has nonzero error, its best property forces its denominator quadrance to be strictly below that terminal level. Taking the least denominator among the finite best table strictly above c produces the consecutive successor. If c has zero error, a later best pair would have to improve strictly below zero, contradicting nonnegativity of quadrance. These two finite arguments prove both directions of the empty option criterion. □

Formal verification:

`Purelib.NumberTheory.Approximation.RATBestApproximation.trackResponseState_refines;Purelib.NumberTheory.Approximation.RATBestApproximation.trackResponseState_accurate;Purelib.NumberTheory.Approximation.RATBestApproximation.trackResponseSample_inside;Purelib.NumberTheory.Approximation.RATBestApproximation.trackResponseSample_residual_le;Purelib.NumberTheory.Approximation.RATBestApproximation.mem_trackBestApproximations_iff;Purelib.NumberTheory.Approximation.RATBestApproximation.trackBestApproximations_nonempty;Purelib.NumberTheory.Approximation.RATBestApproximation.trackNextBestApproximation?_sound;Purelib.NumberTheory.Approximation.RATBestApproximation.trackNextBestApproximation_or_exactTerminal;Purelib.NumberTheory.Approximation.RATBestApproximation.respondToTrack;Purelib.Papers.RFAP.Approximation.respondToRAT.`

Corollary 9.7 (RFAP-II best-response strengthening of FTA and RFAP)**.** *Every output branch of the RFAP-1 finite constructive FTA carries the following additional RFAP-II arithmetic capability.*

(i) A degree-one exact terminal z has a nonempty executable table containing all and only its best Gaussian-rational approximants.

(ii) For every one of the degree-indexed RATs in a higher-degree output, every displayed current state and every positive rational accuracy request produce the complete proof-bearing response of theorem 9.6.

The same alternatives apply to every signed numerator or denominator site in the RFAP-1 rational-function action principle: an exact linear site uses its displayed terminal point, while every finite call to every higher-degree site RAT returns the refined box, residual-bound anchor, all-and-only-best table, and total next-best-or-terminal step. Existing multiplicity signs, winding certificates, separation, ownership, and maximality data are retained unchanged.

Proof. Case-split the RFAP-1 positive-degree FTA result. In the linear branch, canonical common-denominator conversion of the exact displayed point followed by the finite all-best calculation gives the first alternative. In the higher branch, each output is already a degree-indexed `WindingOneTrack`; apply theorem 9.6 to the selected track, current state, and request. This is the finite construction packaged by `strengthenFTAResult`.

Each signed rational-function site has the identical exact-or-higher shape. The exact branch uses the point construction, and the higher branch applies the generic track construction to each track of the focused squarefree factor. This is `strengthenRationalSite`. Both constructions only append RFAP-II arithmetic records, so none of the Paper-I certificates is altered and no dependency from RFAP-1 back to RFAP-2 is introduced. □

Formal verification:

`Purelib.NumberTheory.Approximation.RATBestApproximation.PointBestResponse;Purelib.NumberTheory.Approximation.RATBestApproximation.respondToPoint;Purelib.NumberTheory.Approximation.RATBestApproximation.TrackBestResponse;Purelib.NumberTheory.Approximation.RATBestApproximation.respondToTrack;Purelib.Papers.RFAP.Approximation.FTAApproximationCapability;Purelib.Papers.RFAP.Approximation.strengthenFTAResult;Purelib.Papers.RFAP.`

`Approximation.RationalSiteApproximationCapability;Purelib.Papers.RFAP.Approximation.strengthenRationalSite`.

Lemma 9.8 (coefficient primitivity and Cramer certificate)**.** *A best pair has no nonunit common Gaussian-integer scale: if $c = \lambda d$ coefficientwise and $q_d \neq 0$, then $\mathrm{N}(\lambda) = 1$.*

For arbitrary displayed pairs u, v, x, put $\delta = \det(u, v)$. The following coefficientwise identity is an executable Cramer certificate:

$$\det(x, v)u + \det(u, x)v = \delta x. \tag{85}$$

Multiplying the two displayed coefficients on the left by $\bar{\delta}$ gives

$$\bar{\delta}\det(x, v)u + \bar{\delta}\det(u, x)v = \mathrm{N}(\delta)\ x. \tag{86}$$

In particular, if $\mathrm{N}(\delta) = 1$, the displayed coefficients in (86) express every x as an integral combination of u, v. If $\mathrm{N}(\delta) = 2$, the same coefficients express $2x$ as such a combination, giving the finite annihilation certificate behind the centered index-two alternative.

Proof. If $c = \lambda d$, multiplication of Gaussian quadrances gives

$$\mathrm{N}(q_c) = \mathrm{N}(\lambda)\,\mathrm{N}(q_d)\,, \qquad \mathcal{E}_z(c) = \mathrm{N}(\lambda)\,\mathcal{E}_z(d).$$

The nonzero denominator makes $\mathrm{N}(\lambda)$ positive. If it exceeded one, d would have strictly smaller denominator while its error was no greater, contradicting the strict lower-height clause for c. Hence $\mathrm{N}(\lambda) = 1$.

Both coordinates of (85) are obtained by expanding two products and cancelling the cross terms. Multiplication by $\bar{\delta}$ and the integral identity $\bar{\delta}\delta = \mathrm{N}(\delta)$ give (86). When the norm is one, coefficientwise multiplication by the real Gaussian integer 1 leaves x unchanged. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.scalePair;Purelib.NumberTheory.Approximation.BestApproximation.errorQuadrance_scalePair;Purelib.NumberTheory.Approximation.BestApproximation.denominatorQuadrance_scalePair;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.scalingCoefficient_norm_eq_one;Purelib.NumberTheory.Approximation.BestApproximation.linearCombination_cramer;Purelib.NumberTheory.Approximation.BestApproximation.linearCombination_cramer_norm;Purelib.NumberTheory.Approximation.BestApproximation.linearCombination_cramer_of_index_one;Purelib.NumberTheory.Approximation.BestApproximation.linearCombination_cramer_of_index_two`.

Theorem 9.9 (constructive index-one/index-two dichotomy)**.** *For consecutive best pairs $c_- = (p_-, q_-)$ and $c_+ = (p_+, q_+)$, put*

$$I(c_-, c_+) = \mathrm{N}(p_-q_+ - p_+q_-)\,. \tag{87}$$

Then

$$I(c_-, c_+) \in \{1, 2\}. \tag{88}$$

Proof. The determinant-error identity is

$$p_-q_+ - p_+q_- = q_-E_z(c_+) - q_+E_z(c_-).$$

If the determinant vanished, multiplicativity of quadrance would give

$$\mathrm{N}(q_-)\,\mathcal{E}_z(c_+) = \mathrm{N}(q_+)\,\mathcal{E}_z(c_-).$$

But c_+ is best and $\mathrm{N}(q_-) < \mathrm{N}(q_+)$, so $\mathcal{E}_z(c_+) < \mathcal{E}_z(c_-)$. Multiplying this strict inequality by the positive $\mathrm{N}(q_-)$, and the denominator inequality by the nonnegative $\mathcal{E}_z(c_-)$, makes the left side strictly smaller than the right. Thus the displayed Gaussian-integer determinant is nonzero and its quadrance is positive.

It remains to prove the sharp upper bound without importing a lattice index theorem. Run the finite extended Gaussian Euclidean algorithm on p_-, q_-. It returns displayed g, s, t, p_0, q_0 satisfying

$$g = sp_- + tq_-, \qquad (p_-, q_-) = g(p_0, q_0).$$

Primitivity of c_- forces $\mathrm{N}(g) = 1$. Multiplying the Bézout coefficients by $\bar{g}$ therefore gives $a, b \in \mathbb{Z}[i]$ with $ap_- + bq_- = 1$. The displayed complement $V = (-b, a)$ satisfies $\det(c_-, V) = 1$, so the Cramer certificate gives an exact normal form

$$c_+ = \beta c_- + \delta V, \qquad \delta = \det(c_-, c_+). \tag{89}$$

Apply one nearest-Gaussian-integer division to β/δ. Writing $\beta = k\delta + r$ and $V' = kc_- + V$ turns (89) into

$$c_+ = rc_- + \delta V', \qquad 2\,\mathrm{N}(r) \le \mathrm{N}(\delta) = I(c_-, c_+). \tag{90}$$

The half-open square supplies a little more than this usual half-quadrance bound. If $\mathrm{N}(\delta) \ge 4$, direct integral calculation of the dot and cross coordinates of the remainder gives the alternative

$$2(\mathrm{N}(r) + 1) \le \mathrm{N}(\delta) \quad \text{or} \quad \delta = r(-1 + i). \tag{91}$$

Indeed, failure of the first inequality leaves only $\mathrm{N}(\delta) = 2\,\mathrm{N}(r)$ or $2\,\mathrm{N}(r) + 1$. The odd case is incompatible with the two scaled half-open coordinate bounds and Lagrange's identity. In the even case equality forces both scaled coordinates to be the included lower endpoint, so multiplication out gives the displayed diagonal factorization. This is a finite integer argument; no volume, compactness, or quotient cardinality occurs.

In the first case of (91), $r \neq 0$ by primitivity of c_+. The two-term quadrance estimate and the strict endpoint inequalities give

$$\begin{aligned}
\mathrm{N}(\delta)\ \mathcal{E}_z(V') &\le 2\big(\mathcal{E}_z(c_+) + \mathrm{N}(r)\ \mathcal{E}_z(c_-)\big) < 2(\mathrm{N}(r) + 1)\mathcal{E}_z(c_-),\\
\mathrm{N}(\delta)\ \mathrm{N}(q_{V'}) &\le 2\big(\mathrm{N}(q_+) + \mathrm{N}(r)\ \mathrm{N}(q_-)\big) < 2(\mathrm{N}(r) + 1)\,\mathrm{N}(q_+)\,.
\end{aligned}$$

Thus V' lies strictly inside the mixed cylinder, contradicting lemma 9.3. In the second case, (90) reads $c_+ = r(c_- + (-1 + i)V')$. Primitivity of c_+ forces $\mathrm{N}(r) = 1$, and hence $\mathrm{N}(\delta) = 2$, contrary to $\mathrm{N}(\delta) \ge 4$.

Therefore $0 < I(c_-, c_+) < 4$. An integral Gaussian quadrance below four is 0, 1, or 2; positivity excludes zero and proves (88). □

Formal verification:

`Purelib.NumberTheory.Approximation.GaussianEuclidean.extendedGCD;Purelib.NumberTheory.Approximation.GaussianEuclidean.remainder_dot_cross_bounds;Purelib.NumberTheory.Approximation.GaussianEuclidean.remainder_clearance_or_diagonal_factor;Purelib.NumberTheory.Approximation.BestApproximation.coefficientDeterminant;Purelib.NumberTheory.Approximation.BestApproximation.coefficientIndexQuadrance;Purelib.NumberTheory.Approximation.BestApproximation.toGaussian_coefficientDeterminant;Purelib.NumberTheory.Approximation.BestApproximation.determinant_ne_zero_of_ordered_best;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.coefficientIndexQuadrance_pos;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.euclideanGCD_norm_eq_one;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.coefficientDeterminant_bezoutComplement_eq_one;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.exists_reducedCoefficientBasis;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.coefficientIndexQuadrance_le_two;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.coefficientIndexQuadrance_eq_one_or_two;Purelib.Papers.RFAP.Approximation.consecutiveBest_coefficientIndexQuadrance_pos;Purelib.Papers.RFAP.Approximation.consecutiveBest_coefficientIndexQuadrance_eq_one_or_two.`

Lemma 9.10 (finite index-two error chamber)**.** *Let u, v occupy consecutive best levels for the displayed source z. If $\mathrm{N}(\det(u, v)) = 2$, then*

$$\mathcal{E}_z(u) < 4\mathcal{E}_z(v). \tag{92}$$

Proof. For each Gaussian unit $\alpha \in \{1, -1, i, -i\}$, the parity construction produces a displayed nonzero pair c_α satisfying

$$\det(u, v)c_\alpha = v - \alpha u.$$

The mixed cylinder between u and v is empty. Consequently each of these four identities supplies one of two exact exterior inequalities: either the error quadrance of $v - \alpha u$ is at least $2\mathcal{E}_z(u)$, or its denominator quadrance is at least $2\,\mathrm{N}(q_v)$.

The four alternatives are incompatible with $4\mathcal{E}_z(v) \le \mathcal{E}_z(u)$. Two error alternatives attached to opposite units contradict the parallelogram identity. Two attached to adjacent units contradict the corresponding pair of orthogonal-projection identities under the assumed factor-four bound. If at most one alternative is an error alternative, at least three are denominator alternatives, hence two use opposite units and again contradict the parallelogram identity. This finite sixteen-case calculation proves (92); it contains no square root or angular choice. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.exists_centeredUnitCandidate_outsideCylinder;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.centeredUnit_selfObstruction;Purelib.NumberTheory.Approximation.BestApproximation.four_unit_obstructions_force_error_lt_four_mul;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.errorQuadrance_lower_`

`lt_four_mul_upper_of_index_two;Purelib.Papers.RFAP.Approximation.consecutiveBest_indexTwoErrorRatio`.

For Gaussian integers a and δ, write $\delta \mid a$ only when a displayed Gaussian integer k satisfies $a = \delta k$. This witness-bearing notation is important below: no quotient ring is being introduced.

Theorem 9.11 (finite centered-coset and repeated-index classification)**.** *Let u, v be consecutive best pairs and put $\delta = \det(u, v)$. Every displayed coefficient pair $x \in \mathbb{Z}[i]^2$ satisfies one of*

$$x = au + bv \qquad (a, b \in \mathbb{Z}[i]), \tag{93}$$

$$\delta x = au + bv \qquad (a, b \in \mathbb{Z}[i], \delta \nmid a, \delta \nmid b). \tag{94}$$

If $\mathrm{N}(\delta) = 1$, the first alternative always holds. If $\mathrm{N}(\delta) = 2$, the two alternatives give the ordinary span and its centered coset. Conversely, for every a, b with $\delta \nmid a$ and $\delta \nmid b$, the equation (94) has a displayed pair x as a solution.

Moreover, suppose u, v and v, w are consecutive best pairs and both determinants have quadrance two. There are displayed $\varepsilon, k \in \mathbb{Z}[i]$ such that

$$w = \varepsilon u + \delta k v, \qquad \mathrm{N}(\varepsilon) = 1, \qquad \mathrm{N}(k) \in \{1, 2, 4\}. \tag{95}$$

Proof. Every quadrance-two Gaussian integer is one of the four diagonal elements $\pm 1 \pm i$. Direct multiplication therefore proves the exact test

$$\delta \mid a \iff \mathrm{Re}(a) + \mathrm{Im}(a) \text{ is even.} \tag{96}$$

The integer quotient and remainder on division by two give the parity split constructively. In particular, the nondivisible class is the odd class; it is closed under multiplication, while the sum of two members of that class is divisible by δ.

Complete u to the explicit Bézout basis (u, V) used in the proof of theorem 9.9. Write

$$v = \beta u + \delta V, \qquad x = \gamma u + \eta V.$$

If $\delta \mid \beta$, the first identity gives v a common factor of quadrance two, contrary to primitivity; hence $\delta \nmid \beta$. If $\delta \mid \eta$, substitution gives integral coefficients for x in u, v. Otherwise

$$\delta x = (\delta\gamma - \beta\eta)u + \eta v.$$

Both displayed coefficients are nondivisible: if the first were divisible, then so would be $\beta\eta$, contradicting closure of the odd class under multiplication. This proves the forward classification. Conversely, when a, b are nondivisible, so is βb, and therefore $a + \beta b = \delta q$ for a computed $q \in \mathbb{Z}[i]$. The pair $x = qu + bV$ then satisfies $\delta x = au + bv$.

For the last assertion, first apply the classification to w. Its centered alternative would force a nondivisible cleared coefficient to have quadrance two after taking determinants. But every quadrance-two Gaussian integer is divisible by δ, a contradiction. Hence $w = \varepsilon u + bv$; taking the determinant with v and using both index-two hypotheses gives $\mathrm{N}(\varepsilon) = 1$.

It remains to show $\delta \mid b$. The centered construction for the second pair supplies the displayed half-sum h with $\det(v, w)h = v + w$. Classifying h relative to u, v, its integral alternative is ruled out immediately by multiplicativity of quadrance: it would give $2(2\,\mathrm{N}(r)) = 2$ for an integral coefficient r. Comparing the two coefficient expressions in the centered alternative shows that $1 + b$ is nondivisible by δ. Since -1 is also nondivisible,

their sum $b = (1 + b) + (-1)$ is divisible by δ. The resulting quotient is the displayed k in (95).

The strict increase from the denominator level of u to that of w shows $k \neq 0$. It remains to bound its quadrance. Apply lemma 9.10 to u, v. In the error identity obtained from (95), the first summand has quadrance $\mathcal{E}_z(u) < 4\mathcal{E}_z(v)$, whereas the second has quadrance $2\,\mathrm{N}(k)\ \mathcal{E}_z(v)$. If $\mathrm{N}(k) \geq 5$, the latter is therefore at least $10\mathcal{E}_z(v)$. The exact two-coordinate Cauchy identity

$$(x_1y_1 + x_2y_2)^2 + (x_1y_2 - x_2y_1)^2 = (x_1^2 + x_2^2)(y_1^2 + y_2^2)$$

and the expansion of the quadrance of their sum then show that the error of w cannot be smaller than $\mathcal{E}_z(v)$. This contradicts the strict best comparison between the consecutive levels v, w. Hence $0 < \mathrm{N}(k) \leq 4$; the displayed coordinate-square classification leaves exactly $\mathrm{N}(k) = 1, 2$, or 4.

This proves the finite counterpart of the coefficient bound in Chevallier's repeated-index lemma and supplies the bounded input for the exclusion below. □

Formal verification:

`Purelib.NumberTheory.Approximation.GaussianCoset.coordinateEven_or_coordinateOdd;Purelib.NumberTheory.Approximation.GaussianCoset.divides_iff_coordinateEven_of_norm_two;Purelib.NumberTheory.Approximation.GaussianCoset.divides_of_norm_two;Purelib.NumberTheory.Approximation.GaussianCoset.not_divides_mul_of_norm_two;Purelib.NumberTheory.Approximation.GaussianCoset.divides_add_of_not_divides_of_norm_two;Purelib.NumberTheory.Approximation.BestApproximation.InCoefficientSpan;Purelib.NumberTheory.Approximation.BestApproximation.InCenteredCoefficientCoset;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.inCoefficientSpan_or_inCenteredCoefficientCoset_of_index_two;Purelib.NumberTheory.Approximation.BestApproximation.IsBest.exists_centeredCoefficientPair_of_index_two;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.coefficientSpan_or_centeredCoset;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.integralTransition_with_divisible_middleCoefficient;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.repeatedIndexTwo_boundedNormalForm;Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.repeatedIndexTwoNormalForm;Purelib.Papers.RFAP.Approximation.consecutiveBest_coefficientSpan_or_centeredCoset;Purelib.Papers.RFAP.Approximation.consecutiveBest_indexTwoTransition;Purelib.Papers.RFAP.Approximation.consecutiveBest_indexTwoEnumeratedTransition`.

Theorem 9.12 (no successive index-two transitions)**.** *Let u, v and v, w occupy consecutive best levels for the displayed source z. If*

$$\mathrm{N}(\det(u, v)) = 2,$$

then

$$\mathrm{N}(\det(v, w)) = 1.$$

Proof. Suppose instead that both determinant quadrances are two. By theorem 9.11, write

$$w = \varepsilon u + \delta k v, \qquad \mathrm{N}(\varepsilon) = 1, \qquad \mathrm{N}(k) \in \{1, 2, 4\}, \qquad \delta = \det(u, v).$$

Put $a = \delta k$. Rotate the lower error and denominator by the displayed unit ε, and divide by the two displayed nonzero reference coordinates. This constructs exact Gaussian-rational coordinates x, y for which

$$\mathrm{N}(x) < 1, \qquad 1 < 4\,\mathrm{N}(x)\,, \qquad \mathrm{N}(1+ax) < \mathrm{N}(x)\,, \tag{97}$$

$$\mathrm{N}(y) < 1, \qquad 1 < \mathrm{N}(y+a)\,. \tag{98}$$

The first and last strict inequalities are just the error decrease and denominator increase through the two consecutive levels; the middle bound in (97) is lemma 9.10. The two affine expressions are the coefficientwise error and denominator identities obtained from the displayed normal form.

For every displayed Gaussian unit ζ, the centered candidate outside the first mixed cylinder gives

$$2 \le \mathrm{N}(x-\zeta) \quad \text{or} \quad 2 \le \mathrm{N}(1-\zeta y)\,, \tag{99}$$

while the same candidate outside the next mixed cylinder gives

$$2\,\mathrm{N}(x) \le \mathrm{N}(x-\zeta) \quad \text{or} \quad 2\,\mathrm{N}(y+a) \le \mathrm{N}(1-\zeta y)\,. \tag{100}$$

These are homogeneous consequences of the displayed cylinder certificates; normalization cancels only positive rational quadrances.

Multiplicativity gives $\mathrm{N}(a) = 2\,\mathrm{N}(k) \in \{2, 4, 8\}$. The bounded Gaussian-integer classification makes a a displayed unit associate of, respectively, $1+i$, 2, or $2+2i$. Rotating x by that unit and y by its displayed conjugate inverse preserves every quadrance above and merely permutes the four choices of ζ. It therefore suffices to exclude those three canonical coefficients.

For each coefficient, substitute $\zeta \in \{1, -1, i, -i\}$ into (99)–(100) and expand the quadrances coefficientwise. For $a = 2$, the opposite real unit tests force the boundary equality for the real projection; together with (97) and $1 < 4\,\mathrm{N}(x)$ this gives both $5\,\mathrm{N}(x) < 1$ and $1 < 4\,\mathrm{N}(x)$. For $a = 1+i$ and $a = 2+2i$, the real and imaginary tests leave only the corresponding diagonal crossing patterns. Their strict transition alternatives force a coordinate of x or y outside its open unit disc; the sole weak boundary pattern would give a rational square whose value is three, which is impossible by the reduced numerator–denominator parity argument. Thus all three canonical coefficients are impossible.

The index-one/index-two dichotomy theorem 9.9 leaves quadrance one as the only value for $\det(v, w)$. □

Formal verification:

`Purelib.NumberTheory.Approximation.RepeatedIndex.NormalizedCylinderData;`
`Purelib.NumberTheory.Approximation.RepeatedIndex.normalizeCylinderData;`
`Purelib.NumberTheory.Approximation.RepeatedIndex.transportNormalizedData;`
`Purelib.NumberTheory.Approximation.RepeatedIndex.diagonalStep_`
`obstructions_impossible;Purelib.NumberTheory.Approximation.RepeatedIndex.`
`doubledAxisStep_obstructions_impossible;Purelib.NumberTheory.`
`Approximation.RepeatedIndex.doubledDiagonalStep_obstructions_impossible;`
`Purelib.NumberTheory.Approximation.RepeatedIndex.repeatedIndexTwo_`
`impossible;Purelib.NumberTheory.Approximation.RepeatedIndex.`
`nextCoefficientIndexQuadrance_eq_one_of_eq_two;Purelib.Papers.RFAP.`
`Approximation.consecutiveBest_repeatedIndexTwo_impossible;Purelib.Papers.`
`RFAP.Approximation.consecutiveBest_nextIndex_eq_one_of_indexTwo.`

Theorem 9.13 (bounded coefficient-local successor generation)**.** *Let u, v occupy consecutive best levels for $z = N/D$, and put*

$$C = 4\big(\mathrm{N}(D) + \mathrm{N}(q_u)\big), \qquad \delta = \det(u, v).$$

Form a finite candidate list by taking each of the eight $a \in \mathbb{Z}[i]$ with $\mathrm{N}(a) \in \{1, 2\}$, every displayed $b \in \mathbb{Z}[i]$ whose two coordinate magnitudes are at most C, and both expressions

$$au + bv, \qquad \mathrm{half}\big(\bar{\delta}(au + bv)\big). \tag{101}$$

Here half *divides both displayed integral coordinates by two; the second expression is a total calculation, and is used as exact division only when the centered identity proves those coordinates even.*

Call a candidate improving *when its denominator quadrance is strictly greater than that of v but no greater than $\mathrm{N}(D)$, and its error quadrance is strictly smaller than that of v. Retain an improving candidate c when every improving candidate in the same local list has denominator quadrance at least $\mathrm{N}(q_c)$ and every one at the same denominator level has error quadrance at least $\mathcal{E}_z(c)$. This coefficient-local test returns exactly the pairs w for which v, w occupy consecutive best levels. In particular, the local generator skips no next best level. Before filtering, its length is at most*

$$64(C + 1)^2. \tag{102}$$

Proof. Let w occupy the next best level above v. If $\mathrm{N}(\delta) = 1$, the coefficientwise Cramer identity gives

$$w = au + bv.$$

Taking the determinant with v shows $\mathrm{N}(a) = \mathrm{N}(\det(v, w)) \in \{1, 2\}$.

If $\mathrm{N}(\delta) = 2$, theorem 9.12 gives $\mathrm{N}(\det(v, w)) = 1$. The integral-span branch would instead make that quadrance equal to $2\,\mathrm{N}(a)$, so it is impossible. The centered-coset branch is therefore forced:

$$\delta w = au + bv.$$

Taking determinants now gives $\mathrm{N}(a) = 1$. Since $\bar{\delta}\delta = 2$, multiplying this identity by $\bar{\delta}$ proves that the second expression in (101) computes w coefficientwise; no quotient is selected.

Both branches have the common denominator identity

$$bq_v = sq_w - aq_u, \qquad \mathrm{N}(s) \le 2, \qquad \mathrm{N}(a) \le 2,$$

where $s = 1$ in the integral case and $s = \delta$ in the centered case. The elementary difference bound $\mathrm{N}(x - y) \le 2(\mathrm{N}(x) + \mathrm{N}(y))$, the source bound $\mathrm{N}(q_w) \le \mathrm{N}(D)$, and $q_v \neq 0$ give

$$\mathrm{N}(b) \le \mathrm{N}(b)\,\mathrm{N}(q_v) \le 2\big(2\,\mathrm{N}(D) + 2\,\mathrm{N}(q_u)\big) = C.$$

Thus a and b occur in the displayed coefficient lists, and w occurs in the local candidate list.

For completeness, let w be an actual next pair. Emptying the mixed cylinder between v and w excludes every improving local contender of smaller denominator quadrance. Bestness of w excludes a smaller error at the same level. Hence w passes the local test.

For soundness, let c pass that test. Finite descent below c produces a best pair r with no greater denominator or error. Since c improves on the best pair v, the denominator

of r is strictly greater than that of v. Select the least denominator quadrance among the finite complete list of best pairs strictly above v; call a selected pair w. Then v, w are consecutive and

$$\mathrm{N}(q_w) \le \mathrm{N}(q_r) \le \mathrm{N}(q_c)\,.$$

The completeness argument already places w in the local candidate list, so the local test for c gives the reverse inequality. Thus c and w have the same denominator quadrance. The same-level clause of the local test and bestness of w give equality of their error quadrances. Bestness is unchanged under these two exact ties, so c occupies the same next level as w.

The finite all-best list is used here only to prove soundness of the local calculation. Evaluating the successor update itself enumerates and compares only the displayed coefficient candidates. This proves soundness and completeness. There are eight leading coefficients, $4(C+1)^2$ records in the signed coordinate square for b, and two forms in (101); multiplication gives (102). □

Formal verification:

`Purelib.NumberTheory.Approximation.LocalSuccessor.halfGaussianInt;Purelib.NumberTheory.Approximation.LocalSuccessor.divideByNormTwo;Purelib.NumberTheory.Approximation.LocalSuccessor.divideByNormTwo_mul;Purelib.NumberTheory.Approximation.LocalSuccessor.trailingCoefficientBound;Purelib.NumberTheory.Approximation.LocalSuccessor.candidates;Purelib.NumberTheory.Approximation.LocalSuccessor.next_mem_candidates;Purelib.NumberTheory.Approximation.LocalSuccessor.selectDenominatorMinimum;Purelib.NumberTheory.Approximation.LocalSuccessor.exists_consecutiveBest_below;Purelib.NumberTheory.Approximation.LocalSuccessor.PassesLocalSuccessorTest;Purelib.NumberTheory.Approximation.LocalSuccessor.passesLocalSuccessorTest_of_consecutive;Purelib.NumberTheory.Approximation.LocalSuccessor.consecutive_of_passesLocalSuccessorTest;Purelib.NumberTheory.Approximation.LocalSuccessor.passesLocalSuccessorTest_iff;Purelib.NumberTheory.Approximation.LocalSuccessor.successors;Purelib.NumberTheory.Approximation.LocalSuccessor.mem_successors_iff;Purelib.NumberTheory.Approximation.LocalSuccessor.candidates_length_le;Purelib.Papers.RFAP.Approximation.localSuccessorCoefficientBound;Purelib.Papers.RFAP.Approximation.localConsecutiveBestSuccessors;Purelib.Papers.RFAP.Approximation.passesLocalSuccessorTest;Purelib.Papers.RFAP.Approximation.passesLocalSuccessorTest_iff;Purelib.Papers.RFAP.Approximation.mem_localConsecutiveBestSuccessors_iff;Purelib.Papers.RFAP.Approximation.localSuccessorCandidates_length_le`.

Theorem 9.14 (finite Dirichlet-window certificate)**.** *For the supplied source $z = N/D$, put $B = \mathrm{N}(D)$. Let c_- and c_+ occupy consecutive best levels and define their exact squared Dirichlet product by*

$$\mathcal{D}_z(c_-, c_+) = \mathrm{N}(q_+)\,\mathcal{E}_z(c_-). \tag{103}$$

For every natural quadrance request H satisfying

$$\mathrm{N}(q_-) < H \le \mathrm{N}(q_+)\,, \tag{104}$$

the lower pair has $q_- \neq 0$ and supplies the certificate

$$H\mathcal{E}_z(c_-) \le \mathcal{D}_z(c_-, c_+) \le \frac{B}{2}, \qquad \mathcal{D}_z(c_-, c_+) \le \frac{1189}{102} I(c_-, c_+) \le \frac{1189}{51}. \tag{105}$$

The exact product also satisfies the universal strict lower bounds

$$\frac{1}{4} < \mathcal{D}_z(c_-, c_+), \qquad I(c_-, c_+) = 2 \implies \frac{1}{2} < \mathcal{D}_z(c_-, c_+). \tag{106}$$

Writing

$$x = \frac{E_z(c_+)}{E_z(c_-)}, \qquad y = \frac{q_-}{q_+}, \qquad G = \mathrm{N}(1 - yx)\,,$$

the finite coordinate-octant calculation gives

$$\frac{205}{256} \le G, \qquad I(c_-, c_+) = 1 \quad \Longrightarrow \quad \mathcal{D}_z(c_-, c_+) \le \frac{256}{205}. \tag{107}$$

The index-two centered-coset calculation strengthens the exact product to

$$I(c_-, c_+) = 2 \quad \Longrightarrow \quad \frac{15}{56} \le \mathrm{N}(x)\,, \qquad \frac{15}{56} \le \mathrm{N}(y)\,, \qquad \frac{205}{128} \le G, \qquad \mathcal{D}_z(c_-, c_+) \le \frac{256}{205}. \tag{108}$$

Consequently the index split gives the improved universal bound

$$\mathcal{D}_z(c_-, c_+) \le \frac{256}{205} < \frac{112}{71} < \frac{28}{15} < 2 < \frac{1189}{102} < \frac{1189}{51} < \frac{816}{35}. \tag{109}$$

More exactly, every displayed rational τ satisfying

$$\frac{3}{4} \le \tau < 1, \qquad \tau^2 - 12\tau + 9 \ge 0 \tag{110}$$

gives the index-specific gap bounds

$$\begin{aligned} I(c_-, c_+) = 1 \quad &\Longrightarrow \quad \tau \le G, \\ I(c_-, c_+) = 2 \quad &\Longrightarrow \quad 2\tau \le G, \end{aligned} \tag{111}$$

and hence the universal division-free certificate

$$\tau \mathcal{D}_z(c_-, c_+) \le 1. \tag{112}$$

The concrete value $\tau = 205/256$ recovers (109). The cut itself, rather than a limiting choice of τ, is the stronger theorem. The earlier one-coefficient calculation is retained as the separately checkable root-free algebraic boundary

$$\frac{1}{2} \le G \quad \textit{or} \quad G^2 - 3G + \frac{1}{4} < 0. \tag{113}$$

Consequently every displayed rational t satisfying

$$t < \frac{1}{2}, \qquad t^2 - 3t + \frac{1}{4} \ge 0 \tag{114}$$

is a certified lower bound $t \le G$ and gives the division-free window bound

$$t \mathcal{D}_z(c_-, c_+) \le I(c_-, c_+) \le 2. \tag{115}$$

The displayed threshold $102/1189$ satisfies (114); its integer certificate is

$$4 \cdot 102^2 - 12 \cdot 102 \cdot 1189 + 1189^2 = 1. \tag{116}$$

Thus one exact finite output pair answers every integer squared-denominator request in its whole consecutive window.

Proof. The nonzero denominator and the strict lower inequality in (104) are already part of the best-pair and request data. Multiplying $H \le \mathrm{N}(q_+)$ by the nonnegative $\mathcal{E}_z(c_-)$ gives the first inequality in (105). For the second, the global rectangle gives $\mathrm{N}(q_+) \le B$, while the equal-height nearest-integer comparison in (75) gives $\mathcal{E}_z(c_-) \le 1/2$. Multiplying the two nonnegative inequalities proves the claim.

For the universal upper bound, use the displayed x, y, G from the statement. The lower error and upper denominator are nonzero: otherwise the strict error or denominator ordering of consecutive best levels would fail. Thus these are displayed Gaussian-rational calculations, and denominator ordering also gives $\mathrm{N}(y) < 1$. The empty-mixed-cylinder criterion, applied to the one coefficient candidate $c_+ - ac_-$, says for every $a \in \mathbb{Z}[i]$ that

$$1 \le \mathrm{N}(x - a) \quad \text{or} \quad 1 \le \mathrm{N}(1 - ay)\,. \tag{117}$$

More generally, the same empty-cylinder calculation applied to $gc_- + hc_+$ gives, for every displayed $g, h \in \mathbb{Z}[i]$ not both zero,

$$1 \le \mathrm{N}(g + hx) \quad \text{or} \quad 1 \le \mathrm{N}(gy + h)\,. \tag{118}$$

The one-coefficient alternative above is the special case $(g, h) = (-a, 1)$.

Choose a by coordinatewise nearest-Gaussian-integer rounding and put $r = x - a$, so $\mathrm{N}(r) \le 1/2$. The elementary weighted quadrance estimate

$$60\,\mathrm{N}(u + v) \le 204\,\mathrm{N}(u) + 85\,\mathrm{N}(v) \tag{119}$$

is just the nonnegativity of $\mathrm{N}(12u - 5v)$. The weights were chosen so that

$$204\frac{35}{408} + 85\frac{1}{2} = 60.$$

Suppose instead that $G < 35/408$. If $a = 0$, then $x = r$ and $\mathrm{N}(yx) < 1/2$; applying (119) to $1 = (1 - yx) + yx$ would give $\mathrm{N}(1) < 1$. If $a \neq 0$, the first alternative in (117) is impossible because $\mathrm{N}(r) \le 1/2$. Hence the second holds. But $\mathrm{N}(yr) < 1/2$ and

$$1 - ay = (1 - yx) + yr,$$

so (119) again gives $\mathrm{N}(1 - ay) < 1$, a contradiction. This proves $G \ge 35/408$ using one explicitly computed coefficient.

The exact Cauchy calculation gives the stronger structural statement (113). If $G \ge 1/2$ there is nothing to prove. Otherwise put $R = \mathrm{N}(yr) < 1/2$. In the $a = 0$ branch $1 = (1-yx)+yr$; in the other branch (117) gives $1 \le \mathrm{N}((1 - yx) + yr)$. If d is the displayed coordinate dot product of these two summands, expansion and the two-coordinate Cauchy identity give

$$1 \le G + R + 2d, \qquad d^2 \le GR.$$

Since $G + R < 1$, squaring the positive margin gives $(1 - G - R)^2 \le 4GR$. With $\delta = 1/2 - R > 0$, direct expansion yields

$$(1 - G - R)^2 - 4GR = \left(\frac{1}{2} - G\right)^2 - 2G + \delta(1 + 2G + \delta).$$

The left side is nonpositive and the last term is positive, so $(1/2 - G)^2 - 2G < 0$, which is exactly the second alternative in (113). This records the boundary by rational coefficient arithmetic rather than by naming either root of its quadratic. To obtain the last assertion

of the theorem, suppose a displayed t satisfies (114) but $G < t$. The first branch of (113) is then impossible. In the second branch, $G + t < 1 < 3$ and direct factorization gives

$$G^2 - 3G + \frac{1}{4} = t^2 - 3t + \frac{1}{4} + (t - G)(3 - G - t) > 0,$$

contradicting that branch. Thus sharper rational thresholds can be certified by two finite comparisons, without changing the proof route.

Direct coefficient arithmetic factors the determinant identity as

$$\mathcal{D}_z(c_-, c_+)G = I(c_-, c_+). \tag{120}$$

Multiplying $t \le G$ by the nonnegative exact window gives (115). Combining this exact identity with $G \ge 35/408$ gives $\mathcal{D}_z(c_-, c_+) \le (408/35)I(c_-, c_+)$, the direct output of the weighted estimate. For the sharper displayed value $t = 102/1189$, multiplying $t^2 - 3t + 1/4$ by the positive rational $4 \cdot 1189^2$ gives the integer certificate in (116), hence $t^2 - 3t + 1/4 > 0$. The quadratic boundary therefore gives $t \le G$, and exact reciprocal cancellation gives

$$\mathcal{D}_z(c_-, c_+) \le \frac{1189}{102} I(c_-, c_+) \le \frac{1189}{51}.$$

The additional coefficient tests improve that one-coefficient estimate. Multiplication by a Gaussian unit, followed if necessary by coordinate conjugation, places x in one of the eight closed coordinate octants. Multiply y by the inverse unit and conjugate it whenever x is conjugated. This preserves yx up to conjugation, hence preserves G. Multiplying g by the matching unit, or conjugating both g and h, shows directly that (118) is preserved as well. We may therefore write

$$\begin{aligned} x &= a + ib, \qquad y = c + id, \qquad 0 \le b \le a, \\ X &= \mathrm{N}(x), \qquad Y = \mathrm{N}(y), \qquad P = ac - bd. \end{aligned}$$

Here $X, Y < 1$ and

$$G = 1 + XY - 2P. \tag{121}$$

The three instances of (117) with coefficients $1, i, 1 + i$ expand, coefficient by coefficient, to

$$2a \le X \quad \text{or} \quad 2c \le Y, \tag{122}$$

$$2b \le X \quad \text{or} \quad -2d \le Y, \tag{123}$$

$$2(a + b) \le X + 1 \quad \text{or} \quad c - d \le Y. \tag{124}$$

If the first alternative of (122) holds, then $2a \le X = a^2 + b^2 \le 2a^2$. Since $0 \le a < 1$, this forces $a = 0$, and then $b = 0$; thus $x = 0$ and $G = 1$. In every other case we have $2c \le Y$. Three elementary square expansions will also be used:

$$2a \le 1 + X, \qquad 2(a + b) \le 1 + 2X, \qquad -2d \le 1 + Y. \tag{125}$$

It remains only to inspect the two other displayed alternatives. The elementary square identity

$$0 \le Y\big((u - \tfrac{1}{2})^2 + v^2\big) + (1 - Y)\frac{1}{4}$$

shows, whenever $u^2 + v^2 = U$ and $0 \le Y \le 1$, that $uY \le UY + 1/4$. If $-2d \le Y$, the two denominator-axis obstructions and $Y < 1$ force $c \le 0$ or $d \ge 0$. In either sign case the

preceding component bound gives $2P \le XY + 1/4$. The same conclusion follows when $2b \le X$ and $c - d \le Y$: the diagonal denominator obstruction forces the corresponding sign alternative, and one applies the component bound to either a or $-d$. Hence both of these chambers satisfy

$$G = 1 + XY - 2P \ge \frac{3}{4}.$$

The sole remaining chamber has both non-real exclusions on the error side:

$$2b \le X, \qquad 2(a+b) \le X + 1, \qquad 2c \le Y.$$

The first two inequalities imply

$$2b - b^2 \le a^2, qquad 2b - b^2 \le (1-a)^2.$$

As $0 \le a \le 1$, one of a and $1 - a$ is at most $1/2$; consequently

$$b^2 - 2b + \frac{1}{4} \ge 0, qquad 0 \le b \le \frac{1}{2}.$$

At the displayed rational $r = 13/97$, exact arithmetic gives $r^2 - 2r + 1/4 < 0$. Factoring the difference of these two quadratic values therefore forces $b \le r$.

The extra strength comes from retaining both coefficients in (118). Taking $(g, h) = (1, -(1-i))$ gives

$$1 \le \mathrm{N}(1 - (1-i)x) \quad \text{or} \quad 1 \le \mathrm{N}(y - (1-i)). \tag{126}$$

If $x \ne 0$, then $a > 0$. Since $X < 1$ and $0 \le b \le a$, we have $a^2 < a$ and $b^2 \le b$, hence $X < a + b$. But the error alternative in (126) expands to $a + b \le X$ and is therefore impossible. If $x = 0$, then $G = 1$ already. Thus in the remaining case the denominator alternative supplies the second projection

$$2(c - d) \le Y + 1. \tag{127}$$

Together with $2c \le Y$, it gives

$$2P = 2(a-b)c + 2b(c-d) \le (a-b)Y + b(Y+1) = aY + b,$$

and consequently

$$G \ge 1 + XY - aY - b. \tag{128}$$

As a preliminary, only small rational comparisons remain. Exact cross-multiplication gives

$$\frac{71}{112} \le \frac{3}{4} + r^2 - r, \qquad \left(\frac{3}{4} + r^2 - r\right) - \frac{71}{112} = \frac{13}{1053808} > 0.$$

Since $b \le r \le 1/2$, the nonnegative product $(r-b)(1-r-b)$ gives $r^2 - r \le b^2 - b$. Therefore

$$\frac{71}{112} \le (a - \tfrac{1}{2})^2 + \frac{3}{4} + b^2 - b = 1 + X - a - b. \tag{129}$$

If $X \le a$, subtracting the last expression from the right side of (128) leaves $(1-Y)(a-X) \ge 0$. If $a \le X$, that right side is instead at least $1 - b \ge 1 - r \ge 71/112$. Thus this first four-coefficient calculation gives $G \ge 71/112$.

Retaining the other norm-two leading combination sharpens the same finite octant calculation. In addition to the inequalities above, the coefficient $1+i$ supplies

$$a-b\le X \quad \text{or} \quad 2(c+d)\le Y+1.$$

The five alternatives divide the displayed (c,d) data into four regions:

green:	$Y < c-d$,
red:	$c-d \le Y < -2d$,
exterior:	$-2d \le Y$ and $2(c+d)\le Y+1$,
blue:	$Y+1 < 2(c+d)$.

This is only a decidable split of rational inequalities. In the red and blue regions, the displayed signs and projections give $2P \le XY$, hence $G \ge 1$. In the exterior region they give either $8P^2 \le XY$ or $7P^2 \le XY$; both imply the common dyadic certificate

$$256P^2 \le 51XY,$$

which yields $G \ge 205/256$ by completing a square.

The green region is the extremal one. It forces $d<0$. If $c\le 0$, a direct expansion writes the required remainder as a sum of the nonnegative terms $-2ac$, Xc^2, $(X-2b)d^2$, and $\frac{b}{2}(2d+1)^2$. If $0\le c$, put $e=-d$. The error and denominator rows then have the symmetric form

$$0\le b\le a, \quad 2b\le X, \quad 2(a+b)\le X+1,$$
$$0\le c\le e, \quad 2c\le Y, \quad 2(e+c)\le Y+1.$$

Each row implies its slope quadratic, and an exact homogeneous identity combining the two quadratics with $(be-ac)^2$ gives

$$4(ac+be)^2 \le XY.$$

The remaining radial estimate is the following one-vector certificate:

$$b+X(a-X)\le \frac{51}{256}. \tag{130}$$

First the two error projections give $b\le 35/256$. When $256b\le 13$, the identity $4-27u(1-u)^2=(4-3u)(3u-1)^2$ for $u=1-a$ gives

$$b+X(a-X)\le \frac{13}{256}+\frac{4}{27}\le \frac{51}{256}.$$

For $13\le t=256b\le 35$, put $L=(5t+82)/512$. Splitting once at $t=21$, direct polynomial products prove

$$L^2 \le 2b-b^2, qquadb+\frac{1}{4}(1-L)^2\le \frac{51}{256};$$

the diagonal projection gives $L\le 1-a$, and hence (130). Applying this certificate to the row with smaller norm and using the two linear projections of $ac+be$ gives

$$G=1+XY-2(ac+be)\ge 1-\frac{51}{256}=\frac{205}{256}.$$

Thus all four regions, and therefore all eight finite symmetry choices, satisfy the first inequality in (107); no limiting value or completed complex input enters the calculation. When $I(c_-,c_+)=1$, the exact identity (120) immediately gives $\mathcal{D}_z(c_-,c_+)\le 256/205$.

The same chamber calculation admits an exact rational parameter. Let τ satisfy (110) and put $r = 1 - \tau$. Then $r \geq 0$ and direct expansion gives

$$r^2 + 10r - 2 = \tau^2 - 12\tau + 9 \geq 0. \tag{131}$$

The one-vector projection lemma used in the green region has the exact form

$$0 \leq a, b, \quad X = a^2 + b^2 \leq 1, \quad 2b \leq X, \quad 2(a+b) \leq X + 1 \quad \Longrightarrow \quad b + X(a - X) \leq r$$

for every displayed $r \geq 0$ satisfying $r^2 + 10r - 2 \geq 0$. Its proof first derives $b^2 - 2b + 1/4 \geq 0$ and $b \leq 1/7$. A split at the displayed rational $3/64$ then uses, respectively, the cubic identity already displayed above and a rational linear support to the same coordinate quadratic; after clearing denominators, each remainder is an explicit product of nonnegative rational factors. Thus the lemma identifies the whole polynomial cut, rather than only the point $51/256$.

Apply it to the error row and to the transformed denominator row. The same two projection inequalities used above now give $G \geq 1 - r = \tau$ in the green region. The red and blue regions already give $G \geq 1$, the exterior region gives $G \geq 6/7$, and the nonpositive-real branch of the green region gives $G \geq 13/14$; direct polynomial comparisons put every τ satisfying (110) below those three rational floors. Hence the first line of (111) holds in every one of the eight displayed octants.

It remains to treat coefficient index two. Suppose $I(c_-, c_+) = 2$. For each Gaussian unit ζ, the centered-coset certificate constructed in lemma 9.10, normalized by the positive lower-error and upper-denominator quadrances, gives

$$2 \leq \mathrm{N}(x - \zeta) \quad \text{or} \quad 2 \leq \mathrm{N}(1 - \zeta y)\,. \tag{132}$$

Put $X = \mathrm{N}(x)$, $Y = \mathrm{N}(y)$, $\alpha = 1 - X > 0$, and $\beta = 1 - Y > 0$. The alternatives for opposite units must cross: the two error alternatives for $\zeta, -\zeta$ would have sum $2(1 + X) < 4$, and the two denominator alternatives would have sum $2(1 + Y) < 4$. Expanding the crossed alternatives for $\zeta = 1, -1$ therefore gives

$$4\,\mathrm{Re}(x)\,\mathrm{Re}(y) \leq -\alpha\beta,$$

while the alternatives for $\zeta = i, -i$ give

$$\alpha\beta \leq 4\,\mathrm{Im}(x)\,\mathrm{Im}(y).$$

Adding these inequalities and cancelling the positive rational factor two shows

$$2\,\mathrm{Re}(yx) = 2\big(\mathrm{Re}(x)\,\mathrm{Re}(y) - \mathrm{Im}(x)\,\mathrm{Im}(y)\big) \leq -\alpha\beta.$$

Hence, entirely coefficientwise,

$$G - 1 = XY - 2\,\mathrm{Re}(yx) \geq XY + \alpha\beta. \tag{133}$$

The individual crossed alternatives give a second useful consequence. In the real direction they are exactly one of

$$\alpha \leq -2\,\mathrm{Re}(x), \quad \beta \leq 2\,\mathrm{Re}(y), \qquad \text{or} \qquad \beta \leq -2\,\mathrm{Re}(y), \quad \alpha \leq 2\,\mathrm{Re}(x).$$

In the imaginary direction they are exactly one of

$$\alpha \leq -2\,\mathrm{Im}(x), \quad \beta \leq -2\,\mathrm{Im}(y), \qquad \text{or} \qquad \beta \leq 2\,\mathrm{Im}(y), \quad \alpha \leq 2\,\mathrm{Im}(x).$$

The right sides selected above are nonnegative because $\alpha, \beta \geq 0$. Squaring the two bounds which concern α, adding them, and cancelling two gives

$$(1 - X)^2 = \alpha^2 \leq 2X.$$

The two complementary bounds give the identical result $(1 - Y)^2 \leq 2Y$. Thus each of X, Y satisfies

$$U^2 - 4U + 1 \leq 0. \tag{134}$$

No root of this quadratic is needed. Put $s = 15/56$. Exact arithmetic gives

$$s^2 - 4s + 1 = \frac{1}{3136} > 0.$$

If a displayed U satisfying (134) had $U < s$, then

$$U^2 - 4U + 1 = s^2 - 4s + 1 + (s - U)(4 - U - s) > 0,$$

a contradiction. Hence $s \leq X, Y$.

It remains to bound the balanced term in (133). Write

$$F = XY + (1 - X)(1 - Y).$$

If $Y \leq 1/2$, then $F - Y = (1 - X)(1 - 2Y) \geq 0$, so $F \geq Y \geq s$. If $1/2 \leq Y$, then

$$F - s = (X - s)(2Y - 1) + (1 - Y)(1 - 2s) \geq 0.$$

Thus $F \geq 15/56$, giving the preliminary bound $G \geq 71/56$.

The orientation already chosen for the index-one calculation extracts more from the same four centered alternatives. Write

$$x = a + ib, \qquad y = -u + iv, \qquad \ell(U) = \frac{2(U + 1)}{7}, \qquad h(U) = \frac{1 - U}{2}.$$

The first-octant inequalities and the forced orientation of the unit alternatives give

$$0 \leq b \leq a, \qquad h(X) \leq b, \qquad 0 \leq h(Y) \leq u, v.$$

For every displayed $15/56 \leq U \leq 1$, the exact endpoint factorization proves

$$\ell(U)^2 \leq \frac{U}{2}. \tag{135}$$

Thus $\ell(X) \leq a$ and $\ell(Y) \leq \max(u, v)$, with no square root selected.

There are now only two rational orientation cases. If $u \leq v$, the coordinate products dominate the cross floor

$$XY + \ell(X)(1 - Y) + (1 - X)\ell(Y).$$

If $v \leq u$, they dominate the aligned floor

$$XY + 2\ell(X)\ell(Y) + \frac{(1 - X)(1 - Y)}{2}.$$

Both displayed expressions are increasing from the rational corner $X = Y = 15/56$ in the required directions; clearing denominators at that corner gives the same exact bound $77/128$. Consequently

$$G \geq 1 + \frac{77}{128} = \frac{205}{128}.$$

The exact identity (120) now gives

$$\mathcal{D}_z(c_-, c_+) \leq \frac{2}{205/128} = \frac{256}{205}$$

in index two. The index-one and index-two cases therefore have the same displayed bound and prove (109). Every comparison in this calculation is rational and exact.

To obtain the second line of (111), retain the parameter τ and put

$$s = \frac{\tau}{3}, \qquad m = \frac{s}{1 - s^2}, \qquad \ell_s(U) = m(U + 1).$$

The threshold test gives $1/4 \leq s < 1/3$ and

$$9(s^2 - 4s + 1) = \tau^2 - 12\tau + 9 \geq 0. \tag{136}$$

If a displayed $U < 1$ satisfying $U^2 - 4U + 1 \leq 0$ had $U < s$, then

$$U^2 - 4U + 1 = s^2 - 4s + 1 + (s - U)(4 - U - s) > 0,$$

a contradiction. Hence $s \leq X, Y$.

The denominator $1 - s^2$ is positive. Moreover

$$(1 - s^2)^2 - 12s^2 = (s^2 + 4s + 1)(s^2 - 4s + 1) \geq 0,$$

so $12m^2 \leq 1$. At the lower endpoint the defining cancellation is

$$\ell_s(s)(1 - s) = s. \tag{137}$$

Together with $(1 - s)^2 \geq 2s$, this gives $\ell_s(s)^2 \leq s/2$. For $s \leq U \leq 1$, the exact difference factorization

$$\frac{U}{2} - \ell_s(U)^2 = \left(\frac{s}{2} - \ell_s(s)^2\right) + (U - s)\left(\frac{1}{2} - m^2(U + s + 2)\right) \tag{138}$$

is nonnegative: $U + s + 2 \leq 4$ and $4m^2 \leq 1/2$. Thus $\ell_s(X) \leq a$ and $\ell_s(Y) \leq \max(u, v)$ by the same square comparison as before.

In the cross case $u \leq v$, the resulting floor simplifies to

$$\begin{aligned} &XY + \ell_s(X)(1 - Y) + (1 - X)\ell_s(Y) \\ &\qquad = 2m + (1 - 2m)XY \geq 2m + (1 - 2m)s^2 = s^2 + 2s \geq 6s - 1 = 2\tau - 1. \end{aligned}$$

Here $m \leq 1/2$, and the penultimate inequality is precisely $s^2 - 4s + 1 \geq 0$. In the aligned case $v \leq u$, the floor is increasing from (s, s): its coordinate slope is

$$U + 2m^2(U + 1) - \frac{1 - U}{2} \geq 0$$

for $U \geq s$, using $s \geq 1/4$ and $m \geq s$. At the lower corner the aligned floor exceeds the cross floor by the explicit square

$$2\left(\ell_s(s) - \frac{1-s}{2}\right)^2.$$

It therefore has the same lower bound $2\tau - 1$. Adding the leading one in the determinant-gap expansion proves $G \geq 2\tau$. With $I(c_-, c_+) = 2$, cancellation in (120) gives $\tau \mathcal{D}_z(c_-, c_+) \leq 1$. Together with the index-one calculation, this proves (112).

For the lower bound, put $A = \mathrm{N}(q_+)\,\mathcal{E}_z(c_-)$ and $A' = \mathrm{N}(q_-)\,\mathcal{E}_z(c_+)$. Strict denominator and error ordering give $A' < A$. The coefficient determinant identity and the elementary quadrance difference estimate therefore give

$$\begin{aligned} I(c_-, c_+) &= \mathrm{N}(q_- E_z(c_+) - q_+ E_z(c_-)) \\ &\leq 2(A' + A) < 4A. \end{aligned}$$

The index is one or two, which yields (106). The Lean statements retain the division-free forms $1 < 4A$ and, in the index-two case, $2 < 4A$.

The use of a natural quadrance request is intentional: if a conventional modulus threshold is denoted by Q, then H is its squared counterpart, but the theorem never has to introduce $Q = \sqrt{H}$. The exact product (103) is normally much sharper than $B/2$ and is computed from the returned pairs. The source-independent rational $256/205$ is a convenient fixed instance, while (112) records the whole finite quadratic cut. Conventionally naming its smaller endpoint as $6 - 3\sqrt{3}$ makes the reciprocal endpoint $(2 + \sqrt{3})/3$, Chevallier's optimal squared constant; the theorem itself needs neither name. □

Theorem 9.15 (finite sharpness beyond every scale)**.** *Let $\tau \in \mathbb{Q}$ satisfy*

$$\frac{3}{4} \leq \tau < 1, \qquad \tau^2 - 12\tau + 9 < 0, \tag{139}$$

and let $T \in \mathbb{N}$ be a supplied finite scale. Put

$$\delta = -(\tau^2 - 12\tau + 9), \qquad k = \left\lceil \frac{24}{\delta} \right\rceil + T + 1.$$

Starting from $(m_0, d_0) = (1, 3)$, execute the subtraction-free recurrence

$$(m_{j+1}, d_{j+1}) = (m_j + d_j,\ 2m_j + 3d_j), \qquad n_j = m_j + d_j, \tag{140}$$

through stage k. Define

$$s = \frac{m_k}{n_k}, \qquad x = \frac{1}{2} + i\frac{s}{2}, \qquad y = \frac{s}{2} - i\frac{1}{2},$$

and the two displayed Gaussian-integral pairs

$$c_- = (-2m_k,\ m_k - n_k i), \qquad c_+ = (-m_k - n_k i,\ 2n_k). \tag{141}$$

Finally put

$$e_- = -\frac{1}{q_+(1 - yx)}, \qquad z = \frac{p_- + e_-}{q_-}, \tag{142}$$

where $c_\pm = (p_\pm, q_\pm)$. These are executable displayed rational data. They satisfy

$$\begin{aligned} T < n_k < \mathrm{N}(q_+) = 4n_k^2, \\ c_-, c_+ \text{ occupy consecutive best levels for } z, \\ 1 < \tau \mathcal{D}_z(c_-, c_+). \end{aligned} \tag{143}$$

Thus no displayed rational threshold satisfying (139) can replace the quadratic cut as a universal finite-window bound.

Proof. The recurrence preserves the natural-number identity

$$d_j^2 = 2m_j^2 + 2m_j d_j + 1, \qquad \text{equivalently} \qquad m_j^2 + n_j^2 = 4m_j n_j + 1. \tag{144}$$

It also preserves $2m_j < d_j \le 3m_j$. Hence

$$\frac{1}{4} \le s < \frac{1}{3}, \qquad n_k^2(s^2 - 4s + 1) = 1,$$

and $n_k > k > T$. The second identity in (144) gives $p_- q_+ - p_+ q_- = 1$. Direct substitution into (142) gives

$$E_z(c_-) = e_-, \qquad E_z(c_+) = e_- x, \qquad \frac{q_-}{q_+} = y.$$

The finite critical-cylinder calculation checks every nonzero pair of Gaussian-integral coefficients after an explicit coefficient bound; at $1/4 \le s < 1/3$ with $s^2 - 4s + 1 > 0$, one of its normalized error or denominator coordinates is exterior. The displayed Cramer identity then proves that both pairs are best and that their mixed open cylinder is empty. They therefore occupy consecutive best levels.

Put $G_s = \mathrm{N}(1 - yx)$. The definitions of e_- and $\mathcal{D}$ cancel exactly to

$$G_s \mathcal{D}_z(c_-, c_+) = 1. \tag{145}$$

Writing $P(s) = s^2 - 4s + 1 = 1/n_k^2$, the exact factorization

$$16G_s = 48s + P(s)(s^2 + 4s + 17)$$

gives the convenient strict estimate $G_s < 3s + 2P(s)$. On the sharper side of the threshold boundary, exact difference factorization gives

$$-(\tau^2 - 12\tau + 9) < 12(\tau - 3s).$$

The definition of k, the growth $n_k > k$, and $P(s)n_k^2 = 1$ give

$$24P(s) < -(\tau^2 - 12\tau + 9).$$

Consequently $2P(s) < \tau - 3s$, and hence $G_s < \tau$. The window product is positive, so multiplying this strict inequality by it and applying (145) proves $1 < \tau \mathcal{D}_z(c_-, c_+)$. □

Formal verification:

`Purelib.NumberTheory.Approximation.WindowSharpness.pellState;Purelib.NumberTheory.Approximation.WindowSharpness.pellDifference_sq;Purelib.NumberTheory.Approximation.WindowSharpness.pellDifference_bounds;Purelib.NumberTheory.Approximation.WindowSharpness.pellEquation;Purelib.`

`NumberTheory.Approximation.WindowSharpness.pellParameter_isCritical;`
`Purelib.NumberTheory.Approximation.WindowSharpness.criticalPairs_`
`areConsecutiveBest;Purelib.NumberTheory.Approximation.WindowSharpness.`
`criticalGap_mul_dirichletWindow;Purelib.NumberTheory.Approximation.`
`WindowSharpness.criticalGap_lt_sharperThreshold;Purelib.NumberTheory.`
`Approximation.WindowSharpness.requestedScale_lt_`
`criticalUpperDenominatorQuadrance;Purelib.NumberTheory.Approximation.`
`WindowSharpness.criticalWindowSharpness;Purelib.Papers.RFAP.Approximation.`
`sharpnessStage;Purelib.Papers.RFAP.Approximation.sharpnessSource;Purelib.`
`Papers.RFAP.Approximation.sharpnessLowerPair;Purelib.Papers.RFAP.`
`Approximation.sharpnessUpperPair;Purelib.Papers.RFAP.Approximation.`
`sharpnessDenominatorScale;Purelib.Papers.RFAP.Approximation.`
`sharperGapThreshold_has_finiteCounterexample`.

The preceding source depends on the requested scale. A stronger finite quantifier order is available when the whole desired run length is supplied at once.

Theorem 9.16 (finite same-source robust sharpness run)**.** *Let $\tau \in \mathbb{Q}$ satisfy (139), and supply $T, N \in \mathbb{N}$. A terminating calculation returns one displayed Gaussian-rational source z, one proper rational box B whose lower-left anchor is z, and a list*

$$(c_0^-, c_0^+), \dots, (c_{N-1}^-, c_{N-1}^+)$$

of exactly N pairs. Their upper denominator quadrances obey

$$T < \mathrm{N}\left(q_0^+\right) < \mathrm{N}\left(q_1^+\right) < \cdots < \mathrm{N}\left(q_{N-1}^+\right)$$

when $N > 0$. More importantly, for every displayed $w \in B$ and every $j < N$, the pair (c_j^-, c_j^+) occupies consecutive best levels for w and satisfies

$$1 < \tau \mathcal{D}_w(c_j^-, c_j^+).$$

Thus all data in one supplied finite run share one source description and one uniform rational neighbourhood. The source and box may still depend on T and N.

Proof. Put b for the Pell stage selected by τ and T, and choose the common source at stage $b + N$. For $j < N$, transport the critical determinant-one basis from the backward parameter $b+N-j$ to the forward parameter $b+N+j$. The subtraction-free Pell identities keep every parameter in the critical chamber. The finite mixed-cylinder calculation and its Cramer certificate therefore prove that the transported pair is consecutive best for the one chosen source. The forward stages increase strictly with j, so the corresponding upper denominator quadrances are strictly increasing; their first value already lies beyond T. The exact gap identity and the same threshold estimate used in Theorem 9.15 give the strict violation for every member of the list.

It remains to make these strict conclusions uniform. For one window, first bound every possible mixed-cylinder competitor by the explicit Gaussian coefficient square with coefficient quadrance below 32. This is a finite 20736-entry list. Each strict residual comparison in that list computes a positive rational perturbation budget; the basis residuals and the window inequality compute two further positive budgets. Taking their finite minimum gives a proper rational box on which that window remains consecutive and violating. Taking the finite minimum of these budgets over the N displayed windows gives B. For $N = 0$ the list is empty and the construction uses its explicit default positive radius. In either case all minima and all membership tests are bounded rational calculations. □

Formal verification:

`Purelib.NumberTheory.Approximation.Dirichlet.Critical.Run.criticalBadWindowRun;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.Run.criticalFullyRobustRun;Purelib.Papers.RFAP.Approximation.robustFiniteSharpnessRun`.

Theorem 9.17 (unreserved nested robust sharpness windows)**.** *Let τ satisfy (139). Retain a finite ordered window history W and a proper rational box B on which every window is consecutive best and violates τ. Given a new scale T' and an additional count n, a terminating calculation returns a proper box B' strictly inside B and exactly n additional windows V. The retained window list is exactly $W +\!\!+ V$, with every original window record unchanged. All its windows are consecutive best and satisfy $1 < \tau\mathcal{D}_q$ for every displayed $q \in B'$.*

The upper denominator quadrances of the full list increase strictly. Every new upper quadrance exceeds T' and every earlier upper quadrance. The result is another history of the same kind, so the calculation accepts each request in any supplied finite request list without a reserve endpoint.

Proof. Take the maximum of T' and the actual upper quadrances in W. Choose an interior rational centre of B. Construct one finite critical seed run of length n, and place it near that centre by a determinant-one integral homography. The retained denominator payments enforce the computed maximum, the complete cylinder inequalities, the strict threshold violations and the strict order within the new run. This placement changes the new run's source while preserving its certified windows through the same matrix.

Each placed window computes a positive displacement clearance for its full cylinder and threshold inequality. Their finite minimum gives a proper strict subbox of B. Transport the earlier uniform certificates through that actual containment and append the new window records. The maximum payment orders every new window after every old one. Structural recurrence on the supplied finite request list repeats this construction, retaining the actual earlier segments, matrices, windows and clearances at every step. □

Formal verification:

`Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.ofRobustRun;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.extend;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.extendRequests;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.Extension.windowsEq;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.Extension.nested;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.Extension.earlierConsecutive;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.Extension.earlierViolation;Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.WindowHistory.scalesStrict`.

Corollary 9.18 (two-window refinement)**.** *Let c_0, c_1, c_2 occupy three successive best levels, and abbreviate*

$$I_0 = I(c_0, c_1), \quad I_1 = I(c_1, c_2), \qquad \mathcal{D}_0 = \mathcal{D}_z(c_0, c_1), \quad \mathcal{D}_1 = \mathcal{D}_z(c_1, c_2).$$

For every displayed rational t satisfying (114),

$$t\mathcal{D}_0 \le 1 \quad \text{or} \quad t\mathcal{D}_1 \le 1, \qquad t(\mathcal{D}_0 + \mathcal{D}_1) \le 3. \tag{146}$$

For every displayed τ satisfying (110), the stronger individual window theorem gives

$$\tau\mathcal{D}_0 \le 1, \qquad \tau\mathcal{D}_1 \le 1, \qquad \tau(\mathcal{D}_0 + \mathcal{D}_1) \le 2. \tag{147}$$

In particular,

$$\begin{aligned} &I_0 + I_1 \le 3, && \mathcal{D}_0 \le \frac{256}{205}, && \mathcal{D}_1 \le \frac{256}{205}, \\ &I_j = 2 \implies \mathcal{D}_j \le \frac{256}{205}, && \mathcal{D}_0 + \mathcal{D}_1 \le \frac{512}{205} < 4 < \frac{1189}{34}. \end{aligned} \tag{148}$$

Thus the finite octant and centered index-two calculations bound both individual windows before they are added.

Proof. The index dichotomy gives $I_0 = 1$ or $I_0 = 2$. In the first case the first index is one. In the second, the no-successive-index-two theorem gives $I_1 = 1$. Hence one index is one and in either case $I_0 + I_1 \le 3$. The threshold form of theorem 9.14 gives $t\mathcal{D}_j \le I_j$ for each window. Selecting the index-one inequality and adding the two inequalities proves (146). The certified choice $t = 102/1189$ gives the weaker sum bound $1189/34$. Applying (112) separately to both transitions and adding proves (147). The index-specific bounds in theorem 9.14 give the universal estimate $\mathcal{D}_j \le 256/205$ in either case. Adding the two universal estimates gives $512/205$; the remaining comparisons in (148) are exact rational arithmetic. □

Corollary 9.19 (finite-chain window bound)**.** *Let $c_0, \dots, c_n$ be a supplied finite list in which every adjacent pair occupies consecutive best levels. For $0 \le j < n$, put*

$$I_j = I(c_j, c_{j+1}), \qquad \mathcal{D}_j = \mathcal{D}_z(c_j, c_{j+1}).$$

Then

$$2\sum_{j<n} I_j \le 3n + (n \bmod 2) \le 3n + 1. \tag{149}$$

For every displayed rational t satisfying (114), the same chain satisfies

$$2t\sum_{j<n} \mathcal{D}_j \le 3n + (n \bmod 2) \le 3n + 1. \tag{150}$$

For every displayed τ satisfying (110), the direct universal family instead gives

$$\tau\sum_{j<n} \mathcal{D}_j \le n. \tag{151}$$

For the explicit threshold used above this becomes

$$2\sum_{j<n} \mathcal{D}_j \le \frac{1189}{102}(3n + (n \bmod 2)). \tag{152}$$

The finite-octant and centered-coset estimates give the stronger direct accumulation

$$\sum_{j<n} \mathcal{D}_j \le \frac{256}{205}n. \tag{153}$$

Proof. Group the transitions into adjacent blocks of two. Each complete block has index sum at most three by corollary 9.18; precisely when n is odd, one transition remains and its index is at most two. This finite induction gives (149). Summing the exact inequalities $t\mathcal{D}_j \le I_j$ and multiplying by two gives (150). Taking $t = 102/1189$ and cancelling its exact reciprocal $1189/102$ gives (152). The Lean proof performs precisely this recursion on the supplied list and also checks that its computed index and window lists have the same length. Applying (112) to each displayed transition and recursing over the supplied list proves (151). For the final fixed bound, each transition costs at most $256/205$ by (109); summing those n displayed inequalities gives the result directly. □

Formal verification:

`Purelib.NumberTheory.Approximation.BestApproximation.dirichletWindowQuadrance;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.normalizedCoefficientExterior;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.normalizedLinearCombinationExterior;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticGap_neg_of_sumExterior;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.normalizedDeterminantGap_quadratic_neg_of_lt_half;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.normalizedDeterminantGap_boundary;Purelib.NumberTheory.Approximation.DirichletWindow.threshold_le_of_quadraticBoundary;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.threshold_le_normalizedDeterminantGap_of_quadratic;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.quadraticThreshold_mul_dirichletWindow_le_index;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.quadraticThreshold_mul_dirichletWindow_le_two;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.one_twelfth_le_normalizedDeterminantGap;Purelib.NumberTheory.Approximation.DirichletWindow.refinedGapThreshold_lt_half;Purelib.NumberTheory.Approximation.DirichletWindow.refinedGapThreshold_quadratic_nonnegative;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticGapThreshold_lt_half;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticGapThreshold_quadratic_nonnegative;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticGapThreshold_integerCertificate;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticGapThreshold_mul_quadraticIndexFactor;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticUniversalWindowBound_mul_denominator;Purelib.NumberTheory.Approximation.DirichletWindow.quadraticUniversalWindowBound_lt_refinedUniversalWindowBound;Purelib.NumberTheory.Approximation.DirichletWindow.refinedGapThreshold_mul_denominator;Purelib.NumberTheory.Approximation.DirichletWindow.refinedGapThreshold_budget;Purelib.NumberTheory.Approximation.DirichletWindow.refinedIndexFactor_mul_denominator;Purelib.NumberTheory.Approximation.DirichletWindow.refinedUniversalWindowBound_mul_denominator;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.refinedGap_le_normalizedDeterminantGap;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.dirichletWindow_mul_normalizedGap_eq_index;Purelib.NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_factor_mul_index_of_`

quadraticThreshold;Purelib.NumberTheory.Approximation.DirichletWindow.
AreConsecutiveBest.dirichletWindowQuadrance_le_factor_mul_two_of_
quadraticThreshold;Purelib.NumberTheory.Approximation.DirichletWindow.
AreConsecutiveBest.dirichletWindowQuadrance_le_refinedIndexFactor;Purelib.
NumberTheory.Approximation.DirichletWindow.AreConsecutiveBest.
dirichletWindowQuadrance_le_refinedUniversalBound;Purelib.NumberTheory.
Approximation.DirichletWindow.AreConsecutiveBest.dirichletWindowQuadrance_
le_quadraticIndexFactor;Purelib.NumberTheory.Approximation.
DirichletWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_
quadraticUniversalBound;Purelib.NumberTheory.Approximation.
DirichletWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_twelve_mul_
index;Purelib.NumberTheory.Approximation.DirichletWindow.
AreConsecutiveBest.dirichletWindowQuadrance_le_twentyFour;Purelib.
NumberTheory.Approximation.IndexOneWindow.half_le_determinantGap_of_
firstOctantObstructions;Purelib.NumberTheory.Approximation.IndexOneWindow.
fifteen_twenty_eighths_le_determinantGap_of_firstOctantObstructions;
Purelib.NumberTheory.Approximation.IndexOneWindow.fifteen_twenty_eighths_
le_determinantGap_of_coefficientObstructions;Purelib.NumberTheory.
Approximation.IndexOneWindow.half_le_determinantGap_of_
coefficientObstructions;Purelib.NumberTheory.Approximation.IndexOneWindow.
AreConsecutiveBest.fifteen_twenty_eighths_le_normalizedDeterminantGap;
Purelib.NumberTheory.Approximation.IndexOneWindow.
normTwoIndexOneGapThreshold_mul_denominator;Purelib.NumberTheory.
Approximation.IndexOneWindow.normTwoIndexOneWindowBound_mul_denominator;
Purelib.NumberTheory.Approximation.IndexOneWindow.
normTwoIndexOneWindowBound_mul_gapThreshold;Purelib.NumberTheory.
Approximation.IndexOneWindow.AreConsecutiveBest.
normTwoIndexOneGapThreshold_le_normalizedDeterminantGap;Purelib.
NumberTheory.Approximation.SharpIndexOneWindow.fourRegionSplit_of_
fiveCoefficientObstructions;Purelib.NumberTheory.Approximation.
SharpIndexOneWindow.second_add_norm_mul_first_sub_norm_le_projectionBound;
Purelib.NumberTheory.Approximation.SharpIndexOneWindow.four_crossSquare_
le_normProduct_of_slopeQuadratics;Purelib.NumberTheory.Approximation.
SharpIndexOneWindow.firstRegionGapBound_le_gap_of_greenRegion;Purelib.
NumberTheory.Approximation.SharpIndexOneWindow.firstRegionGapBound_le_gap_
of_redRegion;Purelib.NumberTheory.Approximation.SharpIndexOneWindow.
firstRegionGapBound_le_gap_of_exteriorRegion;Purelib.NumberTheory.
Approximation.SharpIndexOneWindow.firstRegionGapBound_le_gap_of_
blueRegion;Purelib.NumberTheory.Approximation.SharpIndexOneWindow.
firstRegionGapBound_le_gap_of_firstOctantFiveObstructions;Purelib.
NumberTheory.Approximation.OptimalIndexOneWindow.projection_le_of_
firstRegionObstructions;Purelib.NumberTheory.Approximation.
OptimalIndexOneWindow.gapThreshold_le_gap_of_firstOctantFiveObstructions;
Purelib.NumberTheory.Approximation.IndexOneWindow.sharpIndexOneGapBound_
mul_denominator;Purelib.NumberTheory.Approximation.IndexOneWindow.
sharpIndexOneWindowBound_mul_denominator;Purelib.NumberTheory.
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normalizedDeterminantGap;Purelib.NumberTheory.Approximation.
IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_

sharpIndexOneWindowBound_of_indexOne;Purelib.NumberTheory.Approximation.
OptimalIndexTwoWindow.componentLower_sq_le_half;Purelib.NumberTheory.
Approximation.OptimalIndexTwoWindow.twice_gapThreshold_le_determinantGap_
of_firstOctantObstructions;Purelib.NumberTheory.Approximation.
IndexOneWindow.twice_gapThreshold_le_of_centeredUnitObstructions;Purelib.
NumberTheory.Approximation.IndexOneWindow.AreConsecutiveBest.twice_
gapThreshold_le_normalizedDeterminantGap_of_indexTwo;Purelib.NumberTheory.
Approximation.IndexOneWindow.AreConsecutiveBest.gapThreshold_mul_
dirichletWindow_le_one;Purelib.NumberTheory.Approximation.IndexOneWindow.
AreConsecutiveBest.half_le_normalizedDeterminantGap;Purelib.NumberTheory.
Approximation.IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_
le_twentyEight_fifteenths_of_indexOne;Purelib.NumberTheory.Approximation.
IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_
normTwoIndexOneWindowBound_of_indexOne;Purelib.NumberTheory.Approximation.
IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_two_of_
indexOne;Purelib.NumberTheory.Approximation.IndexOneWindow.
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Purelib.NumberTheory.Approximation.IndexOneWindow.AreConsecutiveBest.
dirichletWindowQuadrance_le_normTwoUniversalWindowBound;Purelib.
NumberTheory.Approximation.IndexOneWindow.AreConsecutiveBest.
dirichletWindowQuadrance_le_two;Purelib.NumberTheory.Approximation.
IndexTwoWindow.one_add_balancedNormProduct_le_determinantGap_of_
centeredUnitObstructions;Purelib.NumberTheory.Approximation.
IndexTwoWindow.complement_sq_le_two_norms_of_centeredUnitObstructions;
Purelib.NumberTheory.Approximation.IndexTwoWindow.fifteen_fiftySixths_le_
norms_of_centeredUnitObstructions;Purelib.NumberTheory.Approximation.
IndexTwoWindow.seventyOne_fiftySixths_le_determinantGap_of_
centeredUnitObstructions;Purelib.NumberTheory.Approximation.
IndexTwoWindow.firstOctant_projectionBounds_of_centeredUnitObstructions;
Purelib.NumberTheory.Approximation.IndexTwoWindow.centeredComponentLower_
sq_le_half;Purelib.NumberTheory.Approximation.IndexTwoWindow.
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Purelib.NumberTheory.Approximation.IndexTwoWindow.one_le_determinantGap_
of_centeredUnitObstructions;Purelib.NumberTheory.Approximation.
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NumberTheory.Approximation.IndexTwoWindow.AreConsecutiveBest.
normalizedCenteredUnitObstruction;Purelib.NumberTheory.Approximation.
IndexTwoWindow.AreConsecutiveBest.seventyOne_fiftySixths_le_
normalizedDeterminantGap_of_indexTwo;Purelib.NumberTheory.Approximation.
IndexTwoWindow.AreConsecutiveBest.one_le_normalizedDeterminantGap_of_
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indexTwo;Purelib.NumberTheory.Approximation.IndexTwoWindow.
AreConsecutiveBest.dirichletWindowQuadrance_le_
quadraticIndexFactorUniversal;Purelib.NumberTheory.Approximation.
IndexOneWindow.AreConsecutiveBest.dyadicCenteredGapThreshold_le_
normalizedDeterminantGap_of_indexTwo;Purelib.NumberTheory.Approximation.

IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_
dyadicCenteredWindowBound_of_indexTwo;Purelib.NumberTheory.Approximation.
IndexOneWindow.AreConsecutiveBest.dirichletWindowQuadrance_le_
sharpDyadicUniversalWindowBound;Purelib.NumberTheory.Approximation.
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Approximation.DirichletBlock.refinedIndexFactor_mul_three;Purelib.
NumberTheory.Approximation.DirichletBlock.refinedTwoWindowSumBound_mul_
denominator;Purelib.NumberTheory.Approximation.DirichletBlock.
quadraticIndexFactor_mul_three;Purelib.NumberTheory.Approximation.
DirichletBlock.quadraticTwoWindowSumBound_mul_denominator;Purelib.
NumberTheory.Approximation.DirichletBlock.octantTwoWindowSumBound_mul_
denominator;Purelib.NumberTheory.Approximation.DirichletBlock.
dyadicTwoWindowSumBound_mul_denominator;Purelib.NumberTheory.
Approximation.DirichletBlock.coefficientIndexQuadrance_add_le_three;
Purelib.NumberTheory.Approximation.DirichletBlock.quadraticThreshold_mul_
one_dirichletWindow_le_one;Purelib.NumberTheory.Approximation.
DirichletBlock.quadraticThreshold_mul_dirichletWindowSum_le_three;Purelib.
NumberTheory.Approximation.DirichletBlock.one_dirichletWindowQuadrance_le_
refinedIndexFactor;Purelib.NumberTheory.Approximation.DirichletBlock.one_
dirichletWindowQuadrance_le_quadraticIndexFactor;Purelib.NumberTheory.
Approximation.DirichletBlock.dirichletWindowQuadrance_add_le_
refinedTwoWindowSumBound;Purelib.NumberTheory.Approximation.
DirichletBlock.dirichletWindowQuadrance_add_le_quadraticTwoWindowSumBound;
Purelib.NumberTheory.Approximation.DirichletBlock.
dirichletWindowQuadrance_add_le_quadraticUniversalWindowBound;Purelib.
NumberTheory.Approximation.DirichletBlock.dirichletWindowQuadrance_add_
leOctantTwoWindowSumBound;Purelib.NumberTheory.Approximation.
DirichletBlock.dirichletWindowQuadrance_add_le_dyadicTwoWindowSumBound;
Purelib.NumberTheory.Approximation.DirichletBlock.gapThreshold_mul_
dirichletWindowQuadrance_add_le_two;Purelib.NumberTheory.Approximation.
DirichletBlock.dirichletWindowQuadrance_add_le_four;Purelib.NumberTheory.
Approximation.DirichletChain.dirichletWindowQuadrances_length_eq_
coefficientIndexQuadrances_length;Purelib.NumberTheory.Approximation.
DirichletChain.two_mul_coefficientIndexQuadrances_sum_le_with_parity;
Purelib.NumberTheory.Approximation.DirichletChain.two_mul_
coefficientIndexQuadrances_sum_le;Purelib.NumberTheory.Approximation.
DirichletChain.quadraticThreshold_mul_dirichletWindowQuadrances_sum_le_
indexSum;Purelib.NumberTheory.Approximation.DirichletChain.two_mul_
quadraticThreshold_mul_dirichletWindowQuadrances_sum_le;Purelib.
NumberTheory.Approximation.DirichletChain.two_mul_quadraticThreshold_mul_
dirichletWindowQuadrances_sum_le_with_parity;Purelib.NumberTheory.
Approximation.DirichletChain.two_mul_dirichletWindowQuadrances_sum_le_
factor_mul_parityBound_of_quadraticThreshold;Purelib.NumberTheory.
Approximation.DirichletChain.two_mul_dirichletWindowQuadrances_sum_le_
refinedIndexFactor_mul_parityBound;Purelib.NumberTheory.Approximation.
DirichletChain.two_mul_dirichletWindowQuadrances_sum_le_
quadraticIndexFactor_mul_parityBound;Purelib.NumberTheory.Approximation.
DirichletChain.dirichletWindowQuadrances_sum_le_quadraticIndexFactor_mul_
length;Purelib.NumberTheory.Approximation.DirichletChain.

dirichletWindowQuadrances_sum_le_twentyEight_fifteenths_mul_length;
Purelib.NumberTheory.Approximation.DirichletChain.
dirichletWindowQuadrances_sum_le_normTwoUniversalWindowBound_mul_length;
Purelib.NumberTheory.Approximation.DirichletChain.
dirichletWindowQuadrances_sum_le_sharpDyadicUniversalWindowBound_mul_
length;Purelib.NumberTheory.Approximation.DirichletChain.gapThreshold_mul_
dirichletWindowQuadrances_sum_le_length;Purelib.NumberTheory.
Approximation.DirichletChain.fifteen_mul_dirichletWindowQuadrances_sum_le_
twentyEight_mul_length;Purelib.NumberTheory.Approximation.DirichletChain.
fifteen_mul_dirichletWindowQuadrances_sum_le_with_parity;Purelib.
NumberTheory.Approximation.DirichletChain.dirichletWindowQuadrances_sum_
le_two_mul_length;Purelib.NumberTheory.Approximation.BestApproximation.
AreConsecutiveBest.coefficientIndexQuadrance_lt_four_mul_dirichletWindow;
Purelib.NumberTheory.Approximation.BestApproximation.AreConsecutiveBest.
one_lt_four_mul_dirichletWindow;Purelib.NumberTheory.Approximation.
BestApproximation.AreConsecutiveBest.two_lt_four_mul_dirichletWindow_of_
indexTwo;Purelib.NumberTheory.Approximation.BestApproximation.
AreConsecutiveBest.lower_witnesses_dirichletWindow;Purelib.NumberTheory.
Approximation.BestApproximation.AreConsecutiveBest.
dirichletWindowQuadrance_le_sourceBound;Purelib.Papers.RFAP.Approximation.
consecutiveDirichletWindowQuadrance;Purelib.Papers.RFAP.Approximation.
consecutiveBest_index_lt_four_mul_dirichletWindow;Purelib.Papers.RFAP.
Approximation.one_lt_four_mul_consecutiveDirichletWindow;Purelib.Papers.
RFAP.Approximation.two_lt_four_mul_consecutiveDirichletWindow_of_indexTwo;
Purelib.Papers.RFAP.Approximation.one_twelfth_le_
consecutiveNormalizedDeterminantGap;Purelib.Papers.RFAP.Approximation.
refinedNormalizedDeterminantGapThreshold_lt_half;Purelib.Papers.RFAP.
Approximation.refinedNormalizedDeterminantGapThreshold_quadratic_
nonnegative;Purelib.Papers.RFAP.Approximation.
refinedNormalizedDeterminantGapThreshold_mul_denominator;Purelib.Papers.
RFAP.Approximation.refinedIndexDirichletWindowFactor_mul_denominator;
Purelib.Papers.RFAP.Approximation.refinedUniversalDirichletWindowBound_
mul_denominator;Purelib.Papers.RFAP.Approximation.
refinedTwoWindowDirichletSumBound_mul_denominator;Purelib.Papers.RFAP.
Approximation.quadraticNormalizedDeterminantGapThreshold_lt_half;Purelib.
Papers.RFAP.Approximation.quadraticNormalizedDeterminantGapThreshold_
quadratic_nonnegative;Purelib.Papers.RFAP.Approximation.
quadraticNormalizedDeterminantGapThreshold_integerCertificate;Purelib.
Papers.RFAP.Approximation.quadraticUniversalDirichletWindowBound_mul_
denominator;Purelib.Papers.RFAP.Approximation.
indexAwareUniversalDirichletWindowBound_mul_denominator;Purelib.Papers.
RFAP.Approximation.quadraticUniversalDirichletWindowBound_lt_refined;
Purelib.Papers.RFAP.Approximation.quadraticTwoWindowDirichletSumBound_mul_
denominator;Purelib.Papers.RFAP.Approximation.
octantNormalizedDeterminantGap_mul_denominator;Purelib.Papers.RFAP.
Approximation.octantIndexOneDirichletWindowBound_mul_denominator;Purelib.
Papers.RFAP.Approximation.centeredIndexTwoNormalizedDeterminantGap_mul_
denominator;Purelib.Papers.RFAP.Approximation.
centeredIndexTwoDirichletWindowBound_mul_denominator;Purelib.Papers.RFAP.

Approximation.octantUniversalDirichletWindowBound_mul_denominator;Purelib.
Papers.RFAP.Approximation.octantTwoWindowDirichletSumBound_mul_
denominator;Purelib.Papers.RFAP.Approximation.
dyadicNormalizedDeterminantGap_mul_denominator;Purelib.Papers.RFAP.
Approximation.dyadicUniversalDirichletWindowBound_mul_denominator;Purelib.
Papers.RFAP.Approximation.dyadicTwoWindowDirichletSumBound_mul_
denominator;Purelib.Papers.RFAP.Approximation.refinedThreshold_le_
consecutiveNormalizedDeterminantGap;Purelib.Papers.RFAP.Approximation.
quadraticThreshold_le_consecutiveNormalizedDeterminantGap;Purelib.Papers.
RFAP.Approximation.consecutiveNormalizedDeterminantGap_boundary;Purelib.
Papers.RFAP.Approximation.threshold_le_
consecutiveNormalizedDeterminantGap_of_quadratic;Purelib.Papers.RFAP.
Approximation.quadraticThreshold_mul_consecutiveDirichletWindow_le_index;
Purelib.Papers.RFAP.Approximation.quadraticThreshold_mul_
consecutiveDirichletWindow_le_two;Purelib.Papers.RFAP.Approximation.
consecutiveDirichletWindow_mul_normalizedGap_eq_index;Purelib.Papers.RFAP.
Approximation.consecutiveBest_dirichletWindow_le_twelve_mul_index;Purelib.
Papers.RFAP.Approximation.consecutiveBest_dirichletWindow_le_twentyFour;
Purelib.Papers.RFAP.Approximation.consecutiveBest_dirichletWindow_le_
refinedUniversalBound;Purelib.Papers.RFAP.Approximation.consecutiveBest_
dirichletWindow_le_quadraticIndexFactor;Purelib.Papers.RFAP.Approximation.
half_le_consecutiveNormalizedDeterminantGap;Purelib.Papers.RFAP.
Approximation.octantGap_le_consecutiveNormalizedDeterminantGap;Purelib.
Papers.RFAP.Approximation.dyadicGap_le_
consecutiveNormalizedDeterminantGap;Purelib.Papers.RFAP.Approximation.
dyadicCenteredIndexTwoGap_le_consecutiveNormalizedDeterminantGap;Purelib.
Papers.RFAP.Approximation.consecutiveBest_dirichletWindow_le_
octantIndexOneBound;Purelib.Papers.RFAP.Approximation.consecutiveBest_
dirichletWindow_le_two_of_indexOne;Purelib.Papers.RFAP.Approximation.
centeredIndexTwoNormalizedDeterminantGap_le_consecutive;Purelib.Papers.
RFAP.Approximation.consecutiveBest_dirichletWindow_le_
centeredIndexTwoBound;Purelib.Papers.RFAP.Approximation.one_le_
consecutiveNormalizedDeterminantGap_of_indexTwo;Purelib.Papers.RFAP.
Approximation.consecutiveBest_dirichletWindow_le_two_of_indexTwo;Purelib.
Papers.RFAP.Approximation.consecutiveBest_dirichletWindow_le_
octantUniversalBound;Purelib.Papers.RFAP.Approximation.consecutiveBest_
dirichletWindow_le_dyadicUniversalBound;Purelib.Papers.RFAP.Approximation.
consecutiveBest_dirichletWindow_le_indexAwareUniversalBound;Purelib.
Papers.RFAP.Approximation.consecutiveBest_dirichletWindow_le_
quadraticUniversalBound;Purelib.Papers.RFAP.Approximation.successiveBest_
coefficientIndexQuadrance_add_le_three;Purelib.Papers.RFAP.Approximation.
quadraticThreshold_mul_one_successiveDirichletWindow_le_one;Purelib.
Papers.RFAP.Approximation.quadraticThreshold_mul_
successiveDirichletWindowSum_le_three;Purelib.Papers.RFAP.Approximation.
successiveBest_one_dirichletWindow_le_refinedIndexFactor;Purelib.Papers.
RFAP.Approximation.successiveBest_dirichletWindow_add_le_
refinedTwoWindowSumBound;Purelib.Papers.RFAP.Approximation.successiveBest_
one_dirichletWindow_le_quadraticIndexFactor;Purelib.Papers.RFAP.
Approximation.successiveBest_dirichletWindow_add_le_

quadraticTwoWindowSumBound;Purelib.Papers.RFAP.Approximation.
successiveBest_dirichletWindow_add_le_indexAwareTwoWindowSumBound;Purelib.
Papers.RFAP.Approximation.successiveBest_dirichletWindow_add_le_
octantTwoWindowSumBound;Purelib.Papers.RFAP.Approximation.successiveBest_
dirichletWindow_add_le_dyadicTwoWindowSumBound;Purelib.Papers.RFAP.
Approximation.successiveBest_dirichletWindow_add_le_four;Purelib.Papers.
RFAP.Approximation.consecutiveDirichletWindowQuadrances_length_eq_
indexQuadrances_length;Purelib.Papers.RFAP.Approximation.bestChain_two_
mul_coefficientIndexQuadrances_sum_le_with_parity;Purelib.Papers.RFAP.
Approximation.bestChain_two_mul_coefficientIndexQuadrances_sum_le;Purelib.
Papers.RFAP.Approximation.bestChain_quadraticThreshold_mul_windowSum_le_
indexSum;Purelib.Papers.RFAP.Approximation.bestChain_two_mul_
quadraticThreshold_mul_windowSum_le;Purelib.Papers.RFAP.Approximation.
bestChain_two_mul_quadraticThreshold_mul_windowSum_le_with_parity;Purelib.
Papers.RFAP.Approximation.bestChain_two_mul_windowSum_le_
refinedIndexFactor_mul_parityBound;Purelib.Papers.RFAP.Approximation.
bestChain_two_mul_windowSum_le_quadraticIndexFactor_mul_parityBound;
Purelib.Papers.RFAP.Approximation.bestChain_windowSum_le_
indexAwareUniversalBound_mul_length;Purelib.Papers.RFAP.Approximation.
bestChain_windowSum_le_octantUniversalBound_mul_length;Purelib.Papers.
RFAP.Approximation.bestChain_windowSum_le_dyadicUniversalBound_mul_length;
Purelib.Papers.RFAP.Approximation.bestChain_fifteen_mul_windowSum_le_
twentyEight_mul_length;Purelib.Papers.RFAP.Approximation.bestChain_
fifteen_mul_windowSum_le_with_parity;Purelib.Papers.RFAP.Approximation.
bestChain_windowSum_le_two_mul_length;Purelib.Papers.RFAP.Approximation.
mem_consecutiveBestSuccessors_witnesses_dirichletWindow;Purelib.Papers.
RFAP.Approximation.mem_consecutiveBestSuccessors_dirichletWindow_le_
sourceBound;Purelib.Papers.RFAP.Approximation.mem_
consecutiveBestSuccessors_dirichletWindow_le_twentyFour;Purelib.Papers.
RFAP.Approximation.mem_consecutiveBestSuccessors_dirichletWindow_le_
refinedUniversalBound;Purelib.Papers.RFAP.Approximation.mem_
consecutiveBestSuccessors_dirichletWindow_le_quadraticUniversalBound;
Purelib.Papers.RFAP.Approximation.mem_consecutiveBestSuccessors_
dirichletWindow_le_indexAwareUniversalBound;Purelib.Papers.RFAP.
Approximation.mem_consecutiveBestSuccessors_dirichletWindow_le_
octantUniversalBound;Purelib.Papers.RFAP.Approximation.mem_
consecutiveBestSuccessors_dirichletWindow_le_dyadicUniversalBound;Purelib.
NumberTheory.Approximation.Dirichlet.Critical.Run.criticalBadWindowRun;
Purelib.NumberTheory.Approximation.Dirichlet.Critical.Robust.Run.
criticalFullyRobustRun;Purelib.NumberTheory.Approximation.
SuccessorDynamics.regeneratedTransition_atom;Purelib.NumberTheory.
Approximation.SuccessorDynamics.stoppedBlockHitCertificate;Purelib.
NumberTheory.Approximation.RobustSuccessorSharpness.
robustSuccessorReturnProbability_eq_one;Purelib.NumberTheory.
Approximation.RobustSuccessorSharpness.chevallierAnalogue;Purelib.
NumberTheory.Approximation.DirichletGlobal.globalDirichletRequest;Purelib.
NumberTheory.Approximation.DirichletGlobal.chevallierBoundary.

Lemma 9.20 (the terminal Hurwitz pair is best)**.** *The terminal current pair of every generated finite Hurwitz trace satisfies the full best-approximation predicate. Every displayed*

rational trace has such a terminal stage, so AllBest(z) *is nonempty.*

Proof. At a terminal stage, the current error is zero, so the weak comparison with every competitor is immediate. For strictness, suppose a competitor $c = (p, q)$ with $0 < \mathrm{N}(q) < \mathrm{N}(q_n)$ also had zero error. The cross-error identity

$$\Delta(c_n, c) = q_n E_z(c) - q E_z(c_n)$$

would give $\Delta(c_n, c) = 0$. In the adjacent-continuant basis of lemma 5.12, its previous coefficient is $t = \Delta(c_n, c_{n-1})\Delta(c_n, c) = 0$. Hence the denominator identity reduces to $q = sq_n$. Since $q \neq 0$, also $s \neq 0$, and multiplicativity of integral quadrance gives $\mathrm{N}(q) = \mathrm{N}(s)\,\mathrm{N}(q_n) \geq \mathrm{N}(q_n)$, a contradiction. Every lower-height competitor therefore has positive error, which is strictly larger than the terminal zero error.

The stage recursion decreases the natural quadrance of its complete denominator and stops exactly at zero remainder. Structural recursion therefore supplies a terminal member of every computed stage list. Its current pair is best by the preceding argument and hence belongs to AllBest(z) by (83). □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzTrace.exists_terminal_mem_stages;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.current_error_eq_zero_of_terminal;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.isBest_of_generated_of_terminal;Purelib.NumberTheory.Approximation.HurwitzGeometry.allBest_nonempty;Purelib.Papers.RFAP.Approximation.terminalHurwitzConvergent_isBest;Purelib.Papers.RFAP.Approximation.allBestGaussianApproximations_nonempty`.

10 Explicit arithmetic envelopes

This section puts finite bounds around the RFAP-2 calculations named in the glossary. It first gives the normalized rational working layer used by the repeated geometric constructions, then treats the square-threshold request, grid record, homogeneous integral evaluation, finite Hurwitz recursion, continuants, and protected quotient target. These are bounds computed from the displayed input, not maxima extracted after running the calculation. The later envelopes begin after RFAP-1 has supplied the Newton-band anchor and radius; they include the normalized Horner and Newton arithmetic performed from that anchor. Eliminant, Sturm, winding, and complete repeated-subdivision costs still belong to the larger RFAP-1 resource problem. The optional all-best selector is bounded separately by the exact rectangular count in theorem 9.4; it is not consumed by the Newton or Hurwitz trace calculations bounded in this section.

Definition 10.1 (component width)**.** *For $n \in \mathbb{N}$, put*

$$\ell(n) = \lfloor \log_2 n \rfloor + 1,$$

where the executable natural logarithm has $\log_2 0 = 0$. Thus zero still occupies one displayed bit. For $x \in \mathbb{Z}$, put $\ell_{\mathbb{Z}}(x) = \ell(|x|)$, and for $u = a + ib \in \mathbb{Z}[i]$ put

$$\ell_{\mathbb{Z}[i]}(u) = \max\{\ell_{\mathbb{Z}}(a), \ell_{\mathbb{Z}}(b)\}.$$

The width of $(a+ib)/D$ is the maximum of the two numerator-coordinate widths and $\ell(D)$. Since $|a|, |b| \le \mathrm{N}(a+ib)$, an upper bound K on the quadrance of a Gaussian integer gives the component-width bound $\ell(K)$.

The Lean definition of ℓ is primitive natural recursion rather than a real logarithm. Its monotonicity and the component bounds are proved by integer arithmetic.

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.natWidth;Purelib.NumberTheory.Approximation.CalculationSize.gaussianIntWidth;Purelib.NumberTheory.Approximation.CalculationSize.normNat_eq_component_squares;Purelib.NumberTheory.Approximation.CalculationSize.gaussianIntWidth_le_budget`.

10.1 Normalized working rationals

Definition 10.2 (normalized rational operations)**.** *A working rational is a displayed record a/b with $b > 0$ and $\gcd(|a|, b) = 1$. This canonical record is the rational scalar used by the algebraic statements: two such records are equal exactly when their numerators and denominators are structurally equal, equivalently when their cross products are equal. Given reduced records a/b and c/d, put*

$$g = \gcd(b, d), \qquad R_+ = \frac{a(d/g) + c(b/g)}{(b/g)d}.$$

The working sum is the constructive gcd reduction of R_+. For multiplication put

$$g_1 = \gcd(|a|, d), \qquad g_2 = \gcd(|c|, b), \qquad R_\times = \frac{(a/g_1)(c/g_2)}{(b/g_2)(d/g_1)},$$

and again reduce the resulting record. Thus certain cancellation occurs before either product is formed. The reciprocal of a nonzero record swaps $|a|$ and b and moves the sign to the numerator; zero maps to zero. In particular it avoids the squared-denominator intermediate produced by a generic total-inverse construction.

Negation changes only the numerator sign; subtraction is normalized negation followed by normalized addition. Division is normalized reciprocal followed by the cross-cancelled multiplication. These six operations are the complete scalar layer used below.

For the logical reference implementation of the natural gcd, define

$$G(0, n) = 1, \qquad G(m, n) = 1 + G(n \bmod m, m) \quad (m > 0).$$

This counts the terminal clause and satisfies the elementary finite bound $G(m, n) \le m + 1$.

Lemma 10.3 (normalized-operation envelopes)**.** *Erasing each operation in definition 10.2 gives exactly the ordinary rational sum, product, or reciprocal. If A_+ and B_+ are the absolute numerator and positive denominator of the reduced sum, then*

$$A_+ \le |a|d + |c|b, \qquad B_+ \le bd.$$

For the reduced product,

$$A_\times \le |a|\,|c|, \qquad B_\times \le bd.$$

Consequently the returned records have respective bit-width bounds

$$\ell(\max\{|a|d + |c|b, bd\}), \qquad \ell(\max\{|a|\,|c|, bd\}).$$

The exact logical gcd-clause counts are one denominator gcd plus the final reduction for addition, and two cross-cancellation gcds plus the final reduction for multiplication. Each term is bounded by the displayed $G(m,n) \le m+1$ recurrence. Negation and reciprocal have one final normalization; subtraction and division add that cost to the corresponding binary operation. Their output-component envelopes are respectively $\max\{|a|, b\}$*, the addition envelope,* $\max\{|a|, b\}$*, and the multiplication envelope.*

Proof. Write $b = (b/g)g$ and $d = (d/g)g$ and cross-multiply R_+ with the ordinary sum; the two integer expressions agree coefficientwise. The multiplication identity follows in the same way after reconstructing each cancelled factor. Constructive gcd normalization preserves these values and can only decrease the absolute numerator and positive denominator. The triangle inequality for integer absolute value gives the addition bound; exact divisibility and monotonicity of natural multiplication give the remaining component bounds. Induction on the first argument of the displayed Euclidean recursion proves the clause bound, since $n \bmod m < m$ whenever $m > 0$. □

Formal verification:

`RFAP.NormalizedFraction.addRaw;RFAP.NormalizedFraction.mulRaw;RFAP.NormalizedFraction.normalizedAdd_eq;RFAP.NormalizedFraction.normalizedMul_eq;RFAP.NormalizedFraction.normalizedSub_eq;RFAP.NormalizedFraction.normalizedDiv_eq;RFAP.NormalizedFraction.add_componentMagnitude_le;RFAP.NormalizedFraction.mul_componentMagnitude_le;RFAP.NormalizedFraction.subSteps_le_envelope;RFAP.NormalizedFraction.divSteps_le_envelope;RFAP.NormalizedFraction.gcdClauses_le;Purelib.Papers.RFAP.Approximation.normalizedCoordinateAdd_width_le_budget;Purelib.Papers.RFAP.Approximation.normalizedCoordinateMul_width_le_budget;Purelib.Papers.RFAP.Approximation.normalizedCoordinateSub_width_le_budget;Purelib.Papers.RFAP.Approximation.normalizedCoordinateDiv_width_le_budget`.

Safe-cut candidates, dyadic square-bracket midpoints, grid scaling and division, and the rational Newton centre now use this layer. The Newton centre first reduces the two coordinates of the polynomial and derivative values, performs the Gaussian division through normalized scalar operations, and persists a reduced pair. The corresponding erasure theorem states the same Newton formula used throughout RFAP-1. Hurwitz continuants are not normalized after each recurrence: their integral recurrence is representation-sensitive, and the adjacent unit determinant already supplies the required Gaussian coprimality without introducing a choice of unit.

Formal verification:

`Purelib.RootIsolation.Isolation.SafeCandidates.interiorCandidate;Purelib.Polynomial.Geometry.DyadicBracket.SquareInterval.midpoint;Purelib.Polynomial.HeightApproximation.gridCoordinate;Purelib.RootIsolation.Isolation.LocalExtraction.anchorNewtonCenter_eq;Purelib.Data.Gaussian.Normalized.normalizedDiv_eq`.

Definition 10.4 (normalized Horner and Newton envelopes)**.** *For a reduced scalar record let m be its largest numerator-or-denominator magnitude, and for a reduced Gaussian*

record take the maximum of its two scalar values. Put

$$A(x,y) = 2xy, \qquad M(x,y) = xy,$$

and define the Gaussian envelopes

$$G_+(x,y) = A(x,y), \qquad G_\times(x,y) = A(M(x,y), M(x,y)),$$

$$G_N(x) = A(M(x,x), M(x,x)), \quad G_{\text{inv}}(x) = M(x, G_N(x)), \quad G_/(x,y) = G_\times(x, G_{\text{inv}}(y)).$$

These are deliberately uniform coordinate bounds; cancellation may make the actual records much smaller.

For a high-to-low coefficient list $c_0, \ldots, c_d$ evaluated at z, start with $H_0 = m(0)$ and set

$$H_{j+1} = \max\{H_j, G_+(G_\times(H_j, m(z)), m(c_j))\}.$$

Alongside it, sum the multiplication and addition clause envelopes from definition 10.2. Call the resulting recurrences $H(z; c_0, \ldots, c_d)$ and $T_H(z; c_0, \ldots, c_d)$.

For the Newton update, compute these two recurrences for f and its displayed coefficientwise derivative. Extend the magnitude recurrence through one Gaussian division and one subtraction, retaining the maximum of the two Horner bounds, quotient bound, and returned-centre bound. Extend the step recurrence by the corresponding division and subtraction clause envelopes. Call the final magnitude maximum $E_N(f,z)$, and put $B_N(f,z) = \ell(E_N(f,z))$; call the step sum $T_N(f,z)$.

Lemma 10.5 (normalized Horner and Newton traces)**.** *Normalized Horner evaluation erases to ordinary coefficient-list evaluation. Every accumulator in its executed trace has component width at most $\ell(H)$, and its accounted arithmetic-clause count is at most T_H. The normalized Newton record erases to*

$$z - \frac{f(z)}{f'(z)},$$

every accumulator in both Horner traces and the returned centre has component width at most $B_N(f,z)$, and the combined clause count is at most $T_N(f,z)$.

Proof. At one Horner clause, the multiplication envelope bounds both reduced coordinate products. The addition envelope then bounds both coordinate sums, while the outer maximum retains the preceding accumulator. Induction on the displayed coefficient list proves both the trace bound and the sum of the per-clause costs. Applying the same induction to the coefficientwise derivative gives the second trace. The Gaussian division and subtraction lemmas then give the final two bounds. The erasure identities at each clause prove the two semantic formulae by the same finite induction. □

Formal verification:

`Purelib.Polynomial.NormalizedEvaluation.value_eval;Purelib.Polynomial.NormalizedEvaluation.trace_magnitude_le_envelope;Purelib.Polynomial.NormalizedEvaluation.steps_le_envelope;Purelib.Polynomial.NormalizedEvaluation.value_newtonCenter;Purelib.NumberTheory.Approximation.CalculationSize.normalizedNewtonCenter_width_le_budget;Purelib.NumberTheory.Approximation.CalculationSize.normalizedNewton_steps_le_budget;Purelib.Papers.RFAP.Approximation.normalizedNewtonCenter_eq.`

Lemma 10.6 (target-scan and grid-record envelopes)**.** *Let $C, \delta > 0$ be displayed rationals and put*

$$n = \left\lceil \frac{C}{\delta} \right\rceil, \qquad m = \min\{k \in \mathbb{N}_{>0} : n < k^2\}.$$

The bounded square scan uses at most $n+1$ comparison-bearing clauses and satisfies

$$(m-1)^2 \le n < m^2.$$

For a displayed centre $c \in \mathbb{Q}(i)$ and radius r, put

$$D = \left\lfloor \frac{1/2}{r} \right\rfloor + 1, \qquad h = \frac{1}{2D}, \qquad q = \frac{U}{D}$$

where $U \in \mathbb{Z}[i]$ is obtained by coordinatewise nearest rounding of Dc. Then

$$\mathrm{N}(q-c) \le 2h^2, \qquad \mathrm{N}(U) \le K_{\mathrm{grid}} := \left\lceil D^2\, 2\big(\mathrm{N}(c) + 2h^2\big) \right\rceil.$$

Consequently every component of the returned fraction record has at most

$$\max\{\ell(K_{\mathrm{grid}}), \ell(D)\}$$

bits.

Proof. Minimality of the successful square candidate gives the first bracket; the implementation has fuel $n+1$. Nearest rounding gives a coordinate error in $[-h, h]$, so its two squared errors sum to at most $2h^2$. The two-leg quadrance inequality gives $\mathrm{N}(q) \le 2(\mathrm{N}(c) + 2h^2)$. Finally, clearing the displayed denominator gives the exact identity $\mathrm{N}(U) = D^2\,\mathrm{N}(q)$, and taking the rational ceiling produces the natural bound above. □

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.targetDenominator_bracket;Purelib.NumberTheory.Approximation.CalculationSize.targetComparisonBudget;Purelib.NumberTheory.Approximation.CalculationSize.gridFraction_offset_normSq_le;Purelib.NumberTheory.Approximation.CalculationSize.gridFraction_numerator_norm_le_budget;Purelib.Papers.RFAP.Approximation.gridFraction_width_le_budget`.

Definition 10.7 (homogeneous-evaluation envelope)**.** *For $U, V \in \mathbb{Z}[i]$ and a high-to-low integral coefficient list, define*

$$E_{U,V}([]) = 0, \qquad E_{U,V}(A :: L) = 2\left(\mathrm{N}(A)\ \mathrm{N}(U)^{|L|} + \mathrm{N}(V)\ E_{U,V}(L)\right).$$

Let $E^{\max}_{U,V}(L)$ be the maximum of $E_{U,V}$ over all suffixes of L. This is itself a finite recurrence, computed before evaluation. The displayed implementation uses $|L|$ coefficient clauses and has the multiplication budget

$$T_\times([]) = 0, \qquad T_\times(A :: L) = |L| + 2 + T_\times(L).$$

The term $|L|$ counts the direct finite power calculation in that clause.

Lemma 10.8 (homogeneous working values)**.** *Every Gaussian integer returned by a recursive suffix of the homogeneous evaluation has quadrance at most $E^{\max}_{U,V}(L)$, hence component width at most $\ell(E^{\max}_{U,V}(L))$.*

Proof. Induct on the coefficient list. Multiplicativity of Gaussian quadrance and

$$\mathrm{N}(x+y) \le 2(\mathrm{N}(x)+\mathrm{N}(y))$$

give exactly the recurrence in definition 10.7. Taking the maximum with the recursively computed suffix bound retains every intermediate suffix value. □

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.homogeneousNormEnvelope;Purelib.NumberTheory.Approximation.CalculationSize.homogeneousWorkingEnvelope;Purelib.NumberTheory.Approximation.CalculationSize.homogeneousTrace_norm_le_workingEnvelope;Purelib.NumberTheory.Approximation.CalculationSize.homogeneousMultiplicationBudget;Purelib.Papers.RFAP.Approximation.homogeneousEvaluation_norm_le_envelope.`

Theorem 10.9 (finite Hurwitz arithmetic envelope)**.** *For a displayed source $z = N/D$ with $D \neq 0$, let*

$$L = \ell(\mathrm{N}(D)), \qquad A = \max\{2(\mathrm{N}(N)+\mathrm{N}(D)), 4\,\mathrm{N}(D)\}.$$

The executable Hurwitz encoding has at most L digit clauses, every digit a_j satisfies $\mathrm{N}(a_j) \le A$, and the numerator and denominator of its final continuant each have quadrance at most

$$\big(2(A+1)\big)^L. \tag{154}$$

Thus L, $\ell(A)$, and the binary width of (154) are explicit bounds for the recursion count, digit components, and final continuant components respectively. The same bound holds for every recursive suffix continuant visited by the calculation. Its multiplication-and-addition count is at most $2L$, in addition to the L encoding clauses.

Proof. If $R = N - aD$ is one nonterminal integral remainder, the fundamental-square bound gives the stronger natural inequality

$$2\,\mathrm{N}(R) \le \mathrm{N}(D)\,.$$

Each transition therefore consumes at least one binary digit of denominator quadrance, proving the bound L. The division identity and the two-leg quadrance inequality give $\mathrm{N}(a) \le 2(\mathrm{N}(N)+\mathrm{N}(D))$. After the first step, the new numerator is the old denominator and the new denominator has smaller quadrance; the second term in A makes the same bound invariant down the complete trace.

For a digit list with $\mathrm{N}(a)_j \le A$, induction on the continuant recurrence and the same two-leg inequality bound both continuant entries by $[2(A+1)]^j$ after j digits. Monotonicity in $j \le L$ gives (154) for every suffix. Each digit clause performs one Gaussian-integer multiplication and one addition, giving $2L$. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.twice_remainder_norm_le_denominator;Purelib.NumberTheory.Approximation.HurwitzRational.encode_length_le_width;Purelib.NumberTheory.Approximation.HurwitzRational.encode_digit_norm_le;Purelib.NumberTheory.Approximation.CalculationSize.`

`continuant_norm_le_envelope;Purelib.NumberTheory.Approximation.CalculationSize.continuantTrace_norm_le_envelope;Purelib.NumberTheory.Approximation.CalculationSize.encodedContinuant_steps_le_budget;Purelib.Papers.RFAP.Approximation.hurwitzEncoding_length_le_budget;Purelib.Papers.RFAP.Approximation.hurwitzContinuant_norm_le_budget`.

Proposition 10.10 (protected quotient target)**.** *At a protected quotient site, replace C in lemma 10.6 by the computed coefficient $2L_1(g, A)/\eta_h$. The same comparison budget and exact predecessor bracket hold. No extra unbounded search is introduced by the numerator or pole branch.*

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.protectedTargetComparisonBudget;Purelib.NumberTheory.Approximation.CalculationSize.protectedTargetDenominator_bracket`.

Theorem 10.11 (RFAP-2 calculation-size certificate)**.** *Suppose the displayed input consists of a polynomial f, an RFAP-1 Newton-band anchor and radius, the already cleared Gaussian-integral data U, V and coefficient list used for homogeneous evaluation, a displayed Gaussian fraction for the finite Hurwitz calculation, and the two rationals C, δ for a target scan. Finite recurrences in precisely this data compute natural numbers T and B such that:*

1. *the normalized polynomial and derivative Horner traces, Newton division, grid construction, homogeneous evaluation, Hurwitz encoding and continuant construction, and target scan use at most T accounted arithmetic or recursive clauses;*
2. *every accumulator in the two normalized Horner traces, the returned Newton record, all three normalized grid-operation results, and the returned grid fraction have component width at most B;*
3. *every homogeneous-Horner suffix value, every Hurwitz digit, and both entries of every recursive continuant suffix have component width at most B; and*
4. *the selected target denominator has width at most B.*

Here T is the sum of the source recurrences already displayed in this section. The number B is the maximum of their Horner–Newton, grid, homogeneous, digit, continuant, and target-width envelopes. In particular, neither is defined by running the calculation and then taking a maximum of the trace.

Proof. Apply lemmas 10.5, 10.6 and 10.8 and theorem 10.9 to the corresponding displayed inputs. Addition of their clause inequalities gives the stated T. Monotonicity of ℓ converts each natural component or quadrance bound into a width bound; inclusion into the explicitly nested maximum gives B. The target-denominator term uses $m \le n + 1$, which follows from the minimal square scan. These finite inequalities are assembled into one proof-bearing record by the implementation. □

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.StageTwoInput;Purelib.NumberTheory.Approximation.CalculationSize.stageTwoStepBudget;Purelib.NumberTheory.Approximation.CalculationSize.stageTwoBitBudget;Purelib.NumberTheory.Approximation.CalculationSize.stageTwo_resource_certificate;Purelib.Papers.RFAP.Approximation.stageTwoResourceCertificate`.

This is the RFAP-2 side of the resource statement: its input boundary is the displayed RFAP-1 Newton-band data and cleared integral coefficient list. The companion paper now supplies the complementary certificate for constructing the normalized eliminant and Bézout data, clearing the coefficients, and performing every Sturm winding, safe cut, and requested subdivision round [1]. Its budgets are `stageThreeStepBudget` and `stageThreeBitBudget`. Consequently the two step budgets add and the two width budgets combine by their maximum. The interface remains explicit—the RFAP-2 theorem does not silently charge its post-processing bound for work performed by RFAP-1—but there is no longer an unbounded eliminant, Sturm, winding, clearing, or subdivision phase between the two certificates.

Formal verification:

`Purelib.Papers.RFAP.Paper.stageThreeStepBudget;Purelib.Papers.RFAP.Paper.stageThreeBitBudget;Purelib.Papers.RFAP.Paper.stageThreeResourceCertificate`.

11 Worked calculations

This section feeds one RFAP-1 track through both branches of the glossary. The cubic calculation exhibits the coefficient scale, height obstruction, Newton band, and grid fraction. Its first displayed sample is then treated as a Gaussian fraction: the nearest-integer program produces its complete finite trace, the continuants give the generated stages, and the finite Lakein inequality compares every listed division-free residual with all smaller-denominator competitors.

11.1 The cubic height calculation

Example 11.1 (displayed cubic data)**.** *Consider the RFAP-1 polynomial*

$$f(Z) = Z^3 - (1+i).$$

Its coefficients are Gaussian integers, so $S_f = 1$. The companion calculation returns three initial winding-one boxes with rational centres

$$q_1 = \frac{139+37i}{128}, \qquad q_2 = \frac{-203+203i}{256}, \qquad q_3 = \frac{-37-139i}{128}.$$

The common-denominator presentations have no common positive-integer scale, and

$$H(q_1) = H(q_3) = 128^2 = 16384, \qquad H(q_2) = 256^2 = 65536.$$

Their canonical Gaussian reductions are

$$q_1 = \frac{51+88i}{64+64i}, \qquad q_2 = \frac{-203}{128+128i}, \qquad q_3 = \frac{51-88i}{64+64i},$$

so their intrinsic heights are respectively

$$8192, \qquad 32768, \qquad 8192.$$

Thus reduction by the Gaussian factor $1+i$, which no positive-integer gcd detects, improves every displayed common-denominator height by a factor of two.

Proposition 11.2 (exact cubic height obstruction)**.** *The three residual quadrances are*

$$\mathrm{N}(f(q_1)) = \mathrm{N}(f(q_3)) = \frac{39952889}{549755813888} \approx 7.27 \times 10^{-5},$$
$$\mathrm{N}(f(q_2)) = \frac{537358761}{35184372088832} \approx 1.53 \times 10^{-5}.$$

Their exact scaled obstruction values are

$$H(q_1)^3\,\mathrm{N}(f(q_1)) = H(q_3)^3\,\mathrm{N}(f(q_3)) = 319623112,$$
$$H(q_2)^3\,\mathrm{N}(f(q_2)) = 4298870088.$$

After canonical Gaussian reduction, the intrinsic obstruction values are

$$h(q_1)^3\,\mathrm{N}(f(q_1)) = h(q_3)^3\,\mathrm{N}(f(q_3)) = 39952889,$$
$$h(q_2)^3\,\mathrm{N}(f(q_2)) = 537358761.$$

In particular, all three displayed values satisfy the lower bound of theorem 2.7.

Proof. For $q = (u/D)$, homogeneous evaluation is the Gaussian-integer calculation

$$W_f(u, D) = u^3 - (1 + i)D^3 = D^3 f(q).$$

At the three displayed inputs it gives

$$W_f(139 + 37i, 128) = 17594 - 3174i,$$
$$W_f(-203 + 203i, 256) = -46362 - 46362i,$$
$$W_f(-37 - 139i, 128) = -3174 + 17594i.$$

Squaring and adding their integer coordinates gives respectively 319623112, 4298870088, and 319623112. The identity $\mathrm{N}(W_f) = H(q)^3\,\mathrm{N}(f(q))$ gives the exact residuals and scaled values. For each input, coefficientwise Gaussian division by $1 + i$, followed by the unit i which places the denominator in the chosen sector, gives the three canonical fractions displayed above. Their heights are half the respective common-denominator heights. Cubing that factor divides the three scaled values by eight, giving 39952889, 537358761, and 39952889. Each residual numerator is nonzero, so its quadrance is at least one. □

Formal verification:

`Purelib.Polynomial.DenominatorHeight.residualNumerator;Purelib.Polynomial.DenominatorHeight.residualNumerator_identity;Purelib.Polynomial.DenominatorHeight.scaled_height_mul_residual_eq_normNat;Purelib.Polynomial.DenominatorHeight.one_le_scaled_height_mul_residual;Purelib.Polynomial.GaussianDenominatorHeight.scaled_intrinsicHeight_mul_residual_eq_normNat;Purelib.Polynomial.GaussianDenominatorHeight.one_le_scaled_intrinsicHeight_mul_residual`.

Equivalently, the universal floors at these displayed denominators are

$$\mathrm{N}(f(q_1)), \mathrm{N}(f(q_3)) \geq \frac{1}{128^6} \approx 2.27 \times 10^{-13}, \qquad \mathrm{N}(f(q_2)) \geq \frac{1}{256^6} \approx 3.55 \times 10^{-15}.$$

Canonical reduction strengthens them to

$$\mathrm{N}(f(q_1)), \mathrm{N}(f(q_3)) \geq \frac{1}{8192^3} \approx 1.82 \times 10^{-12}, \qquad \mathrm{N}(f(q_2)) \geq \frac{1}{32768^3} \approx 2.84 \times 10^{-14}.$$

Example 11.3 (three states of one shrinking track)**.** *The first track has the following finite refinement states. Each centre is the rational midpoint of its displayed box.*

state	*centre*	$\mathrm{diag}^2(B)$	$H(q)$
0	$(139+37i)/128$	$1/2048$	$128^2 = 16384$
1	$(2221+595i)/2048$	$1/524288$	$2048^2 = 4194304$
2	$(142111+38079i)/131072$	$1/2147483648$	$131072^2 = 17179869184$

Their exact residual quadrances are

$$\frac{39952889}{549755813888} \approx 7.27 \times 10^{-5},$$
$$\frac{8767783105145}{9223372036854775808} \approx 9.51 \times 10^{-7},$$
$$\frac{546115072853076822625}{63382530011411470074835160 2688} \approx 8.62 \times 10^{-10}.$$

The corresponding nonzero-residual floors are

$$2^{-42} \approx 2.27 \times 10^{-13}, \qquad 2^{-66} \approx 1.36 \times 10^{-20}, \qquad 2^{-102} \approx 1.97 \times 10^{-31}.$$

Every displayed residual in example 11.3 lies above its obstruction floor, as it must. The substantial gap is useful intuition: the height obstruction is universal, not a claim that a particular box midpoint is optimal.

Proposition 11.4 (the sufficient-height invocation)**.** *For $A = [-7,7]^2$, the RFAP-1 coefficient calculation gives $L_1(f,A) = 460992$. For every displayed $\delta > 0$, a finite rational scan can choose $\varepsilon > 0$ satisfying*

$$921984\varepsilon^2 \le \delta.$$

Newton extraction then computes a band radius r and returns a grid fraction q_r satisfying

$$H(q_r) = \left(\left\lfloor \frac{1/2}{r} \right\rfloor + 1\right)^2, \qquad \mathrm{N}(f(q_r)) \le 921984\varepsilon^2 \le \delta.$$

Proof. Here $2L_1(f,A) = 921984$. For the chosen δ, compute

$$n = \left\lceil \frac{921984}{\delta} \right\rceil$$

and scan upward for the least positive m with $n < m^2$; then put $\varepsilon = 1/m$. By corollary 3.7, this gives $921984\varepsilon^2 \le \delta$. Theorem 3.6 computes r and the exact nearest-grid denominator. The returned grid fraction need not be any of the three box midpoints above. □

Formal verification:

`Purelib.Polynomial.HeightApproximation.approximate;Purelib.Polynomial.HeightApproximation.Output.height_eq;Purelib.Polynomial.HeightApproximation.Output.residual_le;Purelib.Polynomial.HeightApproximation.approximateTarget;Purelib.Polynomial.HeightApproximation.approximateTarget_residual_le`.

11.2 A finite Hurwitz trace of the first sample

Definition 11.5 (cleared error for the sample)**.** *Regard the first midpoint as the displayed Gaussian fraction*

$$z = \frac{139 + 37i}{128}.$$

For an integral continuant (P_j, Q_j) *define*

$$E_j = (139 + 37i)Q_j - 128P_j \in \mathbb{Z}[i].$$

Then coefficientwise cross multiplication gives

$$\mathrm{N}(zQ_j - P_j) = \frac{\mathrm{N}(E_j)}{128^2}. \tag{155}$$

Proposition 11.6 (the computed five-stage trace)**.** *The deterministic nearest-Gaussian-integer program returns the digit list*

$$[1,\ 1 - 3i,\ -2 + 5i,\ 2 - i,\ 1 - 2i].$$

Its generated continuants, Gaussian denominator heights, and cleared errors are as follows.

j	a_j	(P_j, Q_j)	$\mathrm{N}(Q_j)$	$\mathrm{N}(E_j)\,/128^2$
0	1	$(1, 1)$	1	$1490/16384 \approx 9.09 \times 10^{-2}$
1	$1 - 3i$	$(2 - 3i, 1 - 3i)$	10	$52/16384 \approx 3.17 \times 10^{-3}$
2	$-2 + 5i$	$(12 + 16i, 14 + 11i)$	317	$10/16384 \approx 6.10 \times 10^{-4}$
3	$2 - i$	$(42 + 17i, 40 + 5i)$	1625	$2/16384 \approx 1.22 \times 10^{-4}$
4	$1 - 2i$	$(88 - 51i, 64 - 64i)$	8192	0

Proof. The complete-quotient numerator–denominator records evolve by the exact Euclidean calculation

$$\begin{aligned}(139 + 37i, 128) &\longmapsto (128, 11 + 37i)\\ &\longmapsto (11 + 37i, 6 - 4i)\\ &\longmapsto (6 - 4i, 3 - i)\\ &\longmapsto (3 - i, 1 + i).\end{aligned}$$

At each arrow, a_j is the coordinatewise nearest Gaussian integer and the new denominator is $N_j - a_j D_j$. At the last record, $(3 - i) - (1 - 2i)(1 + i) = 0$, so the trace terminates. Starting from $(P_{-1}, Q_{-1}) = (1, 0)$ and $(P_0, Q_0) = (1, 1)$, the recurrence (4) gives the five displayed continuants. Squaring the coordinates of each Q_j gives $1, 10, 317, 1625, 8192$; substituting each pair in definition 11.5 gives $\mathrm{N}(E_j) = 1490, 52, 10, 2, 0$. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzRational.encode;Purelib.NumberTheory.Approximation.HurwitzRational.continuant;Purelib.NumberTheory.Approximation.HurwitzTrace.stages;Purelib.NumberTheory.Approximation.HurwitzGeometry.generated_of_mem_stages`.

Arithmetic envelopes for the displayed sample. The new resource recurrences can be evaluated without changing the example. For homogeneous evaluation at q_1, the data are

$$\mathrm{N}(139+37i)=20690, \qquad \mathrm{N}(128)=16384, \qquad [A_0,A_1,A_2,A_3]=[1,0,0,-1-i].$$

Reading the coefficient list from its last suffix to the full list, the values of $E_{139+37i,128}$ are

$$4, \qquad 131072, \qquad 4294967296, \qquad 158451277373328.$$

Thus the certified suffix-record envelope has 48 bits per integral component. The displayed implementation uses four coefficient clauses, four Gaussian-integer additions, and the direct-power multiplication budget $T_\times = 14$. The actual final value $W_f = 17594-3174i$ has 15-bit components; the difference records the deliberate slack in the uniform two-leg recurrence.

For the Hurwitz encoder,

$$L=\ell(16384)=15, \qquad A=\max\{2(20690+16384),4\cdot 16384\}=74148.$$

The theorem therefore certifies at most fifteen digit clauses, 17-bit digit components, and at most 258 bits per component of the final continuant, since

$$(2(A+1))^L = 148298^{15} \approx 3.69\times 10^{77}.$$

The actual trace has five clauses, and the largest absolute integer coordinate displayed in its complete quotients and continuants is 139, an eight-bit value. These gaps are useful: the certified figures are source-only worst-case envelopes, while the table records this particular execution.

Formal verification:

`Purelib.NumberTheory.Approximation.CalculationSize.homogeneousWorkingEnvelope;Purelib.NumberTheory.Approximation.CalculationSize.homogeneousMultiplicationBudget;Purelib.NumberTheory.Approximation.HurwitzRational.digitNormBudget;Purelib.Papers.RFAP.Approximation.hurwitzEncoding_length_le_budget;Purelib.Papers.RFAP.Approximation.hurwitzContinuant_norm_le_budget`.

The last row also displays an explicit difference between the two height notions. In $\mathbb{Z}[i]$,

$$139+37i=(1+i)(88-51i), \qquad 128=(1+i)(64-64i).$$

Thus the original common-denominator record has height 16384, while the terminal continuant has displayed Gaussian denominator height 8192. It is already reduced; multiplication of numerator and denominator by i gives the canonical fraction

$$\frac{51+88i}{64+64i}.$$

Hence the intrinsic height of the original sample is 8192. The common factor appears as output of the finite Euclidean trace, the unit normalization is an explicit four-case calculation, and the final cross product is exactly zero. Applying theorem 2.10 sharpens the nonzero polynomial-residual floor for q_1 from $2^{-42}\approx 2.27\times 10^{-13}$ to

$$\frac{1}{8192^3}=2^{-39}\approx 1.82\times 10^{-12}.$$

The actual residual remains the exact value displayed in proposition 11.2; the stronger floor now depends on the exact value through its canonical reduction rather than on a fortuitously chosen presentation.

Corollary 11.7 (the stagewise Lakein comparisons)**.** *For each row j of proposition 11.6, every displayed $(P, Q) \in \mathbb{Z}[i]^2$ satisfying $Q \neq 0$ and $\mathrm{N}(Q) < \mathrm{N}(Q_j)$ obeys*

$$\mathrm{N}(zQ - P) \geq \frac{\mathrm{N}(E_j)}{16384}.$$

For example, if $\mathrm{N}(Q) < 1625$, then

$$\mathrm{N}(zQ - P) \geq \frac{1}{8192} \approx 1.22 \times 10^{-4}.$$

Proof. Every row is a member of the computed stage trace, hence is generated. Apply theorem 8.8 and replace the current residual by the exact identity (155). For $j = 3$ the table gives $\mathrm{N}(E_3) = 2$, so $2/16384 = 1/8192$. At the terminal row the asserted lower bound is zero. □

Formal verification:

`Purelib.NumberTheory.Approximation.HurwitzGeometry.generated_of_mem_stages;Purelib.NumberTheory.Approximation.HurwitzGeometry.Stage.atLeastAsGoodAs_of_generated;Purelib.Papers.RFAP.Approximation.hurwitzConvergent_atLeastAsGoodAs`.

12 Discussion

In the glossary's terms, least coefficient clearing gives exact displayed common- and Gaussian-denominator identities, and executable Gaussian Euclidean reduction turns the latter into an intrinsic height obstruction; nearest-grid Newton bands construct a sufficient height certificate at square-root target scale. For continued-fraction comparison, the finite trace and its continuants give generated stages; successor geometry and the root-form invariant certify every Lakein factor. The source-lattice cylinder then characterises best pairs, and the source-bounded rectangle computes their complete finite list. The main work of the paper was not discovering a new completed object, but turning each customary appeal to a tail, a picture, a symmetry, a lattice sequence, or a finite choice into displayed data and a checkable certificate.

12.1 Replacing the infinite argument by a finite one

Classical accounts of Hurwitz continued fractions naturally discuss an infinite expansion, its complete quotients, and the complex value represented by the expansion [4, 5, 6]. For the rational inputs of this paper those objects are unnecessary, but removing them is not a word-for-word translation. A reference to "the tail" must become one displayed integral remainder and its proof of nonzeroness. A backwards argument along an infinite sequence must become induction on the constructors of a finite generated trace. A limiting identity must become the exact stage-reconstruction identity (38).

This does not dispute the mathematical conclusions of the earlier proofs. It changes their dependency boundary. The result proved here requires no completed source value, infinite digit stream, convergence theorem, or selection of a limiting object. Once those conveniences are unavailable, however, several steps that look routine in prose have to be supplied as substantial pieces of mathematics.

The same issue appears in the height argument. Density of rational points in a completed plane would say too little: it would neither construct the denominator nor relate it to the residual certificate. The grid fraction of definition 3.1 instead records the exact denominator chosen from the Newton-band radius, and lemma 3.2 proves the exact uniform minimality property of that choice. The bounded square-threshold scan turns a residual request into a reciprocal-integer ε without an appeal to a real square root. At the other end, the lower bound is not a distance from an assumed root; it is the quadrance floor of the explicitly computed Gaussian integer $W_f(U, V)$. The necessary obstruction and sufficient construction are therefore two different finite certificates rather than one asymptotic existence statement.

12.2 What the geometric shorthand concealed

The succession part of the classical proof is often conveyed by a diagram: a complete quotient, or its inverse, lies outside one of several discs. A diagram does not identify the certificate needed by the present theorem. For each orientation we had to name the relevant Gaussian unit, clear an explicitly nonzero denominator, expand the resulting quadrance coefficientwise, and prove its sign by rational inequalities. Distances and square roots contribute nothing useful here; quadrances preserve the exact finite algebra.

The deterministic tie rule makes this more than bureaucratic expansion. Our fundamental square is half-open:

$$[-1/2, 1/2)^2.$$

Some successor inequalities are therefore strict while their reflected counterparts are weak. A phrase such as "the remaining cases follow by symmetry" can erase precisely the strict inequality that excludes a boundary digit. The proof in section 6 retains every orientation. When a weak boundary remains, it supplies a second exterior certificate rather than pretending that reflection preserved the tie rule.

Likewise, saying that only finitely many Gaussian integers remain is not yet a constructive classification. The proof first derives coordinate bounds, then displays the bounded integers, and finally assigns every case to an explicit certificate. This is why the twenty-eight forbidden successors in definition 6.9 are mathematical data, not an incidental implementation table.

12.3 The exceptional state machine

The quadrance-five case was the point at which informal reconstruction was most unreliable. Numerical exploration suggested the eight Hermitian root forms and was useful for finding the finite state machine. It was not a proof. In an early exploration, a pulled-back form whose quadratic coefficient vanished was mistakenly treated as infeasible. Such forms are feasible, and some occur in actual finite Hurwitz traces.

The corrected proof never discards those states. It establishes the exact pullback identity, classifies the twenty-four exceptional first pullbacks, and checks the $24 \cdot 32 = 768$ bounded second-step substitutions. A result is accepted only if it is an original root form, an explicitly nonnegative augmentation, or a form with a displayed slack certificate. The remaining branch supplies one of the forbidden adjacent digit pairs and is ruled out by the exact succession law. Induction then transports the eight root inequalities along the finite trace.

This episode gives a useful standard for computer-assisted exploration. Floating-point searches may reveal a plausible list, but they do not belong to the theorem's trust boundary. Every transition used here is justified by an integral coefficient identity; every discarded transition has a rational inequality or a named forbidden-successor witness; and the finite tables are checked by the Lean kernel without a new native-decision axiom.

12.4 Formalisation and the audited boundary

Our broader complaints about Lean's authoring language, unnecessary choice-bearing routes through Mathlib, and the interface exposed by `Quot.sound` are elaborated in the companion paper [1]. That paper also describes the eventual replacement of the quotient interface by canonical reduced records. The complaints need not be repeated here.

The paper-specific point is that the additional Hurwitz and all-best machinery preserves that audited boundary. Executable Gaussian division, continuant recursion, rational square inequalities, finite Hermitian-form classification, cylinder minimality, and bounded candidate filtering leave the public transitive axiom closures empty. Neither `propext`, `Classical.choice`, `Quot.sound`, nor any other axiom appears; the exact finite certificates are checked without native-decision axioms.

12.5 Scope and next questions

The denominator-height obstruction is universal. Its coefficient scale is least for the normalized coefficient list, its scaled residual is exactly an integral quadrance, and theorem 2.11 shows joint sharpness in the displayed scale, height, and degree. The Gaussian-denominator theorem, the two explicit presentation conversions, and executable gcd reduction remove the earlier gap between the common, displayed Gaussian, and intrinsic height records. Unit normalization makes the reduced record canonical, so equal displayed values have equal intrinsic heights without selecting from a quotient.

The Newton denominator is least for the uniform strict half-mesh condition, not necessarily for one particular centre or polynomial. Likewise, lemma 3.5 makes the geometric factor $1/2$ exact under the stated band constraints, but does not claim that a particular eliminant-selected band is widest or that its ambient Lipschitz majorant is sharp. The analogous Horner majorants and Bézout protection margins at a rational-function site can contain further problem-specific slack.

The calculation-size question is now closed at the intended level in both papers. RFAP-I supplies source-computed T and B bounds for its normalized eliminant, Bézout, coefficient-clearing, Sturm, winding, safe-cut, and requested-subdivision calculations; theorem 10.11 does the same for the normalized Newton and RFAP-II post-processing family. The bounds are finite certificates, not sharp complexity estimates. A logarithmic gcd analysis, tighter subresultant and Sturm growth, common-grid subdivision, or a more detailed machine-cost model could reduce them, but no such refinement is needed to bound the intermediate displayed records used by the two papers.

The Hurwitz–Lakein theorem is also precise about what it does not claim. It compares the division-free residual of each generated continuant with every competitor of strictly smaller Gaussian denominator quadrance. The new all-best theorem is a separate result: for a displayed rational source it proves the minimal-vector equivalence, derives a finite search rectangle, and returns every best pair. It also decides the exact consecutive occupied denominator levels by filtering that rectangle. Finite descent recovers the

equivalent empty-mixed-cylinder condition; the Euclidean complement and half-square boundary calculation show that every consecutive determinant quadrance is one or two; and the Cramer certificate explains the determinant algebra without a quotient carrier. The terminal Hurwitz pair is best, but the paper does not claim that the ordinary Hurwitz trace enumerates the rest. Chevallier's algorithm supplies the stronger structural answer for completed complex inputs by advancing through consecutive minimal vectors [9]. Our coefficient-local generator now derives a complete bounded candidate list from one adjacent pair and validates it by comparisons confined to that list. Its executable update is therefore local, although it uses a bounded coefficient square rather than formalising Chevallier's more economical structural update formula. Our finite Euclidean and half-square calculation proves his index-one/index-two determinant classification for displayed rational sources. The coordinate-parity calculation also proves the centered-coset classification and the unit-and-divisible-coefficient normal form for a putative repeated index-two transition. Four centered candidates give the factor-four error estimate, and an exact Cauchy calculation restricts the remaining coefficient to the twelve Gaussian integers of quadrance one, two, or four. Exact normalization, unit-associate reduction, and the three finite cylinder calculations then rule out every value: an index-two transition is necessarily followed by an index-one transition. This completes the displayed-rational form of Chevallier's repeated-index lemma and proves the no-skipped-level property of the coefficient-local generator and validator. The normalized nearest-coefficient calculation in theorem 9.14 gives the generic root-free quadratic gap. Five small coefficient tests, after one of eight finite coordinate symmetries, split into four rational regions and strengthen the gap to $205/256$, bounding an index-one window by $256/205$. Four centered unit certificates force the two normalized coordinate quadrances above $15/56$; retaining their orientation and the affine minorant $2(U+1)/7$ raises the index-two gap to $205/128$ and gives the same window bound. Replacing the two fixed affine parameters by $s = \tau/3$ and $m = s/(1-s^2)$ proves the exact rational-cut family

$$\frac{3}{4} \le \tau < 1, \quad \tau^2 - 12\tau + 9 \ge 0 \quad \Longrightarrow \quad \tau\mathcal{D}_j \le 1, \quad \tau \sum_{j<n} \mathcal{D}_j \le n.$$

In the conventional completed notation, the reciprocal of the smaller quadratic endpoint is Chevallier's optimal squared constant $(2+\sqrt{3})/3 \approx 1.24$. The converse finite theorem now proves that for every displayed rational τ in the same range with $\tau^2 - 12\tau + 9 < 0$, and every natural scale T, the Pell recurrence (140) constructs a displayed rational source and consecutive index-one window beyond T satisfying $1 < \tau\mathcal{D}$. Thus every displayed rational strengthening of the cut has explicit counterexamples at unbounded requested scales. The arbitrary finite same-source construction strengthens this to any supplied run length; one proper rational box preserves the whole run. At every displayed rational source in that box, stopped coefficient-local successor dynamics reaches a selected critical state with probability one in blocks bounded by the source denominator quadrance plus one. The only non-successor boundary operations are the explicit terminal reset and absorbing seed hold. The universal half is now global over displayed rational data as well: for every natural squared request $R > 1$, finite maximization of the best-pair table returns a denominator quadrance below R with $\tau R\,\mathrm{N}(qz-p)^2 \le 1$.

What remains absent is Chevallier's stronger quantifier order: almost every fixed completed complex input defeats every improved constant at arbitrarily large scales. That statement uses the infinite completed setting and ergodic arguments outside the present discipline; none of the finite reset, stopping, or box-content statements is presented as a substitute for Haar invariance.

Within that boundary, the two parts of the paper now complement one another. The height theory measures what a finite root or pole approximation costs; the Hurwitz theory proves that a particular finite Gaussian Euclidean trace uses its denominator budget efficiently in the exact second-kind sense. Both conclusions are calculations on supplied records, and neither requires the completed values that those records may conventionally be said to approach.

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